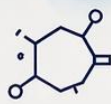


# THE CHEMISTRY OF RNA THERAPEUTICS

FOUNDATIONS, DESIGN,  
AND CLINICAL TRANSLATION



MECHANISMS  
OF ACTION



NUCLEIC ACID  
MODIFICATIONS



DESIGN  
STRATEGIES



DELIVERY &  
CONJUGATES

## JEFF MILTON

— CSO, LA JOLLA LABS —

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## References

**The Chemistry of RNA Therapeutics**  
Foundations, Design, and Clinical Translation

"Discipline is doing what you hate to do but doing it like you love it". --Mike Tyson

## Introduction

Dr Har Gobind Khorana, was an Indian American biochemist and molecular biologist. He shared the 1968 Nobel Prize in Physiology or Medicine with Marshall W. Nirenberg and Robert W. Holley for deciphering how nucleotides in nucleic acids control the synthesis of proteins, a foundational advance in molecular genetics.



For Dr Khorana, the painstaking process of synthesizing short DNA sequences may have seemed slow and tedious. Yet this disciplined, methodical work laid the foundation for a transformation in biology and medicine. In the mid-1950s, in the laboratory of Sir Alexander Todd, Har Gobind Khorana pioneered the use of synthetic DNA and RNA to address fundamental biological questions. He demonstrated that these synthetic molecules could serve as precise templates for DNA and RNA polymerases and developed a technique that would not only help decipher the genetic code but also established a powerful experimental framework for modern synthetic biology.

In the 1960s, Dr. Bob Letsinger, working with Dr. V. Mahadevan, developed the first method for synthesizing DNA on an insoluble polymer, the “phosphotriester method.” The efficiency of this approach enabled the creation of the first bioengineered genes; somatostatin, human insulin, and human growth hormone became early products of a new biotechnology company, Genentech.

Building on this method, Professor Marvin Caruthers advanced the field in the 1970s by developing nucleoside phosphoramidite chemistry for the solid-phase synthesis of DNA and RNA. This innovation enabled the precise and scalable production of oligonucleotides, establishing a method that remains the standard today. At the same time, foundational work by Eckstein and Gindl (1970) highlighted the importance of backbone chemistry in oligonucleotides. They demonstrated that replacing a non-bridging oxygen in the phosphate backbone with sulfur (phosphorothioates) preserved duplex formation and polymerase compatibility while significantly enhancing resistance to nuclease degradation. This work established a key principle: chemical modifications could improve stability without compromising biological function.

These advances in chemistry set the stage for therapeutic applications. Dr. Joseph Walder, founder of Integrated DNA Technologies, demonstrated that antisense oligonucleotides

could inhibit gene expression either by blocking translation or by recruiting RNase H enzymes to cleave target RNA, thereby linking sequence-specific hybridization to RNA degradation (Walder and Walder, 1988); what we now know of as the “gapmer” antisense oligonucleotide. Then, in the 1990s another major breakthrough came with the discovery of RNA interference by Dr. Andrew Fire and Dr. Craig Mello. They revealed a complementary pathway in which double-stranded RNA is processed into small interfering RNAs (siRNAs), which guide the RNA-induced silencing complex (RISC) to degrade target mRNA (Fire et al., 1998; Zamore, 2006).

Alongside the development of these sugar-furanose based technologies, morpholino oligonucleotides (PMOs), developed by James Summerton and colleagues, represented a significant advance in antisense platform engineering. These compounds act via steric blocking rather than enzymatic cleavage and are therefore used to modulate RNA expression through mechanisms such as RNA splicing and translational inhibition. The concept for morpholinos originated from Summerton’s work in the 1980s, largely conducted at AntiVirals Inc. (later Sarepta Therapeutics), where design considerations were strongly influenced by synthetic efficiency, stability and scalability (Summerton, 2017).

In PMOs, the ribose sugar is replaced by a morpholine ring and phosphorodiamidate linkage, enabling a more efficient synthesis from ribonucleoside precursors while eliminating the negatively charged phosphate backbone (Gene Tools, Inc.) This neutral backbone confers increased resistance to nuclease degradation while preserving strong, sequence-specific hybridization to RNA targets. Drawing on a long history of carbamate medicinal chemistry, Summerton and team explored the phosphorodiamidate morpholino chemistry as a practical and safe platform for RNA-targeting therapeutics. This technology was validated through a series of regulatory milestones, beginning with the approval of Eteplirsen in 2016 for Duchenne muscular dystrophy, followed by Golodirsen (2019), Viltolarsen (2020) and Casimersen (2021).

Within the framework of the central dogma, these discoveries converged into modern RNA-targeting therapeutic strategies. RNase H mediated knockdown uses antisense oligonucleotides to direct enzymatic RNA degradation, siRNA-based approaches harness RISC for sequence-specific cleavage and steric-blocking antisense oligonucleotides provide a versatile means of modulating RNA processing and function.

While single-stranded ASOs and siRNA represent the vast majority of oligonucleotide therapeutics in the clinic, a broader and still evolving landscape of alternative nucleic acid platforms is emerging. Among these are aptamers short, single-stranded oligonucleotides that fold into defined three-dimensional structures capable of binding proteins and other targets with high affinity and specificity. Similar to antisense and siRNA modalities, the clinical translation of aptamers is highly dependent on chemical modification strategies, including 2-ribose substitutions such as 2-fluoro (2-F) and 2-O-methyl (2-OMe), as well as terminal modifications like polyethylene glycol (PEG) conjugation to improve stability and pharmacokinetics. These approaches were exemplified in the first FDA-approved aptamer, pegaptanib (Macugen).



More recently, mRNA-based therapeutics propelled into the public by the success of COVID-19 vaccines have demonstrated the potential of nucleic acids as transient protein expression systems. Beyond vaccines, however, the broader therapeutic application of mRNA remains in its early stages, with ongoing efforts focused on overcoming key challenges in delivery, tissue targeting, and the control of immunogenicity. Taken together, these modalities underscore both the versatility and the shared challenges of nucleic acid therapeutics. Across platforms, chemical modification and delivery remain central determinants of clinical success.

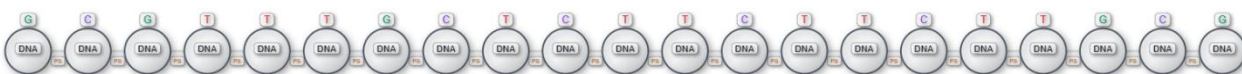
As the field continues to mature, future advances will likely arise not only from entirely new modalities, but from increasingly precise control over stability, specificity, and tissue targeting. Innovations in conjugation chemistry, delivery vehicles, and sequence design are already beginning to expand the therapeutic window of oligonucleotide-based drugs. Ultimately, these improvements will enable broader and more reliable applications across a wide range of diseases, moving nucleic acid therapeutics from specialized interventions toward more routine components of modern medicine.

## Therapeutic Mechanisms

RNA sits at the interface of genetic information and biological function, enabling control over protein expression without permanently altering the genome. Despite its transient nature, targeting RNA can produce robust and often long-lasting clinical effects, making it a versatile platform for a wide spectrum of therapies.

One of the major advantages of RNA therapeutics is their ability to precisely and programmatically regulate gene expression. Depending on the therapeutic objective, these molecules can either enhance protein production through mechanisms such as microRNA inhibition and correction of RNA splicing or suppress the expression of harmful genes using strategies including RNase H-mediated transcript degradation and siRNA-directed RNA cleavage. Unlike many conventional small-molecule drugs, which often rely on broad chemical interactions with proteins, RNA therapeutics function through predictable sequence complementarity, enabling highly selective targeting of specific transcripts. This sequence-driven precision not only improves specificity but also supports a more rational drug discovery process, where candidate therapies can be identified and optimized through precision-based genetic and molecular screening approaches.

Beyond gene silencing and activation, RNA therapeutics can reprogram RNA processing and function. Technologies such as exon skipping, cryptic exon modulation, and isoform switching can alter the proteins produced by cells without permanently changing the genome itself. More advanced strategies employ enzymes such as Adenosine Deaminase Acting on RNA (ADAR) to perform site-specific RNA base editing, enabling transient and potentially reversible correction of pathogenic mutations. Collectively, these approaches highlight how RNA therapeutics move medicine closer to a rational and modular framework for drug development, where treatments are engineered based on precise molecular knowledge of disease pathways rather than discovered primarily through empirical screening.



Fomivirsen (Vitravene™) was the first gene-silencing antisense therapy approved by the FDA, receiving approval in 1998. Originally discovered by a group at the NIH, it was first licensed by Isis Pharmaceuticals and later sublicensed to Novartis. The compound consists of a simple 21-base, DNA oligonucleotide in which a sulfur atom replaces one of the non-bridging oxygen atoms in the phosphate backbone, forming phosphorothioate linkages between nucleosides. Vitravene was developed to treat cytomegalovirus (CMV) retinitis in immunocompromised patients and functions by binding to mRNA transcripts that encode IE2 proteins, essential for viral replication (Geary, Henry, and Grillone, 2002).

Preclinical and in vitro studies demonstrated potent antiviral activity, with mean 50% inhibitory concentrations ( $IC_{50}$ ) ranging from 0.03 to 0.2  $\mu\text{M}$  across multiple cell lines and a median effective concentration ( $EC_{50}$ ) of approximately  $0.34 \pm 0.25 \mu\text{M}$  in human fibroblasts (Azad et al., 1993; Anderson et al., 1996).

|                         |                        |                                |
|-------------------------|------------------------|--------------------------------|
| <b>FOMIVIRSEN (ASO)</b> | 0.03–0.3 $\mu\text{M}$ | Very potent, sequence-specific |
| <b>GANCICLOVIR</b>      | 1–10 $\mu\text{M}$     | Antiviral comparator           |
| <b>IBUPROFEN</b>        | 1–20 $\mu\text{M}$     | Moderate potency               |
| <b>ASPIRIN</b>          | 10–1000 $\mu\text{M}$  | Low potency                    |
| <b>PARACETAMOL</b>      | 100–1000 $\mu\text{M}$ | Very low potency               |
| <b>IMATINIB</b>         | ~0.3 $\mu\text{M}$     | High potency targeted drug     |

Vitravene demonstrated striking potency with  $EC_{50}$  values in the range of 0.03–0.3  $\mu\text{M}$ , highlighting the robustness of a sequence-specific mechanism of action. In comparison, traditional antivirals like ganciclovir showed lower potency ( $EC_{50} \approx 1\text{--}10 \mu\text{M}$ ), while commonly used small molecules such as ibuprofen (1–20  $\mu\text{M}$ ) and aspirin (10–300  $\mu\text{M}$ ) exhibited only moderate to low activity. Paracetamol was even less potent (100–1000  $\mu\text{M}$ ), underscoring the substantial gap between conventional drugs and targeted nucleic acid therapeutics. Thus, reinforcing the promise of rational, sequence-driven drug design. Despite its scientific significance, it was ultimately a commercial failure. The drug was withdrawn in 2000, largely due to the success of highly effective antiretroviral therapies for HIV, which dramatically reduced the incidence of CMV retinitis, the very condition Vitravene was designed to treat.

The clinical landscape of RNA therapeutics has expanded substantially, encompassing a diverse and increasingly sophisticated set of approaches that act across multiple layers of gene regulation and cellular function. Established commercial therapies include RNase H–mediated knockdown (e.g., inotersen for transthyretin amyloidosis), siRNA-driven RISC-mediated degradation (e.g., patisiran and inclisiran), aptamers that bind and inhibit protein targets (e.g., pegaptanib/Macugen), RNA splicing modulation (e.g., nusinersen), and mRNA-based strategies for both protein replacement and vaccination (e.g., mRNA COVID-19 vaccines) (Adams et al., 2018; Ray et al., 2017; Hoy, 2018; Finkel et al., 2017; Polack et al., 2020).

Beyond these established modalities, additional and conceptually distinct mechanisms of action are emerging. Among them, microRNA (miRNA)-based therapies enable network-level regulation by modulating entire gene programs either through inhibition (anti-miRs; e.g., miravirsen targeting miR-122 in HCV) or restoration (miRNA mimics; e.g., MRX34 in oncology) rather than acting on single transcripts (van der Ree et al., 2017; Beg et al., 2017). In parallel, RNA activation (RNAa) has emerged as a clinically validated strategy for upregulating endogenous gene expression using small activating RNAs (saRNAs).

Complementing activation-based approaches, RNA editing therapeutics particularly those leveraging adenosine-deaminases-acting-on-RNA (ADARs) introduce the ability to precisely modify RNA sequences at the transcript level. These systems enable site-specific A-to-I base conversion, which is interpreted as guanosine during translation, allowing correction of pathogenic mutations without permanent genomic alteration. Early clinical programs, such as WVE-006 for alpha-1 antitrypsin deficiency, demonstrate the translational potential of this approach by recruiting endogenous ADAR enzymes to restore functional protein expression in patients (Rainaldi, Mali and Nourredine, 2025).

Beyond editing and activation, additional emerging modalities further broaden the mechanistic scope of RNA therapeutics. Non-degradative antisense oligonucleotides can function through steric blocking to inhibit translation or disrupt RNA–protein interactions without inducing RNA cleavage, as seen in splice-switching therapies. RNA molecules can also act as innate immune agonists, for example synthetic RNAs that activate TLR7/8 or RIG-I pathways in oncology. Advances in expression platforms, including self-amplifying RNA (e.g., saRNA vaccine candidates such as ARCT-021) and circular RNA constructs, enhance the magnitude and durability of protein production (Blakney et al., 2021). Finally, RNA decoys and scaffolds can sequester proteins or regulatory RNAs, expanding RNA therapeutics into modulation of protein–RNA interaction networks.

These developments reflect a clear shift from first-generation paradigms centered on gene silencing or replacement toward a more expansive framework in which RNA therapeutics can activate, edit, regulate, and reprogram cellular behavior with increasing precision. Clinical examples spanning RNAi, antisense, RNAa, miRNA modulation, and RNA editing underscore that this mechanistic diversity is already being realized in the clinic, marking a transition toward a more versatile and programmable class of medicines (Adams et al., 2018; Reebye et al., 2018; Rainaldi, Mali and Nourredine, 2025).

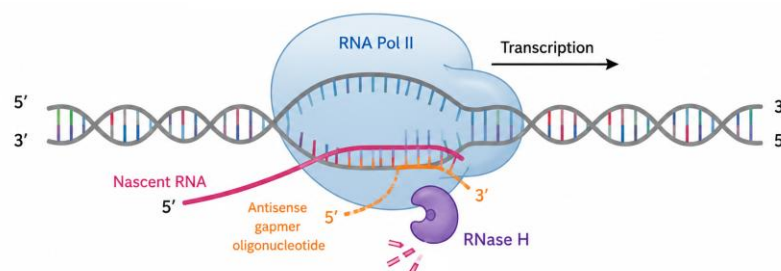
## *RNase H–Mediated, single-stranded antisense oligonucleotides (Gapmers)*

*“Die Nucleinsubstanzen liefern bei ihrer Spaltung verschiedene stickstoffhaltige Spaltungsprodukte, welche zu den wichtigsten Bestandteilen des Zellkerns gehören.”*

*--Kossel, A., 1910. Die chemische Zusammensetzung der Zellkerne*

Albrecht Kossel working at the University of Heidelberg first described nucleic substances in terms of their “cleavage products” (*Spaltungsprodukte*), illustrating that nucleic acids could be broken down into distinct nitrogen-containing bases. For this work Dr. Kossel was awarded the Nobel Prize in Physiology or Medicine. Kossel’s observations laid the foundation for the later discovery of nucleases; enzymes specialized in the cleavage and degradation of DNA and RNA. During the mid-twentieth century, the identification of bacterial restriction endonucleases transformed molecular biology by enabling precise DNA manipulation for cloning, sequencing and recombinant DNA technology (Arber, 1979). Since then, nucleases including TREX1, EXO1 and MRE11 have been recognized as essential regulators of DNA replication, repair, apoptosis and innate immune defense in human systems, while also becoming indispensable tools in biotechnology, genome engineering and CRISPR-based applications (Roberts, 2005; Mazur and Perrino, 2022).

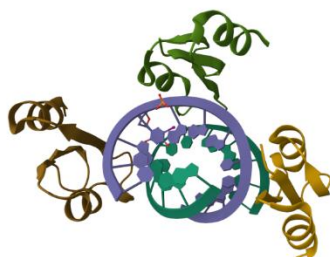
In 1976, venture capitalist Robert Swanson and biochemist Dr. Herbert Boyer founded Genentech to commercialize recombinant DNA methods pioneered by Boyer and Stanley Cohen, demonstrating that human genes could be inserted into bacteria to produce therapeutic proteins. This work led to the first successful production of human proteins in bacteria, including synthetic human insulin, which became the first recombinant pharmaceutical approved for clinical use in 1982. Genentech had established a new model for advancing therapeutics; one that deeply integrated our understanding of biology with the our ability to use it for human therapeutics; a field we call Biotechnology.



RNASE H ACTS AT THE TIME OF TRANSCRIPTION

RNase H is an endonuclease that plays a fundamental role in maintaining genome stability by selectively degrading the RNA strand of RNA–DNA hybrids. During DNA

replication, it removes RNA primers required for initiation, ensuring their replacement with DNA and preserving genomic integrity. It also contributes to DNA repair by removing ribonucleotides, preventing structural instability and mutations (Eder, Walder and Walder, 1993). In addition, RNase H resolves R-loops that form during transcription, which, if accumulated, can interfere with replication and lead to DNA damage. Collectively, these functions establish RNase H as a key enzyme linking nucleic acid metabolism with genomic integrity (White, R.L. and Davis, R.W., 1976).

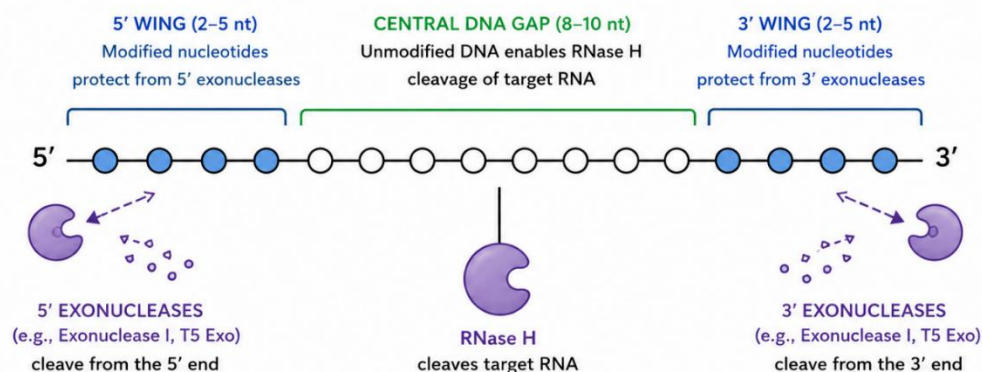


X-RAY DIFFRACTION HYBRID-BINDING DOMAIN OF HUMAN RNASE H1 IN COMPLEX WITH 12-MER RNA/DNA (PDB 3BSU) (NOWOTNY, 2008)

Antisense oligonucleotide (ASO) therapeutics exploit these natural biological mechanisms for therapeutic purposes. When an ASO hybridizes to a complementary RNA transcript, it forms an RNA–DNA duplex that serves as a substrate for RNase H. The enzyme subsequently cleaves the phosphodiester backbone of the RNA strand, generating RNA fragments and preventing translation of downstream protein products.

To apply this mechanism therapeutically, scientists needed to overcome the challenge of delivering DNA-like molecules to their RNA targets while protecting them from rapid nuclease degradation and preserving their ability to recruit RNase H. This led to the development of chemical modifications that maintain the DNA-like properties required for RNase H activity while substantially improving oligonucleotide stability in biological systems. Early unmodified oligonucleotides were highly susceptible to enzymatic degradation, severely limiting their therapeutic potential. This limitation was first addressed through backbone modifications, most notably phosphorothioate linkages, which confer resistance to nuclease-mediated degradation while retaining compatibility with RNase H activity. In parallel, sugar modifications such as 2'-O-methyl and 2'-O-methoxyethyl groups were introduced to enhance binding affinity, increase metabolic stability, and improve overall pharmacological properties.

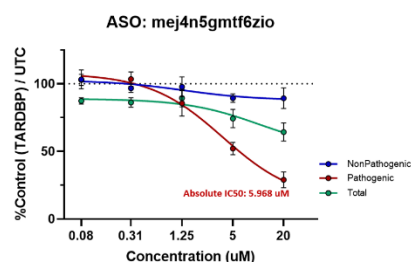
These are often incorporated into “gapmer” designs, where a central DNA region capable of recruiting RNase H is flanked by modified nucleotides for protection. Collectively, these innovations enable antisense oligonucleotides to form stable RNA–DNA hybrids, activate RNase H, and achieve sufficient stability, bioavailability, and specificity for effective therapeutic application.



These advances in antisense chemistry and RNase H-mediated mechanisms have transformed ASOs from experimental molecular tools into precision therapeutics capable of targeting disease-causing transcripts with remarkable specificity. What began with Kossel's early characterization of nucleic substances and the later discovery of nucleases has culminated in a new era of individualized genetic medicine, where pathogenic mutations can be selectively silenced at the RNA level while preserving normal cellular function.

La Jolla Labs, a neurogenetics-focused therapeutics company recently identified and developed an allele-selective antisense oligonucleotide targeting a familial form of amyotrophic lateral sclerosis (ALS). By leveraging advances in ASO chemistry the program moved from mutation identification to clinical therapeutic development in less than six months, an unprecedented timeline in the field of precision neurogenetics. The ASO was specifically engineered to selectively suppress the mutant allele while sparing expression of the wild-type transcript, thereby minimizing the risk of disrupting essential physiological protein function.

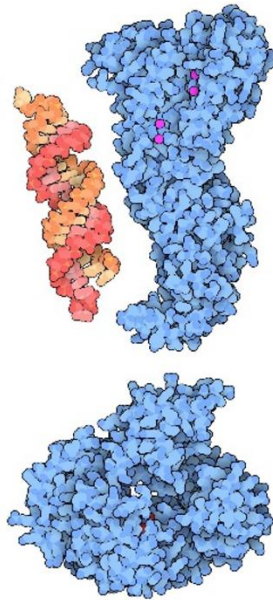
| 1. Risultato Primario                                          |                              |                  |             |           |             |            |
|----------------------------------------------------------------|------------------------------|------------------|-------------|-----------|-------------|------------|
| GENE                                                           |                              | VARIANTE         |             |           |             |            |
| TARDBP<br>1p36.22                                              | c. 1132A>G<br>p. (Asn378Asp) | dbSNP            | -           | Effetto   | Missenso    |            |
|                                                                |                              | Frequenza gnomAD | -           | PhyloP    | -           |            |
|                                                                |                              | SIFT             | 0.2/0.00    | Polyphen2 | 0.025/1.00  |            |
|                                                                |                              | Ref Seq*         | NM_007375.4 | Zigosit   | Eterozigosi | Cadd-Pheed |
|                                                                |                              |                  |             | Coverage  | 334.186.148 |            |
| Variante riportata in letteratura, potenzialmente patogenetica |                              |                  |             |           |             |            |



The rapid development of this therapy illustrates how decades of fundamental discoveries in nuclease biology, RNA-DNA hybrid metabolism, and oligonucleotide engineering are now enabling highly personalized interventions for previously untreatable neurodegenerative diseases. The patient is currently being dosed at the University of Heidelberg, bringing this scientific trajectory full circle: from Kossel's foundational investigations of nucleic acids in Heidelberg more than a century ago to the modern application of RNase H-dependent antisense therapeutics in the treatment of human disease at the same institution. This convergence of historical discovery, molecular biology, and translational medicine highlights the extraordinary progress of nucleic acid therapeutics and underscores the growing potential of allele-selective ASOs as rapid, adaptable treatments for rare genetic disorders.



## RISC-Mediated RNA Interference (siRNA)



DAVID S. GOODSSELL

Small interfering RNA (siRNA) is a powerful cellular defense system that helps protect organisms from viral infections and regulate gene expression. Cells are particularly sensitive to double-stranded RNA, which is often produced during viral replication and signals potential danger. In response, specialized proteins such as dicer (see top blue protein in David Goodsell's watercolor illustration) cut these double-stranded RNA into short fragments. These are then incorporated into argonaut proteins (bottom protein in Goodsell's illustration; PDB 248S), which use one strand as a guide to identify and destroy matching messenger RNA (mRNA) molecules. Through this process, known as RNA interference, cells can selectively silence specific genes or eliminate viral RNA (Fire, A., Xu, S., Montgomery, M.K., Kostas, S.A., Driver, S.E. and Mello, C.C. 1998).

Andrew Z. Fire and Craig C. Mello received the 2006 Nobel Prize in Physiology or Medicine for this discovery of RNA interference (RNAi). Their landmark experiments, published in *Nature* in 1998, fundamentally changed the understanding of gene regulation in eukaryotic organisms and opened entirely new avenues in molecular biology, biotechnology, and medicine.

Fire and Mello's work was conducted using the nematode *Caenorhabditis elegans* (*C. elegans*), a small transparent roundworm widely used as a model organism in genetics. At the time researchers knew that introducing "antisense" RNA could suppress gene expression. However, the effects were weak and inconsistent, and the mechanism remained unclear. Fire and Mello hypothesized that the true silencing trigger might not be single-stranded antisense RNA itself, but rather a "contaminating" double-stranded RNA formed during their experimental preparation.

To test this idea, they designed a series of carefully controlled experiments targeting the *unc-22* gene in *C. elegans*. This gene is responsible for normal muscle function; when it is disrupted, worms exhibit a characteristic twitching movement that can easily be observed under a microscope. The researchers synthesized three types of RNA corresponding to the *unc-22* sequence:

*Sense RNA (matching the mRNA sequence)*

*Antisense RNA (complementary to the mRNA)*

*Double-stranded RNA created by hybridizing the sense and antisense strands*



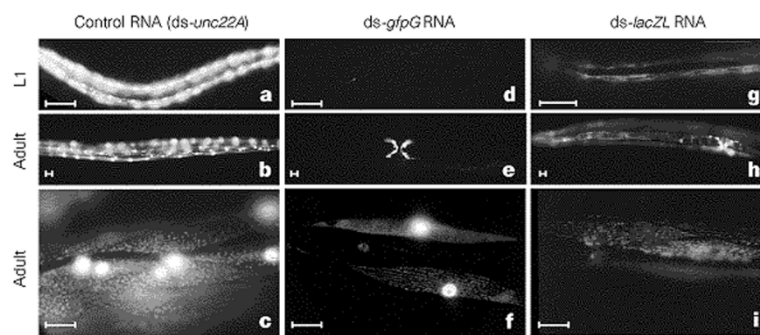
The results were striking; injection of either sense RNA or antisense RNA alone produced little or inconsistent gene silencing. In contrast, injection of tiny amounts of double-stranded RNA caused potent and highly specific suppression of the *unc-22* gene. The treated worms displayed the expected twitching phenotype, indicating that expression of the target gene had been effectively shut down. Importantly, Fire and Mello demonstrated that the effect was:

*Sequence-specific: only genes matching the dsRNA sequence were silenced.*

*Highly potent: extremely small amounts of dsRNA were sufficient.*

*Systemic: the silencing effect spread throughout the organism.*

*Heritable: gene silencing persisted in subsequent generations of worms.*



The figure demonstrates that double-stranded RNA (dsRNA) can specifically silence gene expression in animal offspring through RNA interference. Control animals injected with ds-*unc22A* showed normal GFP expression in both nuclei and mitochondria, while animals injected with ds-*gfpG* displayed a dramatic loss of GFP expression, with only a few cells retaining activity. In contrast, ds-*lacZL* selectively eliminated nuclear GFP-LacZ expression without affecting mitochondrial GFP, showing that RNA interference can precisely target specific genes while leaving others unchanged.

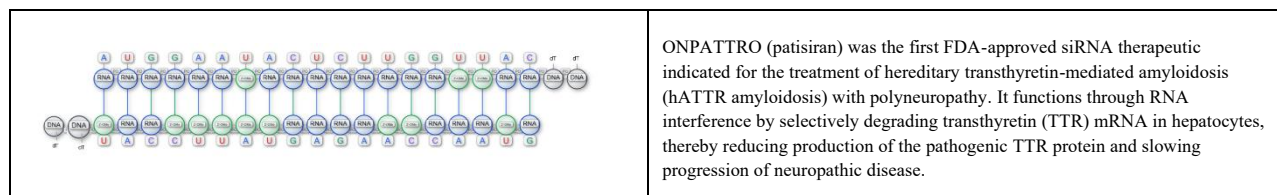
Fire, A., Xu, S., Montgomery, M. *et al.* Potent and specific genetic interference by double-stranded RNA in *Caenorhabditis elegans*. *Nature* **391**, 806–811 (1998).  
<https://doi.org/10.1038/35888>

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They repeated similar experiments with other genes and obtained comparable results, confirming that RNAi represented a general biological mechanism rather than a gene-specific anomaly. Their findings suggested that cells possess machinery capable of recognizing dsRNA and use it as a signal to destroy complementary mRNA molecules. Subsequent research revealed the molecular basis of this pathway. Double-stranded RNA is processed by an enzyme called Dicer into short interfering RNAs (siRNAs), typically around 21–23 nucleotides long. These siRNAs are incorporated into the RNA-induced silencing complex (RISC), which uses the siRNA sequence as a “guide” to identify and cleave matching mRNA targets, thereby preventing protein production.

RNAi has since become an important therapeutic strategy; one that enables precise silencing of disease-associated genes, including targets often considered “undruggable” by conventional therapies (Hu et al., 2020). Since the first FDA approval of Patisiran in 2018 for hereditary transthyretin-mediated amyloidosis, the field has advanced into a clinically validated therapeutic modality. Subsequent approvals of Givosiran, Lumasiran, Inclisiran, Vutrisiran, and Nedosiran expanded its reach into metabolic, cardiovascular, and rare genetic diseases. As of 2025, more than 260 siRNA drug candidates were in preclinical or clinical development. Research now spans major therapeutic areas, including oncology, neurodegenerative disease, infectious disease, and diabetes.

Despite its therapeutic promise, RNA interference (RNAi) therapy continues to face significant challenges related to the efficient delivery of siRNAs to their intended targets. Unmodified siRNAs are highly unstable in vivo and are rapidly degraded by endogenous ribonucleases, resulting in extremely short circulation half-lives. In addition, their negatively charged, hydrophilic nature severely limits passive diffusion across cellular membranes, making intracellular delivery highly inefficient. Even when cellular uptake occurs, most siRNAs become trapped within endosomal compartments, preventing access to the cytoplasm where gene silencing takes place. Consequently, endosomal escape is widely regarded as the rate-limiting barrier restricting broader therapeutic applications of RNAi. Although numerous delivery strategies have been developed, including lipid conjugates, cationic peptides, phosphorothioate modifications, and viral delivery systems, many of these approaches are associated with toxicity caused by nonspecific endosomal disruption (Dowdy, 2023).

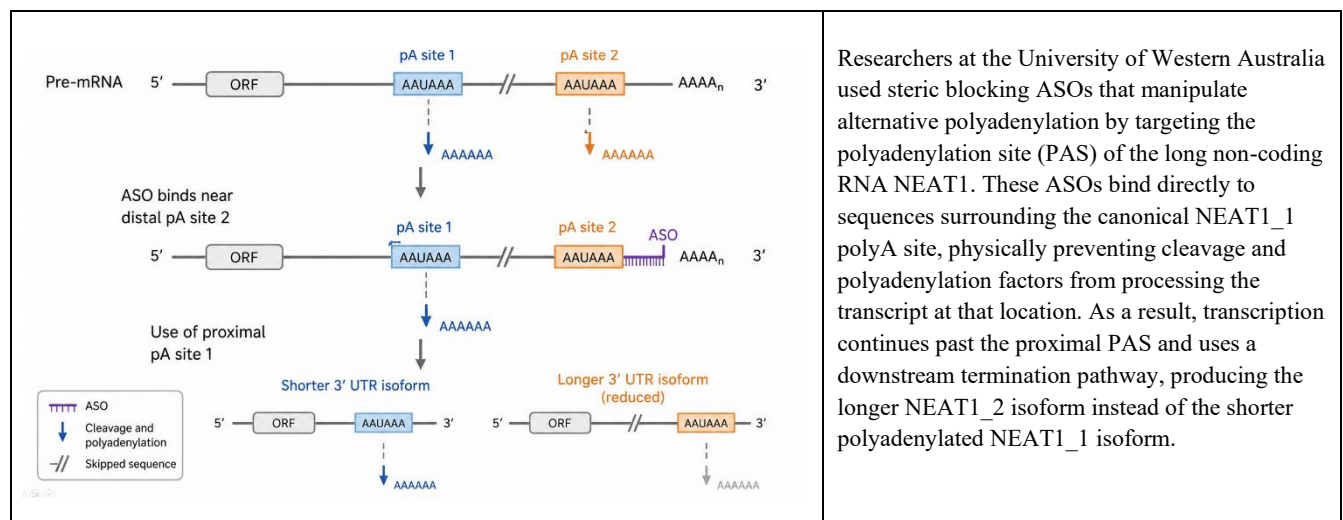
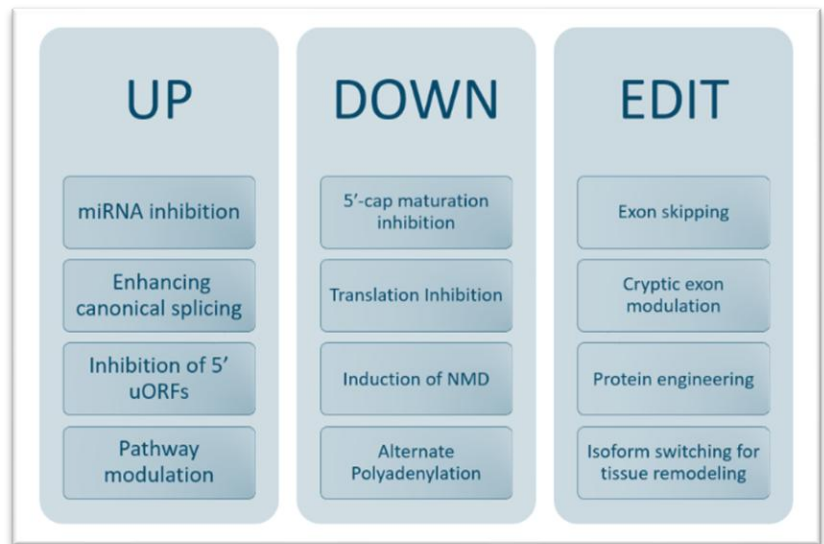


In addition to delivery challenges, naked siRNAs can activate innate immune sensors, triggering inflammatory responses and unintended off-target gene silencing. To overcome these limitations, researchers have developed a range of sophisticated chemical modifications designed to improve siRNA stability, specificity, and pharmacokinetic behavior. Common modifications include 2'-O-methyl, 2'-fluoro, and phosphorothioate substitutions, while more advanced structural designs such as glycol nucleic acids (GNAs) further enhance binding affinity and target selectivity. Terminal conjugations, including cholesterol, GalNAc ligands, antibodies, and aptamers, have also enabled tissue-specific targeting and improved biodistribution. These advances in chemical engineering and delivery technology are transforming siRNA therapeutics into an increasingly versatile and clinically mature platform.

## Steric blocking

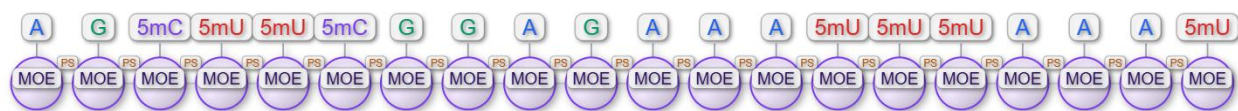
Steric-blocking antisense can offer a broad range of creative therapeutic opportunities. By physically blocking interactions between RNA and cellular proteins or other nucleic acids, these ASOs can alter pre-mRNA splicing, inhibit translation, suppress microRNA activity, or redirect RNA processing events such as polyadenylation. The clinical success of steric-blocking

ASOs has been driven largely by splice altering therapies for neuromuscular disease. Nusinersen, a 2'-MOE phosphorothioate ASO approved in 2016 for spinal muscular atrophy, demonstrated that targeted exon inclusion could restore therapeutically meaningful protein levels by promoting the retention of an unstable exon seven in SMN2. Phosphorodiamidate morpholino oligomer (PMO)-based exon-skipping therapies were also approved for Duchenne muscular dystrophy, including eteplirsen, golodirsen, viltolarsen, and casimersen, each designed to restore the DMD reading frame through selective exon skipping. The first ever personalized therapy was a splice correcting ASO; milasen was designed to suppress a cryptic exon and highlighted the potential for ASOs as a platform for rapid, patient-specific genetic therapy.

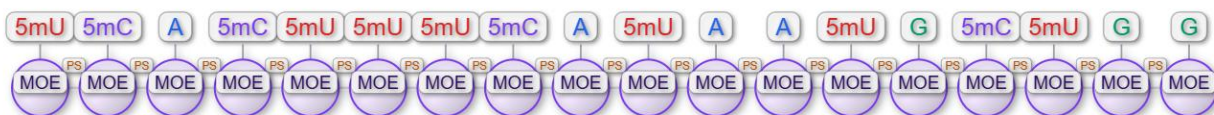


Researchers at the University of Western Australia used steric blocking ASOs that manipulate alternative polyadenylation by targeting the polyadenylation site (PAS) of the long non-coding RNA NEAT1. These ASOs bind directly to sequences surrounding the canonical NEAT1\_1 polyA site, physically preventing cleavage and polyadenylation factors from processing the transcript at that location. As a result, transcription continues past the proximal PAS and uses a downstream termination pathway, producing the longer NEAT1\_2 isoform instead of the shorter polyadenylated NEAT1\_1 isoform.

Naveed, A., Cooper, J.A., Li, R., Hubbard, A., Chen, J., Liu, T., Wilton, S.D., Fletcher, S. and Fox, A.H. (2021) 'NEAT1 polyA-modulating antisense oligonucleotides reveal opposing functions for both long non-coding RNA isoforms in neuroblastoma', *Cellular and Molecular Life Sciences*, 78, pp. 2213–2230. Available at: <https://doi.org/10.1007/s00018-020-03632-6>



Ulteversen (QR-421a) is a 21-mer chemically modified splicing-altering ASO for the treatment of USH2A-associated retinitis pigmentosa and Usher syndrome type 2. The molecule incorporates a fully phosphorothioate backbone together with all nucleosides modified with 2'-O-(2-methoxyethyl; aka MOE) sugars (Girach et al., 2022). Ulteversen binds selectively to USH2A pre-mRNA and induces exon 13 skipping, thereby restoring production of a shorter but functional usherin protein (Girach et al., 2022). This mechanism is particularly relevant for USH2A disease because the large size of the USH2A gene limits conventional AAV-mediated gene replacement strategies. Intravitreal administration allows broad retinal distribution without the need for subretinal surgery or viral vectors; reducing procedural and inflammatory risks associated with gene therapy (Girach et al., 2022). Preclinical studies demonstrated sustained exon skipping, restoration of usherin expression, retinal uptake across photoreceptor layers, and recovery of functional electroretinographic responses (Dulla et al., 2021). Clinically ulteversen demonstrated a favorable tolerability profile with no serious ocular adverse events or treatment discontinuations, alongside trends toward stabilization of retinal structure and visual function compared with sham treatment (Yang et al., 2025). Preservation of ellipsoid zone area, together with stabilization of visual acuity and retinal sensitivity measures, suggests that ulteversen may slow progressive photoreceptor degeneration in USH2A-associated RP, supporting continued evaluation in ongoing Phase 2b/3 studies (Yang et al., 2025).



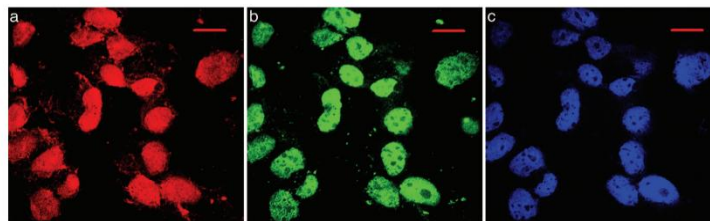
Another splicing example, Spinraza (nusinersen) is also a fully modified antisense oligonucleotide with a phosphorothioate backbone and fully modified 2'-O-methoxyethyl sugar 18mer. This ASO enhances the stability of the SMN2 pseudogene and is clinically indicated for the treatment of spinal muscular atrophy (SMA) in both pediatric and adult patients. The dosing regimen consists of four loading doses (three given at 14-day intervals followed by a fourth after 30 days), followed by a 12 mg maintenance dose every four months (Biogen, 2016) illustrating the therapeutic robustness of this compound in the CNS.

Beyond the approved therapies for SMA and Duchenne muscular dystrophy, many splice-switching antisense oligonucleotides (ASOs) are now in preclinical and clinical development across a wide range of diseases. Active programs target rare neuromuscular and neurodevelopmental disorders, including Dravet syndrome, Batten disease, ataxia-telangiectasia, Huntington's disease, as well as emerging applications in cancer and immune-mediated disorders.

## Short Activating RNAs (saRNA)

Short activating RNAs (saRNAs) are a class of small double-stranded RNA molecules that upregulate gene expression by targeting promoter regions and triggering RNA activation (RNAa). In contrast to siRNAs, saRNAs work within the nucleus, often involving Argonaute proteins, to enhance transcription of specific genes.

A conundrum might seem that siRNAs are understood to function in the cytoplasm, whereas saRNAs, with the same size act within the nucleus to regulate transcription. Early studies of RNA activation infer nuclear localization indirectly, showing that promoter-targeted dsRNAs induce locus-specific chromatin modifications (Li et al., 2006; Janowski et al., 2007). Direct imaging studies demonstrate that siRNAs can enter the nucleus in a selective, target-dependent manner, localizing to nuclear RNAs such as 7SK (RNA polymerase II transcription elongation) and U6 (an RNA that is part of the spliceosome) (Berezhna et al., 2006). However, this is balanced by evidence that small RNAs are not freely distributed and may be actively excluded from parts of the nucleus, indicating regulated trafficking rather than constitutive nuclear access (Ohrt et al., 2006). Together, these findings suggest saRNA activity is context-dependent and maintains selective localization, rather than an absolute compartmental separation.



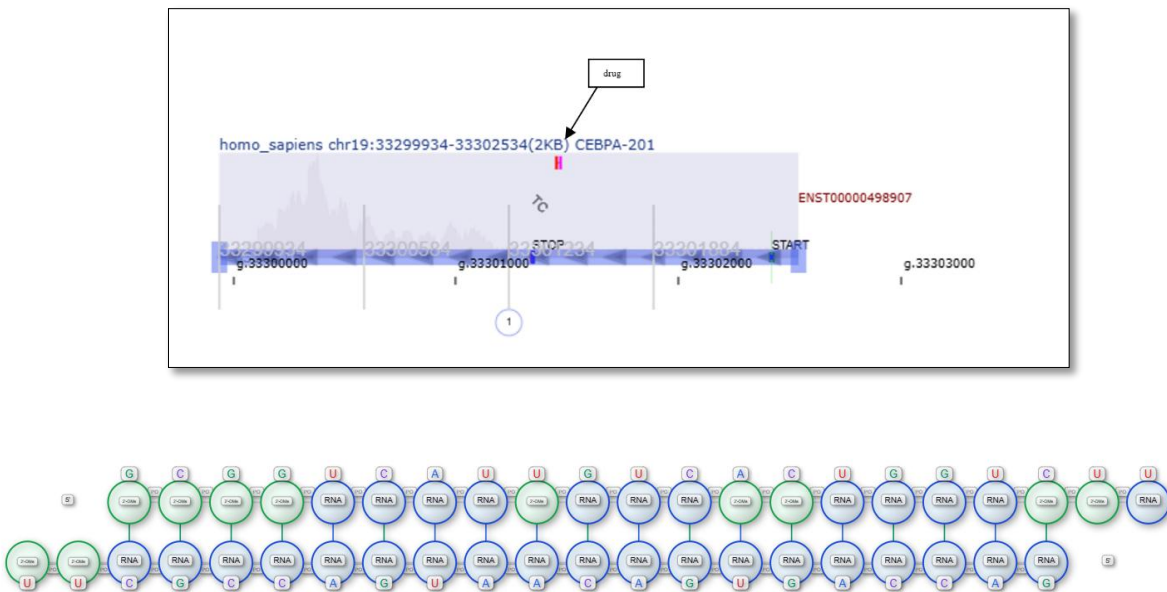
7SK siRNA accumulates in the nucleus in Huh-7 cells 2 h after transfection. siRNA antisense (red) and sense (green) strands colocalize primarily in the nucleus (blue) and seen in the cytoplasm only in small amounts

Berezhna, S.Y., Supekova, L., Schultz, P.G. and Deniz, A.A. (2006) 'siRNA in human cells selectively localizes to target RNA sites', *Proceedings of the National Academy of Sciences*, 103(20), pp. 7682–7687.

Because of this unique ability, saRNAs are emerging as promising tools for therapeutic gene activation (Desaulniers *et al.*, 2025). MTL-CEBPA (CEBPA-51/AW1-51), a liposomal encapsulated 21-nt saRNA targeting the transcription factor CEBPA, which is frequently silenced in hepatocellular carcinoma. In a first-in-human Phase I study (NCT02716012), MTL-CEBPA demonstrated an acceptable safety profile alongside evidence of target gene activation, antitumor activity, including disease stabilization and occasional objective responses. These



results supported advancement into randomized Phase II studies and combination trials with immunotherapies, establishing MTL-CEBPA as a prototype RNAa therapeutic and demonstrating the feasibility of systemic gene activation in humans (Reebye et al., 2018).



Another cancer-targeting RNAa therapeutic is RAG-01 (Ractigen Therapeutics), which targets the tumor suppressor CDKN1A (p21). RAG-01 is delivered via intravesical administration, a method where the drug is delivered directly into the bladder through a catheter, to patients with non-muscle invasive bladder cancer (NCT06351904). This compound attempts to directly activate p21, a historically “undruggable” pathway. Early Phase I data indicate favorable safety, absence of dose-limiting toxicities, and encouraging preliminary efficacy signals, including complete responses in carcinoma in some patients (Kang et al., 2024).

miR-551b-3p is a human microRNA which binds to a promoter and activates the expression of the STAT3 gene. miR-551b-3p activates, rather than represses, gene expression through promoter-associated RNA activation. Experimental studies in ovarian and triple-negative breast cancer models demonstrated that miR-551b-3p translocates to the nucleus and interacts with the STAT3 promoter, leading to recruitment of transcriptional machinery including RNA polymerase II and TWIST1, and increasing STAT3 transcription and downstream oncogenic signaling (Parashar et al., 2016; Parashar et al., 2019). The nuclear localization of miR-551b-3p depends on Importin-8 (IPO8), a member of the karyopherin/importin family previously identified as a nuclear transport mechanism for Argonaute-microRNA complexes (Wei et al., 2014; Parashar et al., 2019). Promoter-directed activation by RNA remains a relatively specialized and emerging area of RNA therapeutics, however the nucleotide chemistry modifications to date have not deviated from those commonly used by other platforms like siRNA.

## Aptamers



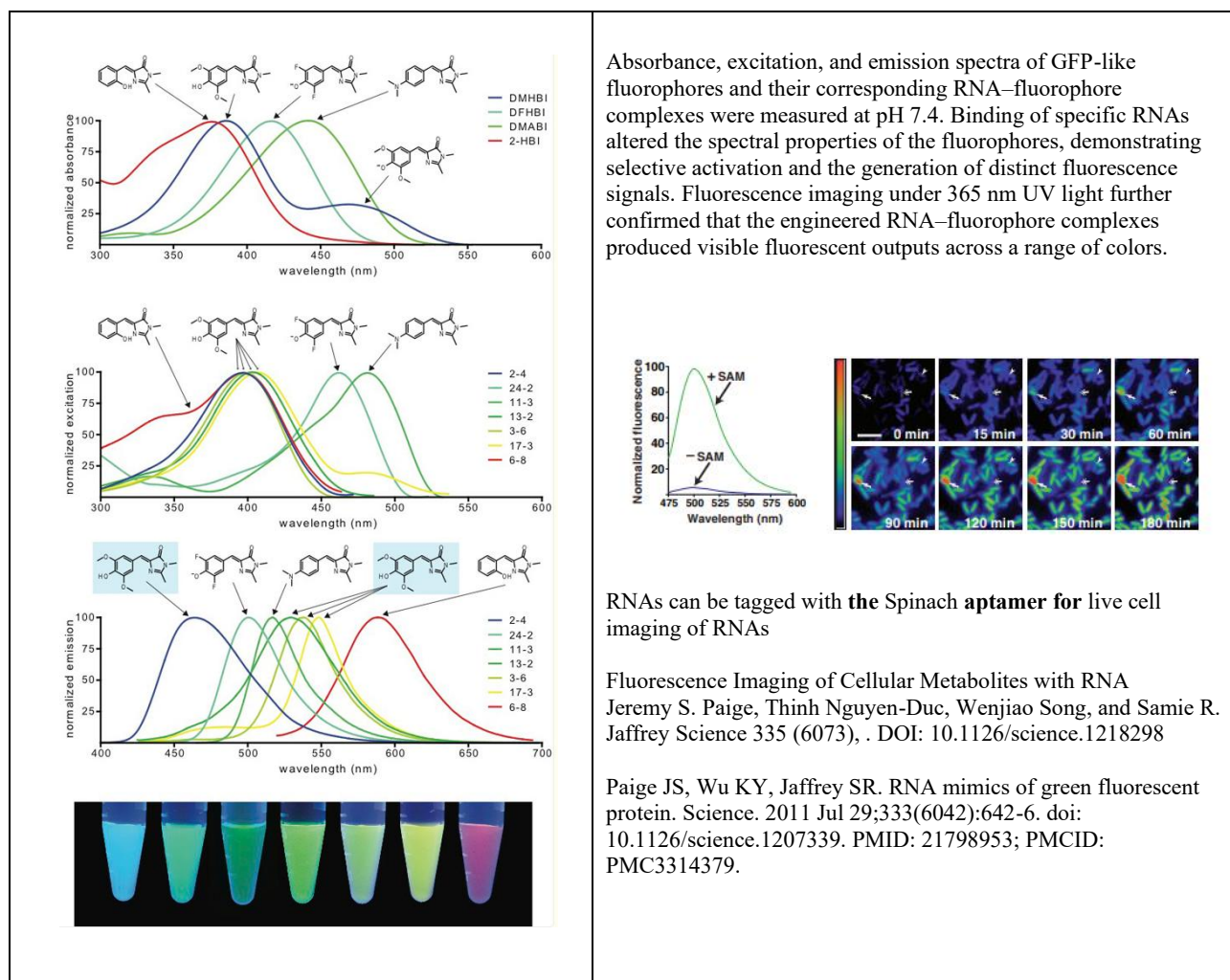
Pegaptanib (Macugen) is a chemically modified RNA aptamer that adopts a defined secondary structure enabling high-affinity and selective binding to the heparin-binding domain of VEGF165. Originally derived from 2-fluoropyrimidine RNA aptamers identified through SELEX, its structure consists of a stabilized folded conformation supported by 2-fluoro and 2-O-methyl substitutions, which enhance nuclease resistance while preserving binding functionality (Ruckman et al., 1998). Although no high-resolution crystal structure of the aptamer–VEGF complex is available, biochemical and structural mapping studies indicate that specific nucleotides within loop regions mediate direct contact with VEGF, defining its inhibitory activity (Ng et al., 2006). The appended polyethylene glycol (PEG) moiety further improves pharmacokinetics without disrupting the core folding architecture. As highlighted in broader aptamer reviews, pegaptanib exemplifies how structured oligonucleotides can achieve antibody-like specificity through well-defined three-dimensional folds, despite being identified and optimized primarily through functional selection rather than structure-first design (Keefe, Pai and Ellington, 2010).



Izervay (avacincaptad pegol) is a pegylated RNA aptamer developed for the treatment of geographic atrophy secondary to age-related macular degeneration (AMD). Unlike earlier retinal therapies that primarily targeted abnormal blood vessel growth, Izervay works by inhibiting complement component C5, a key mediator in the inflammatory complement cascade implicated in retinal cell degeneration. By selectively blocking C5 activation, the drug is designed to reduce downstream inflammatory damage and slow the progressive loss of retinal tissue associated with geographic atrophy. Its approval marked an important advance in ophthalmology, representing one of the first therapies specifically aimed at modifying the complement-driven pathology underlying dry AMD.

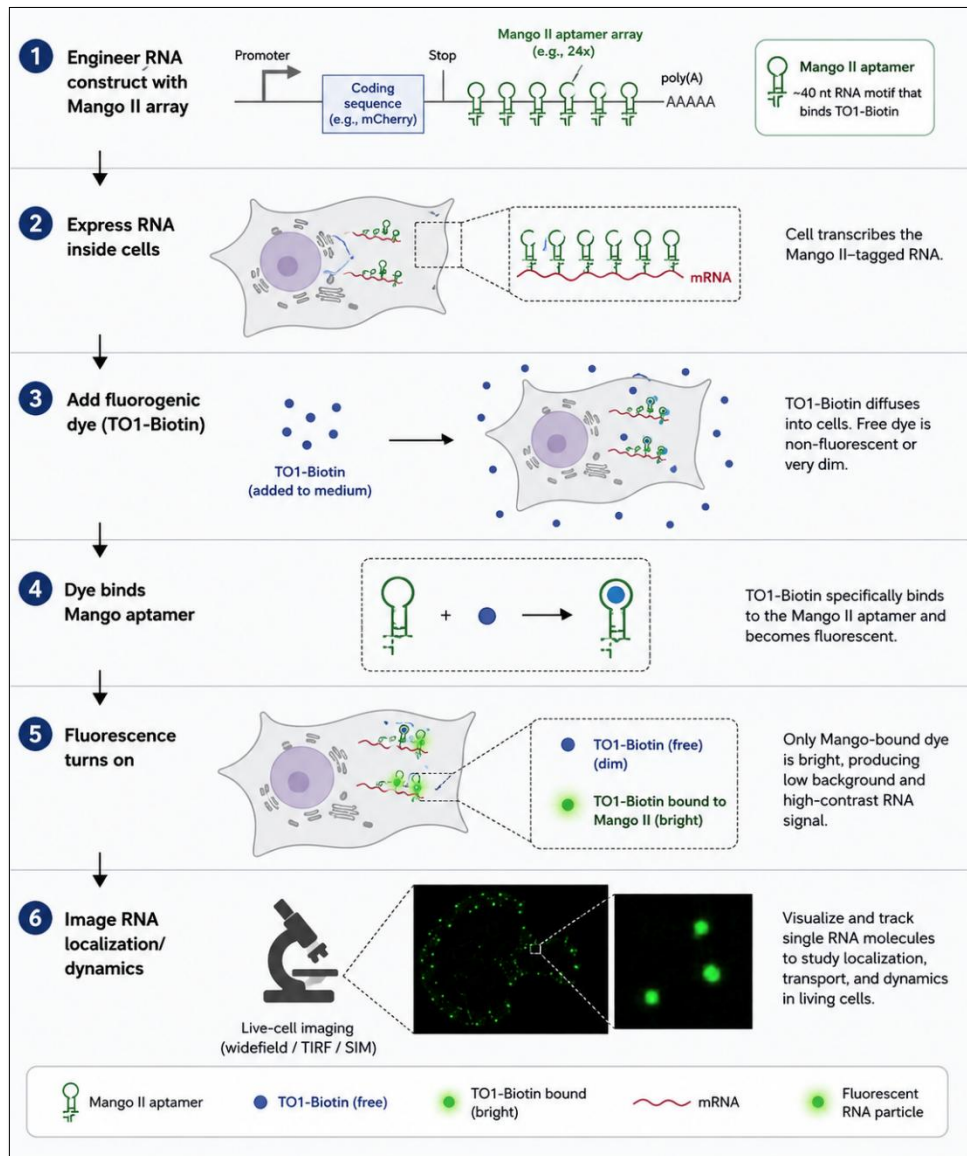


Fluorogenic RNA aptamers have emerged as transformative tools for visualizing RNA localization and dynamics in living cells, enabling direct observation of RNA trafficking, translation, and phase separation with high spatiotemporal resolution. Early studies of mRNA localization using in situ hybridization and live-cell tagging systems established that asymmetric RNA distribution is a conserved mechanism for spatial regulation of gene expression across organisms ranging from yeast to neurons.



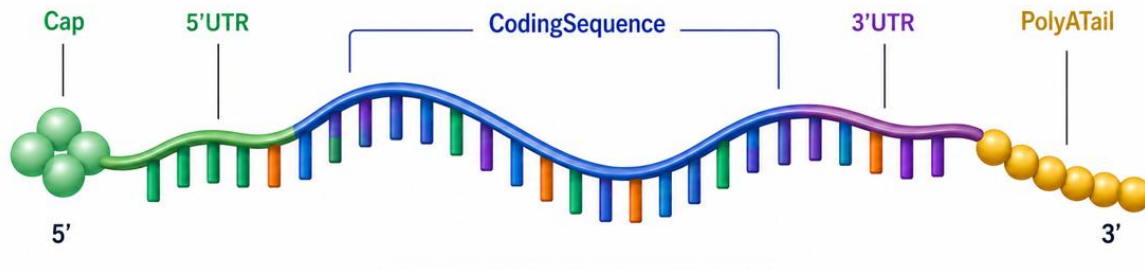
The development of genetically encoded RNA imaging systems, particularly the MS2 bacteriophage platform, enabled the first “live” visualization of endogenous RNA transport dynamics in yeast and mammalian cells, revealing that localized mRNAs undergo directed, motor-dependent transport, anchoring, and translational repression during transit. Subsequent fluorogenic aptamers technology (Spinach and Mango) introduced conditional fluorescence activation, greatly expanding the utility of these tools to specific cellular contexts (Paige et al., 2011; Dolgosheina et al., 2014).





Fluorophore aptamers have also been instrumental in uncovering principles of RNA-mediated phase separation and stress granule biology. Using live-cell imaging approaches, these tools enabled visualization of mRNAs dynamically shuttling into and out of stress granules and demonstrated that RNAs themselves actively contribute to biomolecular condensate assembly rather than passively accumulating within them (Khong et al., 2017). Such studies established that selective RNA recruitment, transport kinetics, and localized translation are tightly coupled to condensate dynamics and cellular stress responses. Importantly, aberrant stress granule assembly and defective RNA trafficking have been directly linked to neurodegenerative diseases including amyotrophic lateral sclerosis (ALS) and frontotemporal dementia through pathological aggregation of RNA-binding proteins such as TDP-43 and FUS, which disrupt normal bidirectional RNA transport and granule dynamics (Williams et al., 2016; Khong et al., 2017).

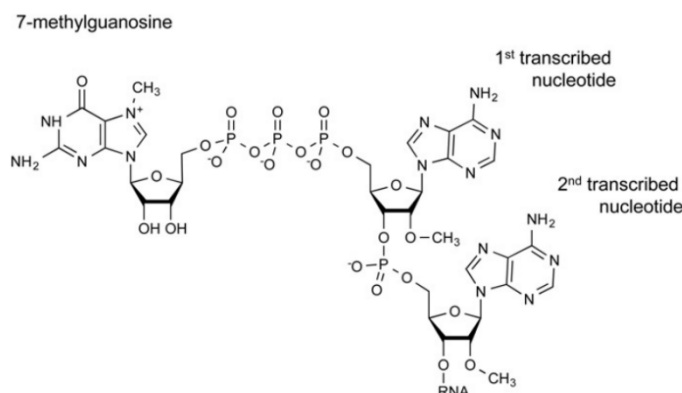
## mRNA vaccines



During the late 1960s and early 1970s, investigations into prokaryotic translation initiation began to reveal that ribosomes recognize specific nucleotide sequences upstream of the AUG start codon. Working at the John Curtin School of Medical Research at the Australian National University in Canberra, John Shine and Lynn Dalgarno published their seminal paper, The 3'-terminal sequence of *Escherichia coli* 16S ribosomal RNA: complementarity to nonsense triplets and ribosome binding sites (Shine and Dalgarno, 1974). In this study, they identified a purine-rich sequence located upstream of bacterial initiation codons that exhibited complementarity to a region near the 3' terminus of the 16S ribosomal RNA. Through comparative sequence analysis and ribosome-binding experiments, they demonstrated that base pairing between the messenger RNA and ribosomal RNA aligned the ribosome precisely at the translational start site, thereby facilitating efficient initiation of protein synthesis.

This discovery established what became known as the Shine–Dalgarno sequence and provided one of the first mechanistic models explaining how ribosomes identify initiation codons in prokaryotes (Shine and Dalgarno, 1974). Because bacterial systems dominated early molecular biology research, the Shine–Dalgarno model strongly influenced assumptions regarding translation initiation across all organisms. However, the absence of equivalent Shine–Dalgarno-like motifs within eukaryotic mRNAs, together with the increasing recognition of long and structurally complex 5' untranslated regions (5' UTRs), suggested that eukaryotic ribosomes employed fundamentally different mechanisms for start codon recognition. These inconsistencies ultimately motivated the experimental work of Marilyn Kozak and others, leading to the development of the ribosomal scanning model and the recognition of the 5' UTR as a critical regulatory component of eukaryotic mRNA.

The discovery of the 5' cap structure on eukaryotic messenger RNA (mRNA) fundamentally changed understanding of post-transcriptional gene regulation. Early work in the 1970s by Shatkin and colleagues at Roche Institute of Molecular Biology in New Jersey, identified an unusual, methylated guanosine residue attached to the 5' end of eukaryotic mRNAs through a 5'–5' triphosphate linkage (Shatkin, 1976). This “cap” structure, later identified as 7-methylguanosine (m7G), was found to be essential for mRNA stability and translation efficiency. Their studies demonstrated that the cap is not merely a passive structural modification, but rather a central regulatory element coordinating transcription, RNA processing, export, translation, and degradation (Furuichi and Shatkin, 2000; Topisirovic et al., 2011).



The canonical 5' cap consists of a 7-methylguanosine linked to the first transcribed nucleotide by a 5'–5' triphosphate bridge, forming the structure m7G(5')ppp(5')N. In higher eukaryotes, additional methylation of the ribose 2'-O position of the first and second nucleotides generates cap 1 and cap 2 structures respectively (Gonatopoulos-Pournatzis and Cowling, 2014).

Cap formation occurs co-transcriptionally during the early stages of RNA polymerase II transcription. Three enzymatic activities are required for cap synthesis: 1) RNA triphosphatase removes the terminal phosphate from the nascent RNA; 2) RNA guanylyltransferase adds GMP through the 5'–5' linkage and 3) RNA guanine-N7 methyltransferase catalyzes methylation of the guanine residue to produce the mature m7G cap (Cowling, 2010). In vertebrates, the cap methyltransferase complex consists of RNMT and its activating partner RAM (Gonatopoulos-Pournatzis et al., 2011). Recruitment of these enzymes depends on phosphorylation of the C-terminal domain (CTD) of RNA polymerase II, thereby coupling transcription directly to RNA capping.

Uncapped RNA molecules are highly susceptible to degradation by 5'→3' prime exonucleases, whereas capped transcripts are stabilized through the recruitment of cap-binding proteins that shield the RNA terminus (Furuichi et al., 1977). The cap also serves as a platform for recruitment of the nuclear cap-binding complex (CBC), composed of CBP20 and CBP80. CBC was first purified from HeLa cell nuclear extracts in the mid-1990s and shown to bind specifically to m7G-capped RNAs (Izaurrealde et al., 1994; 1995). The CBC plays an essential

role in multiple stages of RNA metabolism. During transcription, the cap binding complex associates co-transcriptionally with nascent transcripts and promotes transcription elongation through recruitment of positive transcription elongation factor b (P-TEFb) and associated kinases (Lenasi et al., 2011). CBC is also critically involved in pre-mRNA splicing by facilitating recruitment of components of the spliceosome, particularly U1 snRNP, to the 5' splice site (Lewis et al., 1996). Experimental depletion of CBC from mammalian cell extracts severely impairs spliceosome assembly and pre-mRNA processing.

The 5' cap also interacts with nuclear export factors such as PHAX and TREX to direct the export of U snRNAs and mRNAs from the nucleus to the cytoplasm. The export pathway selected depends partly on transcript length and associated RNA-binding proteins (Cheng et al., 2006; McCloskey et al., 2012). In the cytoplasm, the cap structure is indispensable for efficient translation initiation. The cytoplasmic cap-binding protein eIF4E recognizes the m<sup>7</sup>G cap and recruits the eIF4F translation initiation complex, enabling ribosome loading onto the mRNA. In addition, CBC mediates a specialized “pioneer round” of translation that functions in mRNA surveillance and nonsense-mediated decay (NMD). During this process, CBC-bound transcripts undergo an initial round of translation that allows detection of premature termination codons and elimination of defective mRNAs (Maquat et al., 2010).

More recent work has demonstrated that cap structures are also involved in immune recognition. Properly methylated cap 1 structures help endogenous RNAs evade detection by innate immune receptors such as RIG-I and IFIT proteins, whereas uncapped or improperly capped RNAs are recognized as foreign molecules and trigger antiviral responses. This function has become especially important in the development of synthetic mRNA therapeutics and mRNA vaccines, where engineered cap analogues are used to enhance translation and minimize immune activation.

Overall, the 5' cap is recognized as a master regulator of eukaryotic gene expression. From its historical discovery as a simple terminal modification to its modern understanding as a dynamic signaling and regulatory platform, the mRNA cap is central to RNA stability, processing, export, translation, and immune regulation.

For much of the early development of molecular biology during the 1960s and early 1970s, messenger RNA (mRNA) was viewed primarily as a passive intermediary between DNA and proteins. The dominant focus of gene expression research centered on coding sequences, while non-coding regions of RNA were often regarded as biologically insignificant. In bacterial systems, translation initiation was already relatively well understood through the Shine–Dalgarno model, in which ribosomes bind directly to a conserved sequence upstream of the AUG start codon. Because bacterial models dominated early molecular biology, many researchers initially assumed that eukaryotic translation followed a similar mechanism. However, the discovery that eukaryotic mRNAs possessed long non-coding regions upstream of the coding sequence, later termed the 5' untranslated region (5' UTR), posed important mechanistic questions regarding how ribosomes identified translation start sites.

The emergence of the 5' cap structure in the mid-1970s significantly altered thinking about eukaryotic translation. Studies demonstrated that the addition of a methylated guanosine cap at the 5' terminus dramatically increased translational efficiency, implying that ribosome recruitment was closely associated with the 5' end of the transcript rather than occurring through random internal binding (Shatkin, 1976; Banerjee, 1980; Tahara et al., 1981; Edery et al., 1983; Grifo et al., 1983; Ray et al., 1985; Kozak, 1989). Nevertheless, the exact mechanism by which ribosomes located the AUG initiation codon remained unresolved. Many researchers still considered the 5' untranslated region (5' UTR) largely structural or evolutionarily incidental, rather than functionally regulatory.

A major conceptual breakthrough came through the work of Marilyn Kozak, whose experiments provided some of the earliest direct evidence that the 5' UTR actively participates in translation initiation. In her influential study *Inability of circular mRNA to attach to eukaryotic ribosomes*, Kozak (1979) investigated whether ribosomes required access to the free 5' end of mRNA molecules in order to initiate translation. She experimentally circularized mRNA molecules and compared their ability to recruit ribosomes with that of normal linear transcripts. If ribosomes were capable of binding internally at random positions, circularization should not have substantially affected translation. Instead, Kozak observed that circular mRNAs failed to efficiently bind ribosomes or support translation initiation. These findings strongly suggested that eukaryotic ribosomes must first associate with the 5' end of the transcript before moving downstream toward the initiation codon (Kozak, 1979). This study laid the foundation for what would later become known as the ribosomal scanning model and established the 5' UTR as an essential functional region rather than a passive spacer sequence.

As sequencing technologies improved in the early 1980s, Kozak expanded these ideas by investigating whether conserved sequence motifs existed near eukaryotic translation initiation sites. In *Compilation and analysis of sequences upstream from the translational start site in eukaryotic mRNAs*, Kozak (1984) conducted a large comparative analysis of hundreds of vertebrate mRNA sequences surrounding AUG start codons. Through statistical analysis, she identified a conserved nucleotide pattern surrounding the initiation site, now known as the Kozak consensus sequence:

(gcc)gccRccAUGG

This sequence revealed that nucleotides surrounding the AUG codon, particularly a purine at position -3 and a guanine at +4, were strongly conserved across efficiently translated mRNAs (Kozak, 1984). The discovery was historically important because it demonstrated that translation initiation in eukaryotes depends not simply on the presence of an AUG codon but also on the nucleotide environment created by the surrounding 5' UTR and adjacent coding region.

This shifted scientific understanding of the 5' UTR from a nonfunctional leader sequence to a regulatory domain capable of influencing translational efficiency.

Kozak further strengthened this model through direct experimental validation in her landmark paper Point mutations define a sequence flanking the AUG initiator codon that modulates translation by eukaryotic ribosomes (Kozak, 1986). Using site-directed mutagenesis, she systematically altered nucleotides surrounding AUG codons within expression constructs and measured the resulting effects on protein synthesis. These experiments demonstrated that even single nucleotide substitutions within the flanking sequence could dramatically alter translational efficiency. Specifically, Kozak showed that a purine at the -3 position and a guanine at +4 substantially enhanced ribosomal recognition of the AUG codon, whereas weaker sequence contexts reduced initiation efficiency (Kozak, 1986). Importantly, the experiments also revealed that ribosomes can bypass AUG codons in poor nucleotide contexts, a phenomenon later termed “leaky scanning.” These findings established that the 5' UTR plays an active mechanistic role in directing ribosomal behavior and regulating protein expression.

By the late 1980s, accumulating evidence strongly supported the ribosomal scanning hypothesis. Kozak consolidated these findings in *The scanning model for translation: an update* (Kozak, 1989), which became one of the defining conceptual frameworks for eukaryotic translation initiation. In this model, ribosomes bind near the capped 5' terminus of the mRNA and scan linearly through the 5' UTR until they encounter an AUG codon embedded within a favorable nucleotide context. The paper integrated experimental evidence concerning upstream AUG codons, secondary RNA structures, and inefficient initiation sites, demonstrating that the 5' UTR acts as a highly dynamic regulatory region controlling ribosome movement and initiation fidelity (Kozak, 1989). This represented a major departure from earlier assumptions that non-coding RNA regions were biologically unimportant.

Kozak later broadened the historical and mechanistic significance of these discoveries in *Initiation of translation in prokaryotes and eukaryotes* (Kozak, 1999), where she contrasted bacterial Shine–Dalgarno-mediated initiation with the fundamentally different eukaryotic scanning mechanism. She argued that eukaryotic translation relies heavily on the structural and sequence properties of the 5' UTR, including cap recognition, RNA secondary structure, and AUG context selection. This comparison highlighted the evolutionary divergence between prokaryotic and eukaryotic translational systems and reinforced the importance of the 5' UTR as a defining feature of eukaryotic gene regulation.

In *Pushing the limits of the scanning mechanism for initiation of translation* (Kozak, 2002), Kozak revisited the scanning model in light of increasingly complex observations regarding translational control. She examined noncanonical initiation events, weak AUG codons, upstream open reading frames, and alternative initiation mechanisms, demonstrating that the 5'

UTR can exert highly sophisticated regulatory effects on gene expression. Rather than functioning merely as a passive leader sequence, the 5' UTR emerged as a flexible regulatory environment capable of modulating translation in response to developmental, physiological, and cellular conditions (Kozak, 2002).

Historically, the discovery of the functional importance of the 5' UTR transformed molecular biology's understanding of RNA. What was once regarded as non-coding "space" became recognized as a central regulatory platform governing translation initiation, ribosomal fidelity, and protein expression. Kozak's experimental and conceptual contributions were instrumental in this transformation. Her work established the mechanistic foundations of eukaryotic translation initiation and demonstrated that small variations within non-coding RNA regions could profoundly influence cellular protein synthesis. These discoveries continue to shape modern biotechnology, synthetic biology, and mRNA therapeutic design, where optimization of the 5' UTR and Kozak sequence remains essential for efficient gene expression.



## A-to-I RNA editing oligonucleotide (ADAR)

[TODO: incomplete section ]

### NOTES:

*After becoming the first group to therapeutically edit human RNA in October 2024, Wave Life Sciences has released more data from its ongoing alpha-1 antitrypsin deficiency (AATD) clinical trial—and investors appear less enthusiastic this time around.*

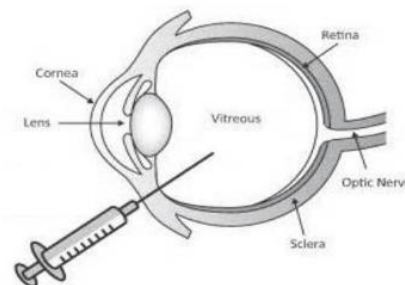
*Patients in the 200-mg single-dose, 200-mg multidose and 400-mg single-dose cohorts of Wave's phase 1b/2a trial all achieved “therapeutically relevant” levels of AAT, the company announced Sept. 3.*

*Despite the trial win, the biotech’s stock toppled from \$9.48 per share at the end of Tuesday to \$7.82 by 3 p.m. ET on Wednesday, Sept. 3, representing an 18.6% decline.*

# PK/PD of RNA Therapeutics

## Distribution

Pharmacokinetic studies showed that Fomivirsen distributes primarily within ocular tissues following intravitreal injection and had sustained concentrations with a vitreous half-life of approximately 55 hours in humans and an estimated retinal half-life of up to 78 hours in animal models (Geary, Henry and Grillone, 2002). Metabolism occurs via endo- and exonuclease-mediated degradation into shorter oligonucleotide fragments, with minimal systemic exposure due to localized administration.



For phosphorothioate (PS) ASOs, which are typically  $\geq 85\text{--}99\%$  protein bound, renal clearance of intact drug is minimal, with studies showing that  $<1\text{--}4\%$  of the administered dose is recovered in urine within 24 hours, reflecting predominant elimination via slow nuclease-mediated metabolism in tissues (Geary et al., 2024; mipomersen data). In contrast, low protein-binding chemistries such as phosphorodiamidate morpholino oligomers (PMOs) undergo substantial renal elimination, with a large fraction of the administered dose excreted unchanged in urine due to limited plasma protein binding and rapid glomerular filtration (Shadida et al., 2021). siRNA shows an intermediate and modality-dependent profile: unmodified siRNA is rapidly cleared by renal filtration, whereas clinically used GalNAc-conjugated siRNAs exhibit reduced renal clearance, with approximately  $12\text{--}25\%$  of the dose recovered unchanged in urine and the remainder undergoing tissue uptake and intracellular metabolism (Habtemariam et al., 2021; vutrisiran data).

## Intracellular

Nucleases are fundamental to life. These enzymes catalyze the cleavage of DNA and RNA, enabling essential processes such as DNA replication (e.g., removal of RNA primers and proofreading), DNA repair and recombination, RNA maturation, and cellular defense against foreign genetic material. For example,  $3'\text{--}5'$  exonucleases such as DnaQ perform proofreading during DNA replication, while RNase H removes RNA primers from DNA–RNA hybrids to ensure proper genome duplication (Yang, 2011 ). In DNA repair, nucleases like Mre11 initiate double-strand break processing, and structure-specific nucleases such as FEN1 remove flap intermediates during lagging-strand synthesis (Yang, 2011 ). In RNA biology, nucleases including RNase E and RNase Z are essential for RNA processing and maturation, whereas Argonaute proteins mediate RNA interference by cleaving target mRNAs (Yang, 2011 ). Additionally, restriction endonucleases and CRISPR-associated nucleases provide defense

against invading genetic elements, highlighting the diverse yet indispensable roles of nucleases across biological systems (Yang, 2011 ).



CRYSTAL STRUCTURE OF HUMAN FLAP ENDONUCLEASE FEN1 (D181A) IN COMPLEX WITH SUBSTRATE 5'-FLAP DNA AND K<sup>+</sup>; HUMAN FLAP ENDONUCLEASE STRUCTURES, DNA DOUBLE-BASE FLIPPING, AND A UNIFIED UNDERSTANDING OF THE FEN1 SUPERFAMILY. TSUTAKAWA, S.E., CLASSEN, S., CHAPADOS, B.R., ARVAL, A.S., FINGER, L.D., GUENTHER, G., TOMLINSON, C.G., THOMPSON, P., SARKER, A.H., SHEN, B., COOPER, P.K., GRASBY, J.A., TAINER, J.A. (2011) CELL 145: 198-211

Antisense oligonucleotides (ASOs) inevitably encounter nucleases immediately upon administration and cellular uptake. In the extracellular space and circulation, they are exposed to DNase I, a nonspecific endonuclease that cleaves both single- and double-stranded DNA into short oligonucleotides as part of normal extracellular DNA clearance and turnover. Following cellular uptake via endocytosis, ASOs enter endosomal and lysosomal compartments, where they encounter degradative enzymes including DNase II, and RNases involved in nucleic acid recycling. Within the intracellular environment, additional nucleases such as 3' exonucleases and endonucleases further metabolize oligonucleotides into shortened fragments, contributing to their overall clearance and activity profile (Jo et al., 2023). For single stranded ASOs that enter the cell nucleus there are numerous nucleases involved in managing various aspects of genomic stability and replication (e.g. FEN1 see figure above). This widespread exposure to therapeutic-damaging enzymes highlights the need for protective strategies when attempting to deliver ASOs to their intracellular targets.

## Endosomal Escape

A central challenge in nucleic acid therapeutics is that although delivery systems can efficiently mediate cellular uptake, only a very small fraction of internalized material reaches the cytosol, where it is functionally active. Multiple orthogonal approaches demonstrate that endosomal escape, rather than uptake, is the primary bottleneck, with estimates consistently converging on ~1% or less of intracellular RNA accessing the cytosol (Dowdy, 2023; Chahal et al., 2026) .

Early studies using electron microscopy with gold-labelled siRNA enabled direct visualization and counting of intracellular particles, revealing that the vast majority localize within endosomes, with only rare cytosolic events, corresponding to ~1–2% escape during a narrow window of endosomal maturation (Gilleron et al., 2013). Quantitative approaches such as nanoscale secondary ion mass spectrometry (NanoSIMS) also provided high-resolution spatial and absolute measurements of oligonucleotide distribution and confirming dense accumulation within endosomal structures and minimal cytosolic presence; again, yielding estimates of ~1–2% productive escape in vivo (He et al., 2021; Dowdy, 2023). Biochemical fractionation methods, which physically separate cytosolic and membrane-bound compartments, further support these findings; despite a known bias toward overestimating cytosolic content due to endosomal disruption during lysis, these studies still report similarly low cytosolic fractions (~1%) (Juliano, 2018). Live-cell fluorescence imaging offers mechanistic insight into this inefficiency, demonstrating that escape is rare and transient (Wittrup et al., 2015).

Functional assays provide an additional layer of validation, as only cytosolic siRNA can engage the RNA-induced silencing complex; the observation that relatively few intracellular molecules are sufficient for robust gene knockdown implies that only a small proportion of the total internalized pool is productively delivered (Wittrup et al., 2015; Dowdy, 2023). Finally, in vivo pharmacokinetic and pharmacodynamic analyses, particularly of GalNAc–siRNA conjugates, reinforce this paradigm, indicating that only 0.3% of internalized siRNA is present in the cytosol at steady state, reflecting slow and continuous leakage from endosomal compartments rather than efficient release (Brown et al., 2020; Dowdy, 2023). Collectively these efforts provide strong evidence that virtually all internalized RNA remains sequestered within endosomal compartments and escape as the rate-limiting step in siRNA delivery.

To address this delivery bottleneck, a wide range of strategies have been developed that aim to enhance endosomal escape by transiently destabilizing membranes or altering intracellular trafficking pathways. Small-molecule agents like chloroquine and other cationic amphiphilic compounds can act by buffering endosomal acidification and inducing osmotic swelling, membrane perturbation, or vesicle rupture, thereby facilitating release of entrapped cargo (Grau and Wagner, 2024; Du Rietz et al., 2020). However, these approaches are inherently non-specific, often disrupting multiple endo-lysosomal compartments, and are limited by dose-dependent cytotoxicity and poor translational performance in vivo, where concentrations required for efficient escape are typically not well tolerated.

Peptide-based approaches such as arginine-rich cell-penetrating peptides and pH-responsive peptides exploit electrostatic interactions and acid-triggered conformational changes to promote membrane fusion, pore formation, or leakage from late endosomal compartments (Grau and Wagner, 2024; Mastrobattista et al., 2002; Wadia et al., 2004), but these systems frequently suffer from serum instability and immunogenicity, as membrane-lytic activity can extend beyond endosomes to other cellular membranes.

Nanoparticle-based delivery systems, particularly ionizable lipid nanoparticles and cationic nanoparticles, are designed to exploit the endosomal environment by undergoing protonation at low pH and interact with anionic lipids to promote formation of non-bilayer structures that destabilize membranes and enable cytosolic release (Cullis and Hope, 2017; Hafez et al., 2001). However, despite their clinical success, these modalities retain intrinsically low escape efficiencies, are largely biased toward hepatic tissue, and can elicit inflammatory or infusion-related responses. These limitations are reflected in the strategic shift by several companies toward ligand–conjugate approaches like GalNAc. For example, Alnylam’s transition away from lipid nanoparticle–formulated siRNA (patisiran/Onpattro) toward GalNAc-conjugated platforms was driven by the need for more predictable pharmacokinetics, improved tolerability, and simpler subcutaneous dosing; therapeutic compatibility and commercial viability often relegate endosomal escape strategies to a lower priority.

Additional approaches, including photochemical internalization and emerging chemical biology strategies that modulate endosomal trafficking machinery, further underscore the breadth of mechanisms being explored to overcome this barrier (Grau and Wagner, 2024; Juliano, 2021). However, these strategies are often constrained by practical limitations, including restricted tissue penetration and the inherent complexity and consequences of perturbing intracellular trafficking networks. These challenges highlight that enhancing endosomal escape is not simply a question of increasing membrane disruption, but rather a fundamentally constrained optimization problem that requires balancing delivery efficiency with cellular integrity and therapeutic compatibility. Despite extensive innovation across multiple modalities, substantial improvements beyond the low baseline escape efficiency observed across systems have remained elusive.

## ASO (Antisense Oligonucleotides)

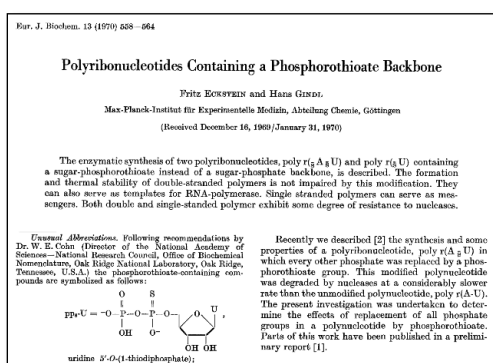
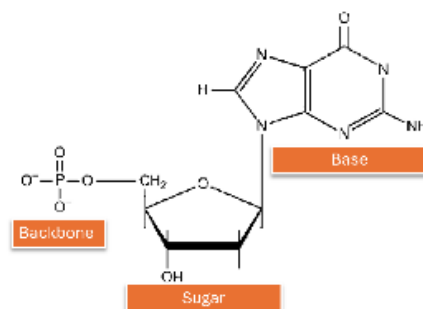
## siRNA (non-conjugated / baseline)

|                                    |                                                                                                                   |                                                                                                                    |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Absorption (SC → plasma)</b>    | Rapid absorption via lymphatics; <b>Tmax 2–6 h</b> ( <i>Geary et al., 2015</i> )                                  | Rapid absorption; <b>Tmax 1–4 h</b> , but often lower bioavailability ( <i>Ayyar &amp; Song, 2023</i> )            |
| <b>Plasma exposure</b>             | Moderate Cmax; declines within <b>hours (≤24 h)</b> due to rapid tissue uptake ( <i>Geary et al., 2015</i> )      | Low and transient plasma exposure; often <b>cleared within hours</b> ( <i>Ayyar &amp; Song, 2023</i> )             |
| <b>Primary tissues (SC dosing)</b> | Liver, kidney, spleen; broad distribution via protein binding ( <i>Geary et al., 2015</i> )                       | Limited distribution without targeting; some liver/kidney uptake but inefficient ( <i>Ayyar &amp; Song, 2023</i> ) |
| <b>Time to tissue distribution</b> | Rapid uptake into tissues within <b>hours post-dose</b> ( <i>Geary et al., 2015</i> )                             | Within hours, but lower fraction reaches productive intracellular compartments                                     |
| <b>Tissue half-life</b>            | Long: <b>1–4 weeks (liver/kidney)</b> due to intracellular retention ( <i>Geary et al., 2015</i> )                | Functional half-life depends on RISC; <b>days–weeks if successfully loaded</b> , otherwise short                   |
| <b>Intracellular trafficking</b>   | Endosomal uptake → gradual release; relatively higher intracellular accumulation                                  | Endocytosis → <b>endosomal escape is limiting (&lt;1% escape)</b> ( <i>Ayyar &amp; Song, 2023</i> )                |
| <b>Onset of PD effect</b>          | Typically, <b>1–3 days</b> post-dose (depends on RNA turnover)                                                    | Typically <b>2–7 days</b> post-dose (processing + RISC loading + RNA turnover)                                     |
| <b>Peak PD effect</b>              | Often <b>~3–7 days</b> post-dose                                                                                  | Often <b>~5–10 days</b> post-dose                                                                                  |
| <b>Duration of PD effect</b>       | Sustained for <b>1–4+ weeks</b> depending on tissue retention ( <i>Geary et al., 2015</i> )                       | Sustained for <b>weeks–months</b> if sufficient RISC loading occurs ( <i>Hammond, 2005</i> )                       |
| <b>PK–PD relationship</b>          | More <b>tissue concentration-driven</b> (liver/kidney levels correlate with effect) ( <i>Geary et al., 2015</i> ) | <b>Decoupled</b> : small intracellular active pool drives prolonged effect ( <i>Hammond, 2005</i> )                |
| <b>Efficiency of tissue use</b>    | Higher fraction of dose becomes pharmacologically active                                                          | Very small fraction (<1%) becomes active (RISC-loaded)                                                             |

# Nucleic Acid Therapeutics

Most RNA therapeutics are nucleic acid-based modalities. In that sense they have a structure that is based on DNA or RNA.

## Backbone Chemistries



The importance of backbone chemistry in oligonucleotides dates to foundational work by Eckstein and Gindl (1970), who demonstrated that replacing a non-bridging oxygen in the phosphate backbone with sulfur (phosphorothioate) produced polyribonucleotide that retained the ability to form stable duplexes and serve as

templates for RNA polymerase, while exhibiting enhanced resistance to nuclease degradation. This early work established the principle that backbone modification could preserve biological function while improving stability.

Today, backbone engineering remains a central parameter in oligonucleotide design, enabling control over stability, protein interactions, and pharmacokinetics. Phosphorothioate (PS) remains universal across ASO and many siRNA designs



FOMIVIRSEN (BRAND NAME VITRAVENE)

Fomivirsen is a simple sequence specific antisense DNA molecule with a modified phosphorothioate backbone (PS); this backbone is the only thing protecting it from nuclease activity.

Intravitreal administration of Fomivirsen significantly prolonged the median time to disease progression in patients with CMV retinitis compared to delayed treatment, supporting its therapeutic efficacy (Geary, Henry and Grillone, 2002). Fomivirsen demonstrated activity against CMV strains resistant to conventional antivirals and exhibits additive effects when used in combination with agents such as ganciclovir or foscarnet (Azad et al., 1995).

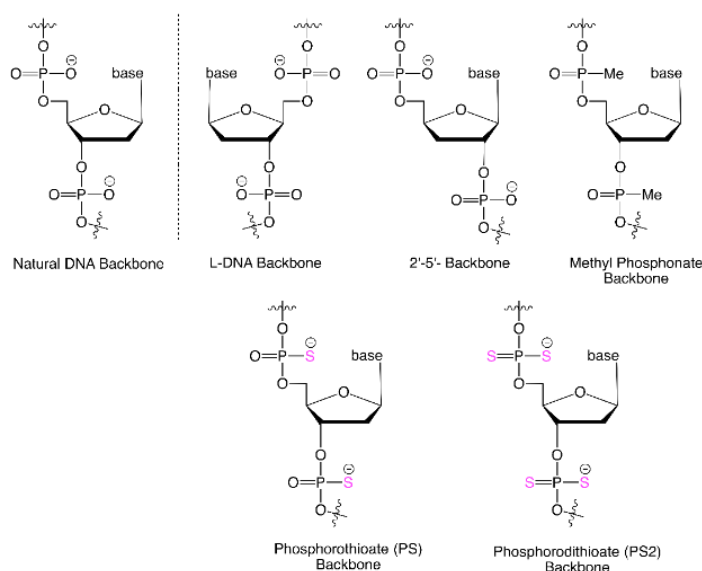


## Charge, Stability and Function

Chemical modification of the oligonucleotide backbone underpins the success of modern nucleic acid therapeutics. While native phosphodiester (PO) linkages provide optimal compatibility with biological machinery, they are intrinsically unstable and pharmacologically limited. A diverse set of backbone and backbone-adjacent chemistries including phosphorothioate (PS), phosphorodithioate (PS<sub>2</sub>), methylphosphonate (MP), phosphoramidate variants, phosphorodiamidate morpholino (PMO), and emerging phosphoramidate diester (PN) chemistries has emerged to address these constraints.

Backbone modifications operate within a constrained design space defined by three interdependent variables: charge, nuclease stability, and biological mechanism compatibility.

First, electrostatic charge governs interactions with proteins, membranes, and biological barriers. The canonical phosphodiester backbone is fully anionic, enabling strong protein binding and efficient recognition by endogenous enzymes, but simultaneously limiting membrane permeability and promoting nonspecific interactions. Phosphorothioate (PS) linkages retain this negative charge while altering its distribution, contributing to increased protein binding and improved pharmacokinetics. In contrast, several alternative chemistries including methylphosphonates, phosphoramidates, phosphorodiamidate morpholino (PMO), and PN linkages reduce or eliminate backbone charge. PMO and PN backbones are largely charge-neutral, fundamentally altering electrostatic interactions, often reducing plasma protein binding and accelerating clearance while potentially improving intracellular distribution. However, reduced charge can also diminish hybridization affinity and weaken protein-mediated uptake pathways that are critical for productive delivery (Crooke, 2017; Juliano, 2016).

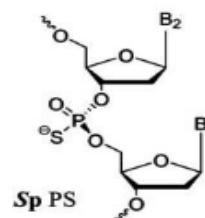
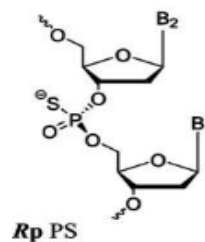


Second, nuclease stability is primarily governed by chemical substitutions at the phosphorus center and by steric protection from sugar modifications. Substitution of non-bridging oxygen atoms with sulfur (e.g., PS linkages) reduces nuclease-mediated hydrolysis and increases resistance to degradation in biological matrices (Lovric et al., 2025) (Wan et al., 2016).

Importantly, stability is not solely dependent on the presence of stabilizing chemistries but also on their distribution and combination. Mixed backbone designs (e.g., PO/PS or PS/PN chimeras) and gapmer architectures allow selective tuning of degradation kinetics and pharmacokinetics. For example, incorporation of PO linkages within a PS backbone reduces stability in a position-dependent manner, while sugar modifications such as MOE can compensate for reduced backbone protection and enable fine-tuning of tissue distribution (Lovrić et al., 2025; Geary, 2001).

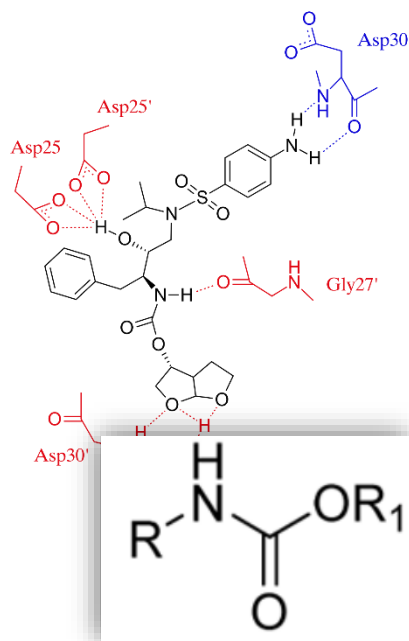
Emerging PN chemistries extend this paradigm by combining altered charge, stereochemical control, and compatibility with chimeric backbone designs. PN-containing oligonucleotides have demonstrated increased potency and durability in both in vitro and in vivo systems, suggesting that their benefits arise not only from nuclease resistance but also from improved cellular uptake and intracellular trafficking.

Taken together, these parameters define a tradeoff triangle in which no single chemistry simultaneously optimizes charge, stability, and function. Backbone chemistries such as PS, PMO, phosphoramidate, and PN, together with sugar modifications such as MOE and LNA, represent different strategies for navigating this space. Rather than replacing earlier chemistries, newer approaches expand the combinatorial design landscape. Modern oligonucleotide therapeutics therefore rely on hybrid architectures—integrating multiple backbone and sugar modifications within a single molecule—to balance pharmacokinetics, potency, and mechanism.



[TODO: incomplete section ]

### [todo] Carbamates



Carbamates are versatile amide–ester hybrid functional groups widely used in medicinal chemistry because they combine chemical stability, tunable conformational properties, and favorable pharmacokinetics. They serve as peptide bond surrogates, prodrug linkers, and key structural motifs in many approved drugs, enabling improved membrane permeability, metabolic stability, and controlled drug release. Their ability to modulate hydrogen bonding and resist enzymatic degradation has made them especially valuable in designing enzyme inhibitors and prodrugs across diverse therapeutic areas. In antisense technologies, carbamates are used to enhance stability and modulating molecular properties for delivery.

Carbamate-based therapeutics exhibit pharmacokinetic and pharmacodynamic profiles that are strongly influenced by their chemical stability and susceptibility to enzymatic cleavage. The carbamate-containing HIV protease inhibitor Darunavir is well absorbed orally with rapid attainment of peak plasma concentrations, and undergoes carbamate hydrolysis-dominated metabolism. PK can be significantly modulated by co-administered agents; for example, ritonavir inhibits CYP3A metabolism, leading to markedly increased systemic exposure and prolonged half-life of darunavir. Excretion occurs via the feces, with minimal renal clearance, reflecting hepatic metabolism and biliary elimination. From a PD perspective, carbamates can either act as stable structural motifs that enhance target binding or as cleavable groups that enable controlled release of active species, thereby linking metabolic processing directly to therapeutic activity. Overall, carbamate functionalities provide a tunable balance between stability and bioactivation, which is central to optimizing both drug exposure and efficacy.

Carbamates are versatile amide–ester hybrid functional groups that are widely used in medicinal chemistry because they combine strong chemical stability with tunable conformational and pharmacokinetic properties. They frequently serve as peptide bond surrogates, prodrug linkers, and key structural motifs in approved drugs. These functional groups improve membrane permeability, metabolic stability, and even controlled drug release. For example, carbamates act as the active binding elements in drugs like Darunavir (HIV protease inhibitor)

and Rivastigmine and provide prodrug strategies in approved drugs Irinotecan, Capecitabine, and Gabapentin enacarbil, where they facilitate solubility, absorption, and targeted activation. Their capacity to modulate hydrogen bonding and resist enzymatic degradation makes them especially valuable in the design of enzyme inhibitors and prodrugs across a wide range of therapeutic areas. Incorporation of carbamates into antisense technologies similarly enhance molecular stability, and improving delivery. Much like charge-masked or backbone-modified oligonucleotides, carbamate-based strategies aim to balance stability with efficient release of the active agent.

The development of phosphorodiamidate linkages (carbamate-like linkages) for morpholino antisense was a natural extension of the broader medicinal chemistry principles. Early antisense oligonucleotides were limited by the instability and charged nature of the phosphodiester backbone, prompting efforts to introduce chemically modified linkages that could enhance durability and biological performance. Building on small-molecule success of this carbamate chemistry, Dr James Summerton and colleagues introduced the phosphorodiamidate linkage as part of the morpholino platform. This modification replaced the negatively charged phosphodiester with a neutral backbone incorporating P–N bonds, conceptually analogous to carbamate functionality, yielding molecules that were highly resistant to nucleases while maintaining strong and specific hybridisation to RNA.

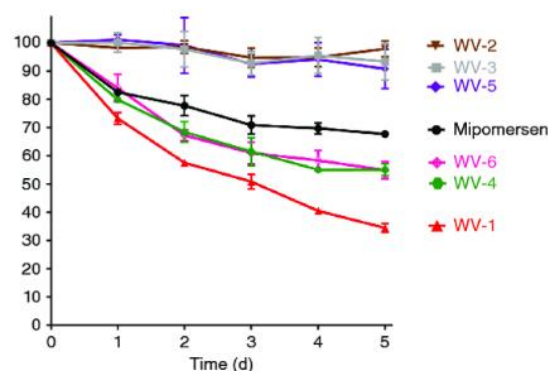
As this chemistry matured, phosphorodiamidate morpholino oligomers (PMOs) emerged as a distinct class of antisense agents that operate through steric blockade rather than enzymatic degradation, marking a shift toward safer and more controllable mechanisms of action (Summerton and Weller, 1997; Summerton, 2017). The introduction of phosphorodiamidates was a key translational step in antisense design.

## Chirality

Substitution at the phosphorus center introduces chirality into the backbone, generating Rp and Sp stereoisomers. In phosphorothioate backbones, each linkage can adopt either configuration, resulting in large mixtures of diastereomers in conventionally synthesized oligonucleotides. Chirality can influence multiple aspects of oligonucleotide behavior. Rp linkages are often associated with improved RNase H recognition and RNA binding, whereas Sp linkages confer greater resistance to nuclease degradation. These opposing effects suggest that stereochemical control could, in principle, enable fine-tuning of backbone properties beyond chemistry alone. Recent advances in stereocontrolled synthesis have enabled the generation of stereodefined oligonucleotides, allowing systematic exploration of these effects.

| ASO        | Sequence | Number of stereoisomer | Tm (°C) | V0 (μM/min) |
|------------|----------|------------------------|---------|-------------|
| Mipomersen |          | 524,288                | 80.2    | 0.05        |
| WV-1       |          | 1                      | 84.7    | ---         |
| WV-2       |          | 1                      | 74.7    | ---         |
| WV-3       |          | 1                      | 78.8    | ---         |
| WV-4       |          | 1                      | 80.0    | ---         |
| WV-5       |          | 1                      | 81.6    | 0.13        |
| WV-6       |          | 1                      | 78.3    | ---         |
| WV-7       |          | 1                      | 68.2    | ---         |

◇ Stereorandom    ▲ Rp    ▼ Sp  
 ○ DNA    ○ 5-Methyl DNA    ● 2'-Methoxyethyl (MOE)    ● 5-Methyl 2'-Methoxyethyl (MOE)



Iwamoto, N., Butler, D.C.D., Svrzikapa, N., Mohapatra, S., Zlatev, I., Sah, D.W.Y., Meena, Standley, S.M., Lu, G., Apponi, L.H., Frank-Kamenetsky, M., Zhang, J.J., Vargeese, C. and Verdine, G.L., 2017. *Control of phosphorothioate stereochemistry substantially increases the efficacy of antisense oligonucleotides*. Nature Biotechnology.

A team at Wave Life Sciences demonstrated the importance of backbone stereochemistry in a 2017 study where they showed that a lack of control masked significant differences in biological activity, as individual Rp and Sp configurations at phosphorus profoundly influenced stability, protein interactions, and RNase H-mediated cleavage (Iwamoto et al., 2017). By developing methods to produce stereopure oligonucleotides with defined chirality patterns, the team demonstrated improved, in vivo, potency, and greater durability compared to conventional stereorandom ASOs (Iwamoto et al., 2017). This helped establish a key platform technology for Wave Therapeutics and laid the foundation for compounds such as their DMD compound WVE-N531.



**WVE-N531** is a stereopure antisense oligonucleotide (ASO) designed to modulate splicing of the dystrophin pre-mRNA for the treatment of Duchenne muscular dystrophy (DMD). It is a 20-mer single-stranded RNA-like molecule built from a precisely controlled pattern of 2'-fluoro (2'-F) and 2'-O-methyl (2'-OMe) sugar modifications, combined with a chirally defined backbone containing mostly “Sp” phosphorothioate linkages and a smaller number of Rp phosphoramidate linkages, along with a phosphodiester. This stereochemical control is a key feature of Wave Life Sciences’ design approach, as it can influence binding affinity, stability, and protein interactions. WVE-N531 binds to dystrophin pre-mRNA and promotes exon 53 skipping, enabling restoration of a shorter but functional dystrophin protein in patients with certain mutations.

Despite its conceptual appeal, experimental evidence indicates that the benefits of stereochemical control is less obvious than it may seem at first. Systematic studies by a group at Ionis pharmaceuticals demonstrate that while PS chirality modulates RNase H1 cleavage patterns in a predictable manner, these changes do not translate into improved potency or therapeutic index. Stereodefined oligonucleotides exhibit target knockdown comparable to stereorandom mixtures, with no configuration consistently outperforming the parent compounds. Indeed, fully Sp-modified backbones can exhibit substantial losses in activity.

At the mechanistic level, introduction of defined Rp and Sp linkages can bias cleavage site selection and, in some cases, produce dominant cleavage positions. Notably, such effects can be achieved with minimal stereochemical intervention, such as incorporation of a single RpSp step within the gap region. However, these alterations in enzymatic processing do not result in enhanced cellular or in vivo efficacy, underscoring a disconnect between biochemical precision and pharmacological outcome (Wan WB, Seth PP., 2016).

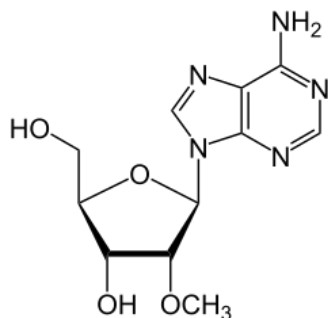
### Backbone modifications in the clinic

| Chemistry   | Pattern             | Charge        | Used by | In approved drugs?           |
|-------------|---------------------|---------------|---------|------------------------------|
| PO          | soooooooooooooos    | full negative | Alnylam | Onpattro and others          |
| PS          | ssssssssssssssss    | full negative | Ionis   | Yes, many including Spinraza |
| PS + PO mix | soooooossssssssooss | mixed         | n-Lorem | Yes, neuro – gapmer          |
| PN          | sssnsssnsssnssss    | reduced       | Wave    |                              |

| Stereopure PS |                       | full negative | Wave |
|---------------|-----------------------|---------------|------|
| Symbol        | Chemistry             |               |      |
| o             | phosphodiester (PO)   |               |      |
| s             | phosphorothioate (PS) |               |      |
| n             | phosphoroamidate / PN |               |      |

## Furanose Sugar Modifications

### 2'-O-Methyl (2'-OMe)

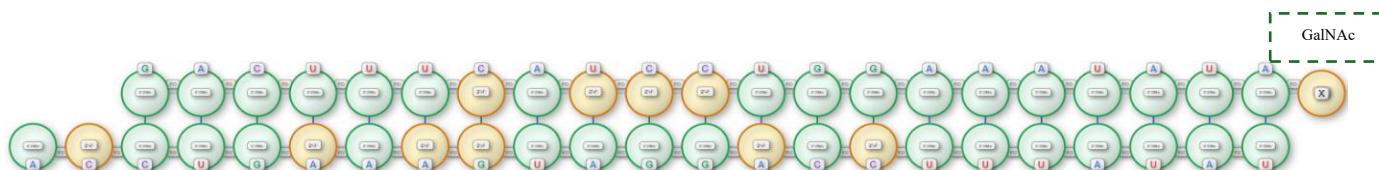


**Benefits:** - Improved nuclease resistance - Reduced immune activation - Moderate affinity enhancement. Clinically validated in more than 100,000 patients via multiple different routes of administration.

Common in both siRNA and single-stranded ASO platforms but more common in siRNA.

Endogenously, 2'-O-methylation is one of the most abundant and conserved RNA modifications, occurring across multiple RNA classes including tRNA, rRNA, mRNA, and small non-coding RNAs where it fine-tunes RNA structure, stability, and function. In tRNAs, 2'-OMe modifications are critical for accurate codon recognition. In rRNAs and mRNAs, 2'-OMe is found at the 5' cap and internally, where it regulates mRNA stability, translation efficiency, and discrimination between self and non-self RNA in innate immunity.

In therapeutics, this chemistry is central to siRNA design in Alnylam's "enhanced stabilization chemistry" (ESC), where 2'-OMe combined with 2'-fluoro modifications improve pharmacokinetics, reduces off-target effects, and minimizes immune activation (Nair et al., 2014; Fitzgerald et al., 2017). These advances are clinically validated by approved drugs including Givosiran for acute hepatic porphyria, Lumasiran for primary hyperoxaluria type 1, and Inclisiran for hypercholesterolemia; all of which incorporate 2'-OMe-containing chemistries to enable durable gene silencing with infrequent dosing (Adams et al., 2020; Garrelfs et al., 2021; Raal et al., 2020).



**Lumasiran** (Oxlumo) is administered via subcutaneous and is an *ESC/GalNAc-conjugated* siRNA, it is designed for targeted delivery to hepatocytes following systemic administration, enabling RNA interference of the HAO1 gene in the liver. Efficacy was successfully achieved in the phase 3 ILLUMINATE-A trial, where Lumasiran caused a 65% mean reduction in 24-hour urinary oxalate compared to a 12% reduction in the placebo group;

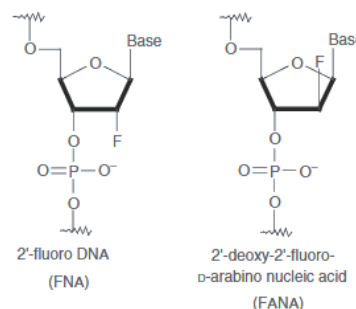


this is a statistically significant difference; 54% ( $p < 0.0001$ ) (FDA, 2020). Approximately half of treated patients reached normalization of urinary oxalate levels by Month 6, indicating a clinically meaningful biochemical response (FDA, 2020). Consistent results were observed in younger patients in the ILLUMINATE-B study, where a 71% reduction in urinary oxalate (measured as oxalate-to-creatinine ratio) was reported, supporting efficacy across age groups (FDA, 2020). Across studies, reductions in urinary oxalate were rapid, occurring within the first few months of treatment and sustained over the study duration (FDA, 2020). The data demonstrates that Lumasiran provides substantial and consistent reductions in urinary oxalate (surrogate endpoint), supporting its efficacy in the treatment of primary hyperoxaluria type (FDA, 2020).

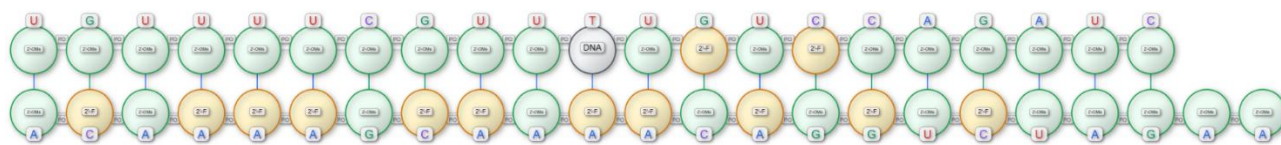
## 2'-Fluoro

**Benefits:** - Increase nuclease resistance and strengthen target binding while maintaining the RNA's natural structural shape for better therapeutic performance.

2'-Fluoro modifications, including 2'F-ANA (fluoro-arabino nucleic acid) and related FNA chemistries are highly potent nucleic acid modifications in which a fluorine atom replaces the 2'-hydroxyl group of the sugar. This change enhances nuclease stability and strengthens target binding, while also supporting steric-blocking mechanisms; in some cases, these modifications can enable gymnotic cellular uptake without delivery systems.



|                         | "Up" (Arabino configuration)         | "Down" (Ribo configuration)  |
|-------------------------|--------------------------------------|------------------------------|
| <b>Sugar Pucker</b>     | Mostly O4'-endo (East) or C2'-endo   | Strictly C3'-endo (North)    |
| <b>Helix Type</b>       | DNA-like (B-form)                    | RNA-like (A-form)            |
| <b>RNase H Activity</b> | Activates RNase H                    | Does not activate RNase H    |
| <b>RNA Affinity</b>     | Higher than DNA; lower than 2'-F-RNA | Very high (extremely stable) |



**Inclisiran (Leqvio)** is a chemically modified, double-stranded small interfering RNA (siRNA) conjugated to GalNAc to tissue-selectively knock down PCSK9 mRNA, leading to reduced hepatic PCSK9 synthesis and increased LDL receptor recycling, thereby lowering circulating low-density lipoprotein cholesterol (LDL-C) (FDA, 2021; EMA, 2020) . It is indicated as an adjunct to diet and lipid-lowering therapy, including statins, for patients with primary hypercholesterolemia or atherosclerotic cardiovascular disease who require additional LDL-C reduction, and is administered subcutaneously with an extended dosing interval of every 6 months; illustrating a robust and long hepatic half-life (FDA, 2021; EMA, 2020).

Nonclinical and clinical studies demonstrate dose-dependent and sustained reductions in PCSK9 and LDL-C, with distribution primarily to the liver, metabolism via nuclease degradation to shorter RNA fragments, and elimination mainly through urine and feces (FDA, 2021). Toxicological findings were largely non-adverse and associated with tissue accumulation, with no evidence of genotoxicity, carcinogenicity, or reproductive toxicity, supporting a favorable nonclinical safety profile (FDA, 2021).

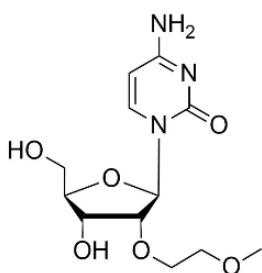


**Givosiran (GIVLAARI)** is a small interfering RNA (siRNA) therapeutic indicated for the treatment of adults with acute hepatic porphyria (AHP). It acts by targeting Aminolevulinate Synthase 1 (ALAS1) mRNA in hepatocytes, thereby reducing the accumulation of neurotoxic intermediates such as aminolaevulinic acid (ALA) and porphobilinogen (PBG). The drug is administered once a month through subcutaneous injection at a dose of 2.5 mg/kg (Alnylam Pharmaceuticals, Inc., 2019) or about 180mg per dose for an average person.

Clinical studies have demonstrated that Givosiran significantly reduces the frequency of porphyria attacks, with patients experiencing approximately 70% fewer attacks compared to placebo in the ENVISION trial (*page 10 on the drug label/US Prescribing Information (USPI)*). Common adverse effects include nausea and injection-site reactions, while more serious risks include hepatic toxicity, renal impairment, and rare anaphylactic reactions (Alnylam Pharmaceuticals, Inc., 2019).

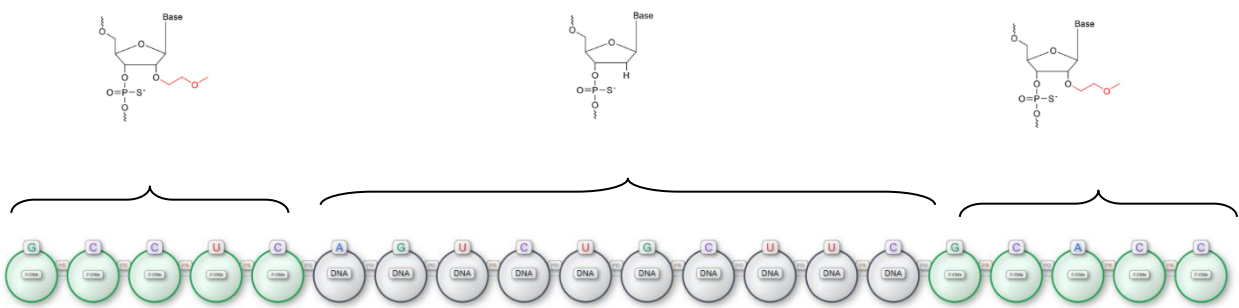
**2'F-ANA** (2'-fluoro-arabino nucleic acid) is a chemically modified nucleotide in which substitution at the 2' position of the sugar enhances nuclease resistance and increases binding affinity to complementary RNA, while maintaining compatibility with RNase H activity in gapmer designs (Souleimanian et al., 2012). These properties have made 2'F-ANA a useful modification in preclinical studies, where it has demonstrated potent and sustained gene silencing. However, despite its favorable biochemical profile, this chemistry has not yet been translated into clinically approved oligonucleotide therapeutics and remains primarily confined to experimental applications.

## 2'-O-Methoxyethyl (2'-MOE) / MOE chemistry

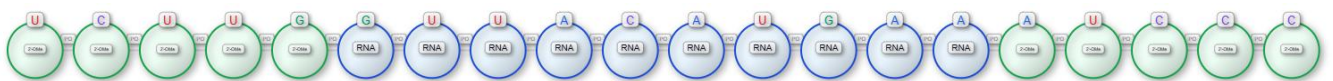


**Benefits:** - High binding affinity - Strong nuclease resistance - Favorable safety profile – long duration of action

The development of 2'-O-methoxyethyl (MOE) chemistry as a therapeutic originated from limitations in antisense oligonucleotide (ASO) technology; unmodified nucleic acids were rapidly degraded and exhibited poor pharmacokinetic properties. In 1995, Pierre Martin, a chemist working at Ciba-Geigy (now Novartis) in Basel, Switzerland, reported the synthesis of 2'-O-alkylated ribonucleosides, including the 2'-O-(2-methoxyethyl) (MOE) modification, in a publication in *Helvetica Chimica Acta*: Martin P. A New Access to 2'-O-Alkylated Ribonucleosides and Properties of 2'-O-Alkylated Oligoribonucleotides. Helvetica Chimica Acta. 1995 Mar;78(2):486-504. DOI: 10.1002/hlca.19950780219. (Hill & Hall, 2023). As antisense technology evolved, MOE modification became a hallmark of second-generation ASOs, building upon phosphorothioate backbone modifications to further enhance potency, tissue residence time, and tolerability (Zanardi et al., 2018). Over time, these combined chemical innovations enabled the successful clinical translation of antisense therapeutics, with MOE-modified oligonucleotides now forming the basis of several approved drugs and numerous candidates in development.



Inotersen is administered as a once-weekly subcutaneous injection and has demonstrated clinical efficacy in phase II/III NEURO-TTR randomized controlled trial. The study showed that inotersen significantly slowed the progression of neuropathy and improved quality of life compared with placebo, as measured by the modified Neuropathy Impairment Score and the Norfolk Quality of Life–Diabetic Neuropathy questionnaire (CADTH, 2020). However, despite achieving statistical significance, the clinical meaningfulness of these improvements remains uncertain due to the absence of clearly defined thresholds for clinically important differences (CADTH, 2020).



Safety remains a key concern, as treatment is associated with a high incidence of adverse events, including thrombocytopenia, renal impairment, and injection-site reactions. Serious adverse events and deaths have been reported in clinical trials, necessitating regular monitoring of platelet counts and renal function (CADTH, 2020). Clinical guidance further emphasizes the importance of ongoing monitoring through regular blood and urine tests to assess both efficacy and toxicity (Mayo Clinic, n.d.).

Inotersen can reduce platelet counts, increasing the risk of bleeding and infection, and patients are advised to take precautions to minimize injury and exposure to infectious agents (Mayo Clinic, n.d.). It has also been associated with serious organ-related complications, including kidney and liver toxicity, with symptoms such as reduced urine output, swelling, jaundice, or abdominal pain requiring immediate medical attention (Mayo Clinic, n.d.). In addition, there is an elevated risk of inflammatory or immune-mediated conditions, stroke, and arterial dissection, highlighting the need for vigilance regarding neurological symptoms (Mayo Clinic, n.d.).

Hypersensitivity reactions, including life-threatening allergic responses, have also been reported, alongside reductions in vitamin A levels that may lead to visual disturbances, necessitating ophthalmologic monitoring in some patients (Mayo Clinic, n.d.). Patients are

further advised to consult their healthcare provider before taking any concomitant medications, including over-the-counter or herbal products, due to potential interactions (Mayo Clinic, n.d.).

Overall, Inotersen represents an important advancement in RNA-targeting therapeutics and offers a more convenient, self-administered alternative to intravenous treatments. However, its use is limited by safety concerns and ongoing uncertainties regarding long-term efficacy and broader clinical benefits (CADTH, 2020; Rüger et al., 2020).

MOE-modified ASOs are widely used in two RNA-targeting strategies: RNase H-mediated RNA degradation and splice modulation via steric blocking. These two approaches enable treatment of a broad range of genetic diseases (Sang et al., 2024). For example, MOE-containing gapmer ASOs such as Inotersen and Eplontersen are used in the treatment of hereditary transthyretin amyloidosis (hATTR), where they bind transthyretin (TTR) mRNA and promote its degradation, thereby reducing pathogenic protein accumulation (Sang et al., 2024). Similarly, Tofersen, a 5-10-5 MOE-modified gapmer ASO approved for amyotrophic lateral sclerosis (ALS), targets mutant SOD1 mRNA, demonstrating the utility of MOE chemistry in neurodegenerative disease through selective reduction of toxic protein expression (Sang et al., 2024). In the context of splice modulation, Nusinersen (Spinraza) discovered at Cold Spring Harbor uses a fully MOE-modified ASO to promoting inclusion of exon 7 in SMN2 mRNA and thereby restoring production of functional SMN protein (Sang et al., 2024). Beyond these established therapies, MOE chemistry has also enabled highly personalized approaches, exemplified by Milasen, a patient-specific ASO designed to correct aberrant splicing in Batten disease, highlighting the adaptability of MOE-modified oligonucleotides for precision medicine (Sang et al., 2024). This chemistry is heavily used by nLore for their nano-rare therapeutics programs.



**5-10-5 MOE GAPMER WITH ESC siRNA OVERLAP IN BOTH APPROVED FOR ATTR AMYLOIDOSIS. THESE FDA APPROVED DRUGS OVERLAP IN SEQUENCE BUT CONTAIN VERY DIFFERENT CHEMISTRIES AND DIFFERENT MECHANISM OF ACTION**

MOE-modified ASOs are widely used in two principal RNA-targeting strategies: RNase H-mediated RNA degradation and splice modulation via steric blocking, enabling treatment of a broad range of genetic diseases (Sang et al., 2024). For example, MOE-containing gapmer ASOs such as Inotersen and Eplontersen are used in hereditary transthyretin amyloidosis (hATTR), where they bind transthyretin (TTR) mRNA and promote its degradation (Sang et al., 2024). Similarly, Tofersen, a MOE-modified gapmer ASO approved for amyotrophic lateral sclerosis (ALS), targets mutant SOD1 mRNA to reduce toxic protein expression (Sang et al., 2024). In the

|                                 |                                       |                                                                 |                   |
|---------------------------------|---------------------------------------|-----------------------------------------------------------------|-------------------|
| <b>Fomivirsen (Vitraven®)</b>   | <b>Phosphorothioate backbone (PS)</b> | <b>Cytomegalovirus (CMV) retinitis</b>                          | <b>1999</b>       |
| <b>Mipomersen (Kynamro®)</b>    | PS & 2'-MOE                           | Homozygous familial hypercholesterolemia                        | 2013              |
| <b>Nusinersen (Spinraza®)</b>   | PS & 2'-MOE                           | Spinal muscular atrophy                                         | 2016              |
| <b>Inotersen (Tegsedi®)</b>     | PS & 2'-MOE                           | Hereditary transthyretin-mediated amyloidosis                   | 2018              |
| <b>Milasen</b>                  | PS & 2'-MOE                           | CLN7 gene mutation associated with Batten disease               | 2018              |
| <b>Volanesorsen (Waylivra®)</b> | 2'-MOE                                | Familial chylomicronemia syndrome                               | EMA approved 2019 |
| <b>Tofersen (Qalsody®)</b>      | PS & 2'-MOE                           | Amyotrophic lateral sclerosis with SOD1 mutation                | 2023              |
| <b>Eplontersen (Wainua®)</b>    | PS, 2'-MOE & GalNAc                   | Polyneuropathy of hereditary transthyretin-mediated amyloidosis | 2023              |

context of splice modulation, Nusinersen (Spinraza) employs a fully MOE-modified ASO to promote inclusion of exon 7 in SMN2 mRNA, restoring functional SMN protein (Sang et al., 2024). Beyond these established therapies, MOE chemistry has enabled highly personalized approaches such as Milasen, a patient-specific ASO designed to correct aberrant splicing in Batten disease (Sang et al., 2024). The continued evolution of MOE chemistry, particularly when combined with advanced delivery strategies and conjugation technologies, underscores its central role in enabling durable, tissue-targeted, and precision RNA therapeutics.

Incorporation of MOE at the 2'-position of the ribose increases conformational rigidity and enhances hybridization while reducing susceptibility to nuclease-mediated cleavage (Zanardi et al., 2018). From a pharmacokinetic and pharmacodynamic (PK/PD) perspective, MOE-modified ASOs exhibit slow plasma clearance, high plasma protein binding, and extensive tissue distribution—particularly to the liver, kidney, and adipose tissue—resulting in long tissue half-lives that can extend from weeks to months (Takakusa et al., 2023). For example, 2'-MOE ASOs such as Mipomersen demonstrate terminal half-lives of 30 days in plasma and similar persistence in liver tissue, reflecting equilibrium between plasma and tissue compartments and supporting infrequent dosing regimens (Takakusa et al., 2023). These prolonged kinetics arise from strong protein binding and intracellular retention following endocytosis and endosomal trafficking, which enable sustained pharmacological activity even after plasma levels decline (Takakusa et al., 2023).

Importantly, pharmacodynamic effects are more closely correlated with tissue concentrations than plasma levels, highlighting the importance of tissue PK in dose optimization and therapeutic efficacy (Christensen et al., 2025). Furthermore, advances in chemical stabilization and conjugation strategies have enabled improved delivery beyond the liver, expanding the therapeutic reach of oligonucleotides to extrahepatic tissues such as muscle, brain, and lung (Anand et al., 2025 ).



**Volanesorsen (Waylivra):** ApoC-III targeting ASO for familial chylomicronemia syndrome (FCS)

Base: A G mC mU mU mC T T G T mC mC A G mC mU mU mU A mU  
 Backbone Full phosphorothioate (PS)  
 Sugar: E E E E d d d d d d d d E E E E  
 legend: **E** = 2'-MOE **d** = DNA (2'-deoxy)

Volanesorsen demonstrated substantial triglyceride-lowering efficacy but is associated with several safety considerations. The most clinically significant risk is thrombocytopenia, including cases of severe platelet reduction. In addition, injection-site reactions are very common, occurring in a substantial proportion of administrations, although these are generally mild to moderate and primarily affect tolerability rather than safety. Rare but serious immune-

mediated events, such as serum sickness, have also been reported, indicating potential for hypersensitivity reactions. (Witztum et al., 2019; Tardif et al., 2016).

Olezarsen, a GalNAc-conjugated antisense oligonucleotide version of Volanesorsen and was developed to solve its relatively poor safety profile. Thrombocytopenia with Volanesorsen was attributed to high systemic exposure and dosing burden. By incorporating N-acetylgalactosamine (GalNAc) for hepatocyte-targeted delivery, Olezarsen matches efficacy with less frequent doses, thereby reducing systemic exposure. Clinical data demonstrate that while some adverse events persist, including mild thrombocytopenia and liver enzyme elevations, these effects are dose-dependent and more pronounced at higher doses, with overall safety outcomes approaching those of placebo. This supports the concept that dose and exposure are central determinants of antisense oligonucleotide toxicity, and that GalNAc conjugation can effectively improve the therapeutic index by decoupling efficacy from systemic toxicity (Marston et al., 2026; Witztum et al., 2019).

|                                  | <b>Volanesorsen</b>                           | <b>Olezarsen (GalNAc)</b>                              |
|----------------------------------|-----------------------------------------------|--------------------------------------------------------|
| <b>Dosing</b>                    | High dose (e.g., 300 mg weekly → biweekly)    | Lower dose (50–80 mg monthly)                          |
| <b>Delivery</b>                  | Systemic (no targeting)                       | Hepatocyte-targeted (GalNAc)                           |
| <b>Thrombocytopenia</b>          | Clinically significant; severe cases reported | Less frequent; dose-dependent, higher at 80 mg         |
| <b>Injection-site reactions</b>  | Very common (~24% of injections)              | Less frequent (~10–16%)                                |
| <b>Liver enzyme elevations</b>   | Not dominant signal                           | Increased, especially at higher dose (80 mg)           |
| <b>Hepatic fat</b>               | No consistent increase                        | Dose-dependent increase observed                       |
| <b>Hypersensitivity / immune</b> | Rare (e.g., serum sickness)                   | Mild hypersensitivity events slightly increased        |
| <b>AE rate</b>                   | Higher discontinuation vs placebo             | Similar to placebo overall                             |
| <b>Key safety driver</b>         | Systemic exposure / high dose                 | Residual dose-dependent toxicity, reduced by targeting |





IONIS-HTTRx (Tominersen, Roche/Ionis) – MOE-modified “gapmer” ASO targeting huntingtin mRNA for Huntington’s disease.

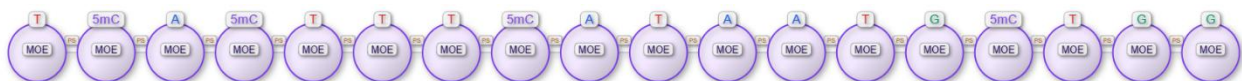
**Base**                    **m5C m5U m5C A G T A A m5C A T T G A m5C A m5C m5C A m5C**

**Backbone**            PS – PO – PO – PO – PS – PS – PS – PS – PS – PS – PS – PS – PS – PS – PS – PS – PO – PO – PO – PS

**Sugar**                E E E E d d d d d d d d d d E E E E → 5 – 10 – 5 MOE gapmer

**Legend**            E = 2'-MOE; d = DNA

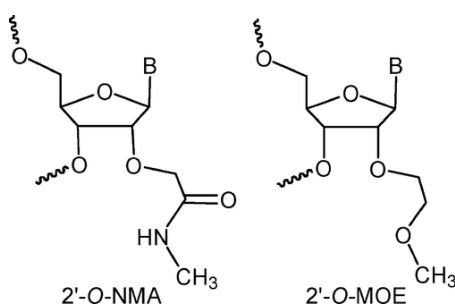
In the Phase III GENERATION HD1 trial, patients received Tominersen via intrathecal injection at doses of 120 mg every 8 weeks or every 16 weeks. Although the drug successfully lowered huntingtin protein levels, it failed to improve clinical outcomes and in some cases worsened disease progression, particularly with more frequent dosing. This lack of efficacy is thought to result from non-selective reduction of both mutant and normal huntingtin along with possible overdosing and intervention occurring too late in the disease course.



**Nusinersen (Spinraza®)** is a fully modified 2'-O-methoxyethyl (MOE) antisense oligonucleotide with a phosphorothioate backbone, designed to modulate SMN2 splicing by promoting inclusion of exon 7. In clinical trials, including ENDEAR (NCT02193074) and CHERISH (NCT02292537), this sterically blocking MOE chemistry demonstrated substantial therapeutic benefit in patients with spinal muscular atrophy (SMA), leading to significant improvements in motor function and survival compared to untreated cohorts (Finkel et al., 2017; Mercuri et al., 2018). The clinical efficacy of nusinersen reflects key properties of fully modified MOE ASOs, including high binding affinity, strong resistance to nuclease degradation, and prolonged tissue residence time within the central nervous system following intrathecal administration (Geary et al., 2015). The durability of response was observed in long-term studies and allowed dosing at four-month intervals.

The study by medicinal chemists at Ionis Pharmaceuticals (Ling et al. 2025) evaluated progressive 2'-prime modifications to increase the pharmacological performance. Using a fully 2'-MOE/fully modified phosphorothioate ASO (MOE-1) as a clinically validated baseline, the

team demonstrate that substitution with the novel NMA chemistry (NMA-1) enhances potency and prolongs duration of action, likely due to improved RNA binding and intracellular stability. Further optimization through sequence shifting (NMA-2) results in increased splice-modulating activity, underscoring the importance of target site selection. Importantly, the introduction of selective phosphodiester linkages into this optimized sequence (NMA-3) maintains high potency while improving tolerability and extending duration, producing the most favorable overall profile, including sustained central nervous system activity. Together, these findings highlight that optimal ASO performance is achieved through the combined effects of advanced sugar chemistry, sequence refinement, and backbone engineering rather than any single modification alone (Ling et al., 2025; Prakash et al., 2008).



The 2'-O-[2-(methyl-amino)-2-oxoethyl] (NMA) ribose modification originated from efforts to improve second-generation antisense oligonucleotide (ASO) chemistries, particularly those based on 2'-O-sugar modifications such as 2'-O-methoxyethyl (MOE).

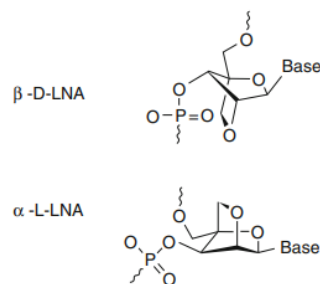
which were developed to overcome the limitations of first-generation phosphorothioate DNA (e.g., low RNA affinity and short in vivo half-life). NMA was rationally designed as an alternative 2'-O-substituent incorporating an amide-containing side chain, enabling enhanced hydrogen bonding and conformational stabilization of the ribose in a C3'-endo geometry, thereby improving RNA binding affinity and nuclease resistance. Early synthetic work demonstrated that NMA-modified nucleosides could be efficiently produced and incorporated into gapmer ASOs, maintaining comparable thermal stability ( $T_m$ ) and potency to MOE analogues while preserving favorable safety profiles. Building on this chemical foundation, newer NMA-modified ASOs exhibit enhanced pharmacological efficiency, achieving similar splice-modulating effects at lower doses and enabling prolonged activity, including sustained central nervous system effects for up to 52 weeks, likely driven by improved intracellular stability rather than increased accumulation. Collectively, the evolution of NMA chemistry from earlier 2'-O-modification strategies underpins its ability to extend ASO duration of action while maintaining potency and tolerability.

## Locked Nucleic Acids (LNA)

Locked nucleic acids were introduced in the late 1990s:

*Obika, Satoshi; Nanbu, Daishu; Hari, Yoshiyuki; Morio, Ken-ichiro; In, Yasuko; Ishida, Toshimasa; Imanishi, Takeshi (1997-12-15). "Synthesis of 2'-O,4'-C-methyleneuridine and -cytidine. Novel bicyclic nucleosides having a fixed C3, -endo sugar pucker". Tetrahedron Letters. 38 (50): 8735–8738. Doi:10.1016/S0040-4039(97)10322-7. ISSN 0040-4039.*

These nucleosides incorporate a 2'-O,4'-C methylene bridge that locks the ribose ring into a C3'-endo conformation characteristic of RNA duplexes (Koshkin et al., 1998; Obika et al., 1998).

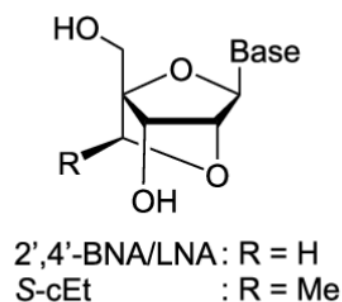


This conformational constraint produces several key effects:

- Strongly enhanced hybridization affinity
- Increased thermal stability of duplexes
- Improved mismatch discrimination
- Terminal LNA incorporation increases stability approximately tenfold over unmodified
- Resist exonuclease degradation when flanking residues are modified (Kurreck et al., 2002)

Unlike earlier modifications, LNA technology was a shift toward structure-driven nucleic acid engineering, where conformational preorganization directly enhances function.

Constrained ethyl (cEt) chemistry is a type of LNA modification that incorporates a 2'-O-ethyl substituent within a constrained bicyclic framework; used primarily by Ionis Pharmaceuticals. Structural analyses show that cEt and LNA adopt highly similar sugar conformations, while cEt positions its substituents closer to the phosphate backbone, contributing to enhanced resistance to enzymatic degradation (Pallan et al., 2012).



|                                   | $\Delta T_m$ per modification ( $^{\circ}\text{C}$ ) | Relative affinity vs DNA |
|-----------------------------------|------------------------------------------------------|--------------------------|
| <b>DNA (unmodified)</b>           | Baseline                                             |                          |
| <b>Phosphorothioate (PS DNA)</b>  | 0                                                    | 0.8–1×                   |
| <b>2'-O-methyl (2'-OMe)</b>       | +0.5 to +1.0                                         | 1.2–1.5×                 |
| <b>2'-fluoro (2'-F)</b>           | +1.0 to +2.0                                         | 1.5–2×                   |
| <b>2'-O-methoxyethyl (2'-MOE)</b> | +0.9 to +1.7                                         | 1.5–2×                   |
| <b>Locked nucleic acid (LNA)</b>  | +3.5 to +8.0                                         | 3–10×                    |

### LNA as Bio-Tools

Locked nucleic acid (LNA) chemistry has become an important component of modern PCR-based technologies, particularly in applications requiring high specificity, sensitivity, and reproducibility. The enhanced hybridization affinity conferred by LNA modifications enables the design of short oligonucleotide primers and probes with elevated and tunable melting temperatures ( $T_m$ ), which improves assay performance across a wide range of targets (Braasch and Corey, 2001; Grunweller and Hartmann, 2007).

Incorporation of LNA residues into PCR primers and fluorescent probes:

- Improved binding to difficult or GC-rich templates
- Shorter primer design without loss of affinity
- Enhanced mismatch discrimination
- Enhanced fluorescence signal upon hybridization
- Enable stable hybridization to short targets

Strategic placement of LNA nucleotides often near the 3'-end can significantly improve allele-specific amplification, making LNA-modified primers particularly useful for single nucleotide polymorphism (SNP) detection and genotyping assays (Di Giusto and King, 2004; Mouritzen et al., 2003). The increased thermodynamic stability also reduces nonspecific amplification by favoring correct primer–template pairing. (Morandi et al., 2007; Valoczi et al., 2004; Castoldi et al., 2006).

## LNA in Therapeutics

**Miravirsen** is a first-in-class LNA-modified antisense oligonucleotide that targets miR-122, a liver microRNA essential for HCV replication, leading to durable, dose-dependent viral suppression with a strong safety profile and no observed resistance. Its success validated LNA chemistry as a viable and safe platform for RNA-targeting therapeutics.

Miravirsen (Santaris/Roche) – anti-miR-122 (HCV)

**L d L d d L L d d L d L d L L** → L = LNA ; d = DNA

Miravirsen is a 15-nucleotide LNA-modified phosphorothioate antisense oligonucleotide designed to bind and sequester miR-122, a liver-specific microRNA essential for hepatitis C virus (HCV) replication. By forming a highly stable heteroduplex with miR-122, miravirsen functionally inhibits the microRNA, thereby destabilizing HCV RNA and suppressing viral replication.

miR-122 binds to two conserved sites in the 5' untranslated region of the HCV genome, enhancing viral stability and replication. Miravirsen blocks this interaction.

Clinical Data (Phase 2a):

Dosing: Five weekly subcutaneous injections (3, 5, or 7 mg/kg)

Demonstrated dose-dependent reductions in HCV RNA:

1.2 log<sub>10</sub> IU/mL reduction (3 mg/kg)

2.9–3.0 log<sub>10</sub> IU/mL reduction (5–7 mg/kg)

Viral suppression persisted beyond the dosing period

Undetectable HCV RNA observed in multiple patients at higher doses

Key Differentiators:

No evidence of viral resistance or escape mutations

Activity across HCV genotypes due to conserved miR-122 binding sites

Favorable pharmacokinetics with prolonged tissue half-life

Lack of cytochrome P450 interactions, reducing drug–drug interaction risk

Safety Profile:

No dose-limiting toxicities observed

Mostly mild to moderate adverse events (e.g., headache, fatigue)

No significant hepatotoxicity or clinically relevant laboratory abnormalities

Reduction in cholesterol levels consistent with miR-122 biology

Miravirsen was a first-in-class microRNA-targeting therapeutic that showed benefit and validated the LNA chemistry as a safe clinical construct for RNA-targeting drugs.

**Cobomarsen** is an LNA-based anti-miR-155 therapy for cutaneous T-cell lymphoma that showed encouraging early clinical results, including a favorable safety profile and disease improvement in most patients, supporting its potential as a novel microRNA-targeted cancer treatment.

**L d L d d L L d L d L L L →** L= LNA ; d = DNA (phosphonothioate backbone)

Phase 1 clinical trial evaluated Cobomarsen in patients with mycosis fungoides (MF), the most common subtype of CTCL, assessed safety, tolerability, pharmacokinetics, and preliminary efficacy across multiple routes of administration, including intralesional, subcutaneous (SC), and intravenous (IV) delivery at doses up to 900 mg.

Cobomarsen demonstrated a favorable safety profile, with no serious adverse events attributed to the drug. The most commonly reported AE was fatigue, neutropenia, injection site pain, nausea, pruritus, and headache. Dose-limiting toxicities were limited to Grade 3 pruritus and tumor flare, and the maximum tolerated dose was not reached.

91% showed improvement in disease burden as measured by the modified Severity Weighted Assessment Tool mSWAT -- the percentage of Body Surface Area affected by patches, plaques, and tumors.

Importantly, improvements in clinical endpoints were accompanied by enhanced quality of life and meaningful patient benefits beyond tumor reduction. These results highlight the potential therapeutic potential of LNA-based anti-miR strategies in oncology. The strategy is not without controversy, however. Targeting microRNA can have unintended consequences as the intended target typically regulates hundreds of other genes. Therefore, it is essential to understand the biological consequences of a potential “on-target” tox effect.



**RG-101** is a GalNAc-conjugated anti-miR-122 therapy that showed strong antiviral activity and good initial tolerability but later revealed safety concerns due to transient hyperbilirubinemia from off-target effects. This led to the development of RG6650, an optimized follow-up compound designed to maintain efficacy while improving safety by reducing interference with bilirubin transport.

|        |             |                                               |
|--------|-------------|-----------------------------------------------|
| RG-101 | LNA/MOE/DNA | Clinically strong antiviral, but safety issue |
|--------|-------------|-----------------------------------------------|

|        |         |                              |
|--------|---------|------------------------------|
| RG6650 | cEt/DNA | selected for improved safety |
|--------|---------|------------------------------|

In early clinical studies (phase 1B study), RG-101 was well tolerated after single-dose administration, with no serious adverse events and mostly mild, self-limiting side effects such as fatigue and injection site reactions (van der Ree et al., 2017). Transient increases in alkaline phosphatase and a single case of cholestasis were observed, but there was no consistent evidence of hepatocellular injury or significant bilirubin elevation across the cohort (van der Ree et al., 2017). Later studies of RG-101 demonstrated strong antiviral efficacy but showed transient hyperbilirubinemia that was attributed to off-target inhibition of the MRP2 transporter (van der Ree et al., 2017; Drygin, 2022). Following this finding, Regulus conducted a focused optimization effort to identify next-generation compounds that retained potent miR-122 inhibition while minimizing interference with bilirubin transport (Drygin, 2022). This work involved a screening and structure–activity relationship (SAR) campaign that ultimately leading to the identification of RG6650 (Drygin, 2022).

|                    | <b>RG-101</b>                                                                                                                                                          | <b>RG6650 / RG7443</b> |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>Base</b>        | A m <sup>5</sup> C A m <sup>5</sup> C m <sup>5</sup> C A m <sup>5</sup> U T G U C A C A C T C C A                                                                      | U C A C A C T C C      |
| <b>Backbone</b>    | Full phosphorothioate                                                                                                                                                  | Full phosphorothioate  |
| <b>Sugar</b>       | E E E E E E E d d L L d L d L d L L d                                                                                                                                  | K K d K d K d K K      |
| <b>Legend</b>      | E = 2'-MOE; d = DNA; L = LNA; A = adenine; T = thymine; G = guanine; C = cytosine; U = uracil; m <sup>5</sup> C = 5-methylcytosine; m <sup>5</sup> U = 5-methyluracil. | K = S-cEt; d = DNA     |
| <b>Conjugation</b> | 3' GalNAc-conjugated                                                                                                                                                   | 3' GalNAc-conjugated   |

[TODO: Incomplete section]



In early clinical development, the LNA gapmer EZN-4176 exemplified the balance between potency and safety that characterizes high-affinity antisense chemistries. In a first-in-human phase I study in patients with castration-resistant prostate cancer, EZN-4176 demonstrated limited clinical activity despite promising preclinical rationale. Treatment was generally tolerated at lower dose levels; however, dose escalation was constrained by hepatotoxicity, with reversible elevations in alanine aminotransferase (ALT) and aspartate aminotransferase (AST) observed as dose-limiting toxicities at 10 mg/kg (Bianchini et al., 2013). Notably, these

transaminase elevations occurred in the absence of concomitant bilirubin increases, suggesting hepatocellular injury rather than cholestatic dysfunction (Bianchini et al., 2013). Despite achieving systemic exposures consistent with pharmacological activity, no meaningful reduction in androgen receptor (AR) expression or sustained antitumor responses were observed, and the study was discontinued due to the combination of toxicity and insufficient efficacy (Bianchini et al., 2013).

Mechanistically, these findings are consistent with the known properties of locked nucleic acid (LNA)–modified gapmers. The increased binding affinity conferred by LNA chemistry enhances target hybridization but also increases the likelihood of hybridization-dependent off-target effects and nonspecific interactions with cellular macromolecules. Both preclinical and clinical evidence suggest that such interactions may drive hepatic accumulation and toxicity, a phenomenon observed across multiple LNA-based oligonucleotides (Swayze et al., 2007; Bianchini et al., 2013). In the case of EZN-4176, its high-affinity binding profile likely contributed not only to hepatotoxicity but also to reduced specificity, limiting its ability to achieve selective and effective AR suppression in patients.

Importantly, although EZN-4176 was designed to target exon 4 of AR mRNA and suppress both full-length and splice variant receptors, clinical biopsy data failed to demonstrate meaningful AR downregulation following treatment (Bianchini et al., 2013). This disconnect between target engagement in preclinical models and clinical outcomes underscores a central challenge in antisense drug development: increasing binding affinity can improve in vitro potency but may also exacerbate off-target effects, narrow the therapeutic window, and ultimately hinder clinical translation.

These limitations parallel observations from other antisense programs targeting the androgen receptor. For example, EZN-4176 can be contrasted with later-generation compounds such as ISIS-AR Rx (AZD5312), which incorporates constrained ethyl (cEt) modifications designed to retain potency while improving tolerability and specificity. While cEt chemistry reduces some of the toxicity risks associated with LNA, clinical outcomes have shown that improved safety alone does not guarantee efficacy, particularly if sufficient target engagement or downstream biological effect is not achieved.

Taken together, these findings highlight a fundamental tension in antisense therapeutics: while increasing binding affinity (as seen with LNA chemistry) can enhance potency, it often comes at the cost of increased off-target interactions and toxicity. Conversely, newer chemistries such as cEt may improve safety margins but do not inherently ensure clinical success. Achieving an optimal balance between affinity, specificity, and tolerability remains a key determinant of successful antisense drug development.



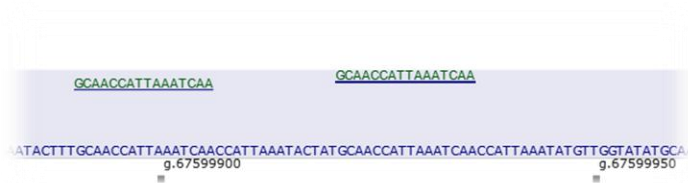
|                  | <b>EZN-4176</b>                  | <b>ISIS-AR Rx (AZD5312)</b>         |
|------------------|----------------------------------|-------------------------------------|
| Developer        | Enzon                            | Ionis / AstraZeneca                 |
| Chemistry        | LNA gapmer                       | cEt gapmer                          |
| Target           | Androgen receptor (AR)           | Androgen receptor (AR)              |
| Backbone         | Phosphorothioate                 | Phosphorothioate                    |
| Sugar mods       | LNA + DNA                        | cEt + DNA                           |
| Clinical outcome | ALT elevations (toxicity signal) | Developed to improve potency/safety |



*The Enzon–Santaris LNA gapmer program progressed into the clinic by targeting oncogenic drivers like AR and PI3K/Akt, but the same compound may have also unintentionally suppressed MCM9, a DNA repair gene linked to cancer predisposition. --a critical failure in bioinformatics off-target analysis that made it all the way into the clinic!*

|                 |                                                        |
|-----------------|--------------------------------------------------------|
| <b>Feature</b>  | ISIS-AR Rx / AZD5312 (AKA ISIS 560131)                 |
| <b>Base</b>     | TTGATTTAATGGTTGC                                       |
| <b>Backbone</b> | Full phosphorothioate                                  |
| <b>Sugar</b>    | k k k d d d d d d d d k k E                            |
| <b>Legend</b>   | k = (S)-cEt; d = DNA; E = 2'-MOE; C = 5-methylcytosine |

**ISIS AR Rx HITS ANDROGEN RECEPTOR TWICE.**



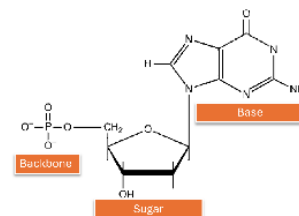
Recent data (2024–early 2026) indicate that AZD5312 shows promising efficacy in prostate cancer with observed PSA declines in some heavily pre-treated patients and enhanced activity in combination with AKT inhibitors (e.g.,

Capivasertib). Its safety profile is consistent with antisense oligonucleotides, with common adverse events including fatigue, injection site reactions, and flu-like symptoms, while more serious but manageable risks include thrombocytopenia, hepatotoxicity, and renal effects requiring routine monitoring (AstraZeneca, 2024; ClinicalTrials.gov, 2025; de Bono et al., 2023).

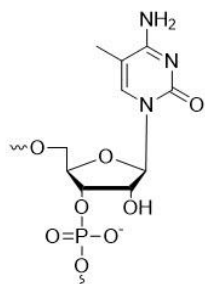
Locked nucleic acid (LNA) chemistry represents one of the most potent antisense modification strategies for RNA targeting due to its conformationally “locked” sugar, which markedly increases binding affinity and duplex stability relative to earlier chemistries such as 2'-O-methoxyethyl (MOE) and 2'-O-methyl modifications (Qassem et al., 2024; Crooke et al., 2018). This enhanced affinity enables shorter oligonucleotides, improved potency, and flexible designs (e.g., LNA/DNA gapmers) that retain RNase H activity while achieving stronger target engagement than MOE-based ASOs (Qassem et al., 2024). However, these advantages come with important trade-offs: compared with MOE chemistries, which generally exhibit lower toxicity and reduced immunogenicity, LNA-modified ASOs show a higher incidence of hybridization-dependent off-target effects due to their increased binding strength and shorter sequence length, as well as a greater risk of hepatotoxicity observed in preclinical models (Qassem et al., 2024; Crooke et al., 2018). Thus, while MOE modifications offer a more balanced safety profile with moderate potency, LNA chemistry shifts the balance toward maximal efficacy at the expense of a narrower therapeutic window, making careful sequence design and dose optimization critical for clinical application.

# Nucleobase modifications

“Base” chemistry is increasingly used to modulate recognition, stability, and enzymatic interactions. While there are numerous one-off chemistries this review will cover some of the more commonly used modifications.

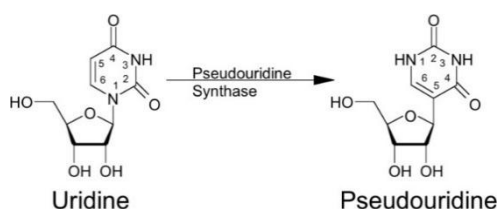


**5-methylcytidine (m5C):** Improves stability and reduces immunogenicity



5-methylcytosine (m5C; 5-methylcytidine, 5-Me-C) is a methylated RNA base analog found across tRNA, other cellular RNAs, where it functions as a post-transcriptional modification contributing to RNA stability, processing, and translation (Motorin and Helm, 2011; Squires et al., 2012). Its best-characterized role is in tRNA, where m5C protects transcripts from endonucleolytic cleavage and preserves structural integrity under stress (Blanco et al., 2014).

Incorporation of 5-methylcytidine into synthetic RNA oligonucleotides enhances base pairing and duplex stability through increased hydrophobicity at the C5 position and raises melting temperature (~1–1.5 °C per modification). This stabilizing effect improves hybridization, supports disruption of target secondary structure, and enables the use of shorter antisense oligonucleotides with improved binding efficiency. At the same time m5C reduces susceptibility to nuclease degradation and dampens activation of innate immune sensors that preferentially recognize unmodified RNA (Karikó et al., 2005; Anderson et al., 2010).



RNA-like

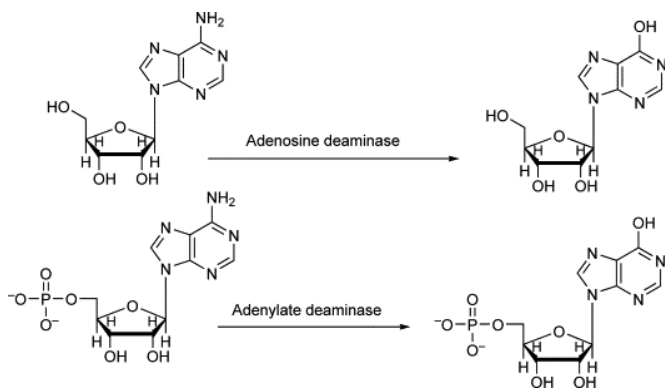
**Pseudouridine (Ψ):** enhances behavior and stability

Used in mRNA-based vaccines, such as those for COVID-19, to enhance stability and reduce immune recognition, Pseudouridine (Ψ), the C5-glycoside isomer of uridine, is the most abundant post-transcriptional RNA modification and is widely distributed across tRNA, rRNA, mRNA, and non-coding RNAs, where it contributes to RNA stability, structure, and translation (Lin et al., 2021). The isomerization of uridine to Ψ introduces a carbon–carbon glycosidic bond

and an additional hydrogen bond donor, enhancing base stacking, strengthening RNA–RNA and RNA–protein interactions, and increasing structural rigidity (Lin et al., 2021). Endogenously,  $\Psi$  plays critical roles in stabilizing tRNA architecture, supporting ribosome function through rRNA modification, and modulating mRNA stability and translation efficiency (Zhao and He, 2015; Lin et al., 2021). These native properties drive their use in RNA therapeutics, where incorporation of  $\Psi$  into synthetic RNAs enhances duplex stability, increases resistance to nuclease degradation, and improves translational output, often yielding significantly higher protein expression compared to unmodified RNA (Lin et al., 2021).

Pseudouridine also reduces immunogenicity by diminishing activation of innate immune sensors and downstream cytokine responses, thereby enabling sustained translation and improved tolerability of exogenous RNA (Karikó et al., 2008; Lin et al., 2021). This combination of structural stabilization, enhanced translation, and immune evasion has made pseudouridine and its derivatives (e.g., N1-methylpseudouridine) foundational to modern RNA therapeutic platforms, including mRNA vaccines and protein replacement therapies, where mimicking endogenous RNA chemistry is key to achieving clinical efficacy.

mRNA COVID-19 vaccines are structurally composed of a nucleoside-modified mRNA encoding the SARS-CoV-2 spike protein, encapsulated within a lipid nanoparticle (LNP) delivery system. The mRNA includes regulatory elements such as 5' and 3' untranslated regions, a coding sequence for a stabilized pre-fusion spike protein, and multiple stop codons to ensure accurate translation (Morais, Adachi and Yu, 2021). A key structural feature is the substitution of uridine with N1-methyl-pseudouridine, which enhances mRNA stability, reduces innate immune activation, and improves protein translation efficiency (Morais, Adachi and Yu, 2021). Surrounding this, the LNP structure protects the mRNA from degradation, facilitates cellular uptake, and promotes delivery to immune cells such as dendritic cells, enabling antigen expression and subsequent immune activation (Morais, Adachi and Yu, 2021)

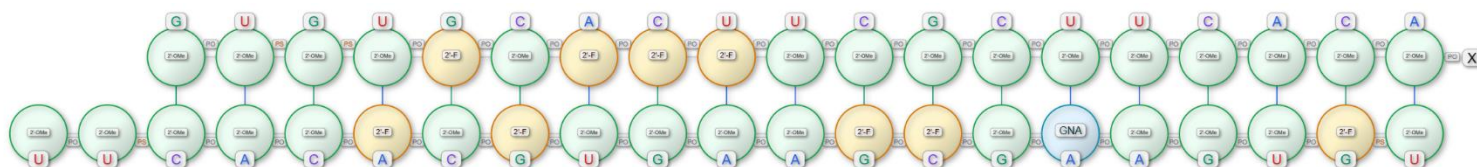


**Inosine:** enables flexible base pairing, critical for editing systems; Inosine is used in ADAR technology (RNA editing) since it is read by cells as a G. Inosine is a functionally

unique RNA modification generated through ADAR-mediated deamination of adenosine, enabling dynamic rewriting of RNA information without altering the genome. Because inosine is interpreted as guanosine during translation and splicing, A-to-I editing can recode proteins, alter splice sites, and reshape RNA structure, thereby modulating transcript stability, localization, and protein output (Srinivasan et al., 2021 ; Eisenberg and Levanon, 2018). In tRNAs, inosine at the wobble position expands codon recognition capacity, allowing a single tRNA to decode multiple synonymous codons and enhancing translational efficiency and proteome diversity (Srinivasan et al., 2021 ). In mRNAs, site-specific editing can produce amino acid substitutions or regulate RNA processing, as exemplified by editing of neurotransmitter receptors and ion channels, with direct implications for neurological function and disease.

This intrinsic property provides the platform for RNA editing therapeutics; programmable A-to-I editing uses natural ADAR enzymes to fix disease-causing mutations at the RNA level, making it a temporary and potentially safer alternative to editing DNA. Inosine also helps reduce immune reactions to therapeutic RNAs by preventing detection by sensors like MDA5, improving their safety. Its ability to pair flexibly with different bases allows better control over RNA structure and protein production, which is useful in designing therapies like antisense, siRNA, and mRNA. However, challenges such as unintended edits, variable efficiency, and delivery issues still limit clinical translation.

**Glycerol nucleic acids (GNA)** Glycol nucleic acid (GNA) is a nucleic acid analogue featuring an acyclic three-carbon glycol backbone that can still support Watson–Crick base pairing and duplex formation, despite lacking the conventional ribose sugar found in DNA and RNA (Zhang et al., 2005). In the context of siRNA therapeutics, GNA is incorporated as a chemical modification to improve pharmacological properties, particularly safety. Specifically, (S)-GNA substitutions—most effectively at position 7 of the antisense strand seed region—introduce controlled thermodynamic destabilization that reduces unintended, seed-mediated off-target interactions while preserving on-target gene silencing (Schlegel et al., 2022). This modification also enhances nuclease resistance and can improve strand selection and loading into the RNA-induced silencing complex (RISC), thereby maintaining or even increasing potency (Schlegel et al., 2017). Overall, GNA is used in siRNA therapeutics because it enables fine-tuning of duplex stability and structure, leading to improved specificity, reduced toxicity, and favorable clinical performance.

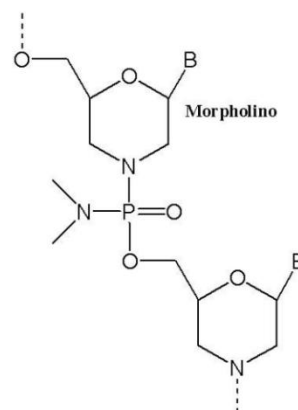


**Elebsiran (ALN-HBV02)**, a GalNAc-conjugated siRNA developed for chronic hepatitis B uses the “enhanced stabilization chemistry” (ESC+), within which a GNA substitution contributes to improved strand specificity and reduced off-target activity.

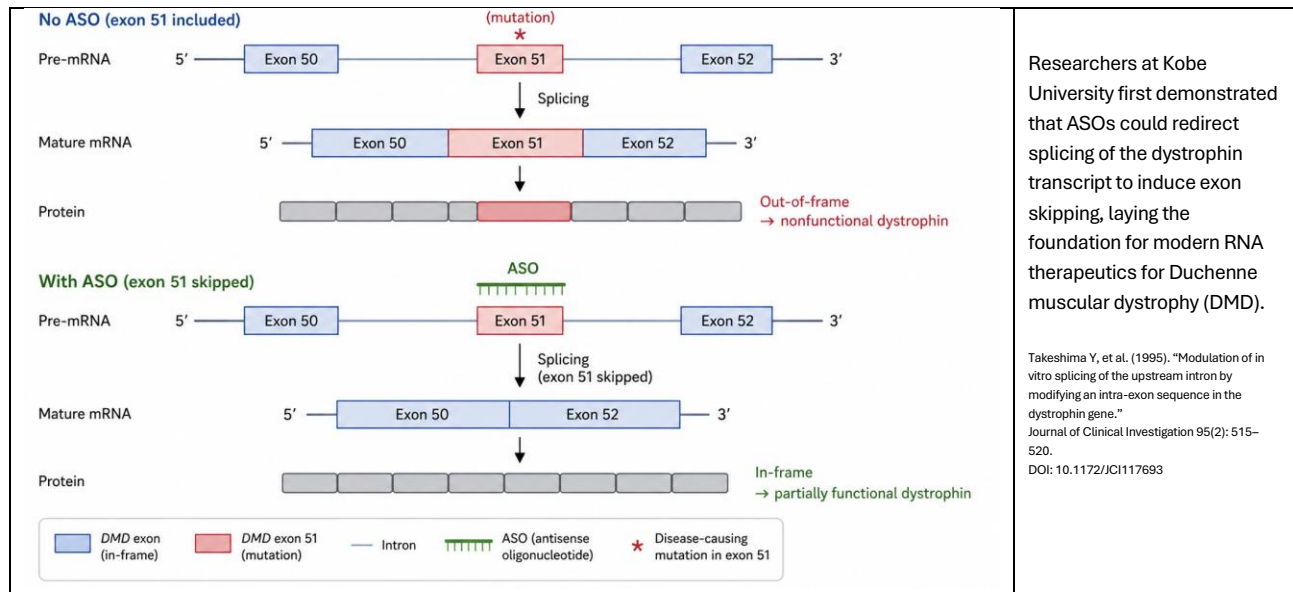
In Phase 1/2 studies, Elebsiran demonstrated knockdown of HBsAg, with reductions often exceeding 1.5–2.0 log<sub>10</sub> IU/mL following subcutaneous dosing at 300 mg every 4–8 weeks, and evidence of sustained activity beyond the dosing interval (Yuen et al., 2022; Gane et al., 2023). Importantly, the safety profile was favorable, with no dose-limiting toxicities and a low incidence of ALT elevations, supporting the hypothesis that chemical refinements such as GNA substitutions mitigate hepatocellular stress linked to off-target effects. Across cohorts, repeated dosing maintained efficacy without evidence of tachyphylaxis, and pharmacodynamic responses were consistent with prolonged hepatic RISC loading. (Yuen et al., 2022; Gane et al., 2023).

# Morpholinos

The development of morpholino oligonucleotides arose from the need to overcome fundamental limitations of early antisense chemistries, particularly the poor stability of natural oligonucleotides in biological systems due to nuclease degradation and the nonspecific interactions caused by their negatively charged phosphodiester backbone (Stein and Cheng, 1993; Summerton, 1999). To address these issues, James Summerton in 1985 working at Antivirals which later became Santarus Inc replaced the ribose sugar with a chemically robust morpholine ring and substituted the charged phosphodiester linkage with a neutral phosphorodiamidate backbone. (Summerton and Weller, 1997; Li, 2017, p. 40). This elegant design preserved the Watson–Crick base pairing and allowed for effective sequence-specific binding to target RNA and subsequent steric blocking for transcript manipulation (Moulton and Moulton, 2017, pp. v–vi, 11).



In the early 1990s, Zbigniew Dominski and Ryszard Kole at the University of North Carolina demonstrated that targeting specific splice elements in  $\beta$ -globin pre-mRNA could redirect the splicing machinery and restore correct mRNA production in vitro. This seminal work laid the foundation for the modern splice-switching therapeutic strategies we have today, showing that RNA splicing is not only mechanistically tractable but also chemically programmable. The favorable safety of morpholinos subsequently enabled the first clinical applications of splicing therapeutics.



AVI-4658, an exon 51–targeting PMO, showed in a proof-of-concept clinical study showed that direct intramuscular administration could induce sequence-specific exon skipping and restore dystrophin expression locally in patients, without significant safety concerns (Kinali et al., 2009). This study provided the first clinical validation of PMOs as a therapeutic platform, establishing proof-of-mechanism in humans and laying the foundation for subsequent systemic PMO-based therapies.



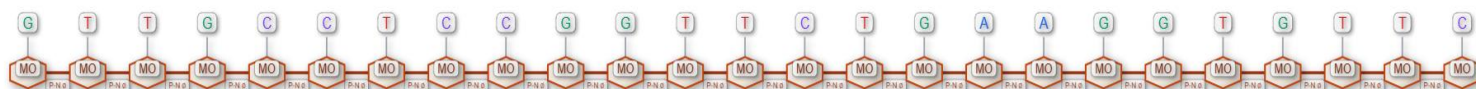
#### ETEPLIRSEN

Eteplirsen, a phosphorodiamidate morpholino oligomer, was evaluated in a small randomized, placebo-controlled trial in boys with Duchenne muscular dystrophy amenable to exon 51 skipping, demonstrating that sustained treatment restores production of truncated but functional dystrophin (Mendell et al., 2013). While no meaningful dystrophin increase was observed at 12 weeks, significant increases in dystrophin-positive muscle fibers were seen by 24 weeks and reached 40-50% by 48 weeks, with corresponding restoration of associated proteins such as nNOS and sarcoglycans (Mendell et al., 2013). Functionally, Eteplirsen-treated patients showed stabilization of walking ability compared with the expected decline in placebo patients, with an approximate 67 m benefit in the 6-minute walk test among evaluable patients (Mendell et al., 2013). The study suggests that treatment duration, rather than dose, is the key driver of



efficacy, and that Eteplirsen is generally well tolerated, though conclusions are limited by the small sample size and variability in disease progression (Mendell et al., 2013).

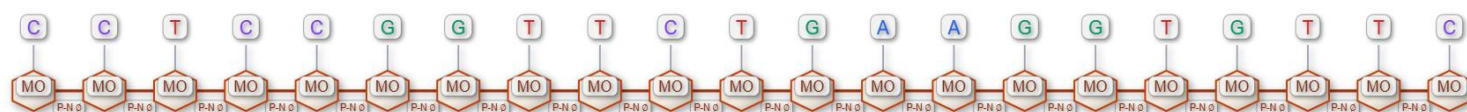
#### GOLODIRSEN (VYONDYS 53):



Golodirsen (Vyondys 53) is a phosphorodiamidate morpholino oligomer (PMO) antisense oligonucleotide indicated for Duchenne muscular dystrophy (DMD) patients with mutations amenable to exon 53 skipping. It binds to exon 53 of dystrophin pre-mRNA, promoting exon skipping and enabling production of a truncated but partially functional dystrophin protein (FDA, 2019). It received accelerated approval in the U.S. based on increased dystrophin production rather than confirmed clinical outcomes (FDA, 2019). The recommended dosing is 30 mg/kg IV weekly (FDA, 2019).

|                   |                                                                                             |
|-------------------|---------------------------------------------------------------------------------------------|
| Exon skipping     | Increased in all patients; 28.9-fold increase at 48 weeks (FDA, 2019; Servais et al., 2022) |
| Dystrophin        | 0.10% → 1.02% of normal (mean +0.92%)                                                       |
| Dose              | 30 mg/kg IV weekly                                                                          |
| PK linearity      | Dose-proportional exposure; minimal accumulation                                            |
| Cmax variability  | 38–72% CV                                                                                   |
| AUC variability   | 34–44% CV                                                                                   |
| Vd (steady-state) | 668 mL/kg                                                                                   |
| Protein binding   | 33–39%                                                                                      |
| Half-life         | 3.4 hours                                                                                   |
| Clearance         | 346 mL/hr/kg                                                                                |
| Metabolism        | Minimal; no metabolites detected                                                            |
| Excretion         | Primarily unchanged in urine                                                                |
| Renal impairment  | AUC ↑ 1.2× (Stage 2 CKD), 1.9× (Stage 3 CKD) (FDA, 2019)                                    |
| DDI potential     | Low (FDA, 2019)                                                                             |

Source: FDA (2019) Vyondys 53 (golodirsen) prescribing information. Sarepta Therapeutics.



## VILTOLARSEN

**Viltolarsen (Viltepso)** is a phosphorodiamidate morpholino oligomer that induces exon 53 skipping in dystrophin pre-mRNA, restoring production of a truncated but functional dystrophin protein (FDA, 2020). Treatment results in measurable increases in dystrophin expression (~5–6% of normal) and consistent exon skipping at the mRNA level, confirming target engagement (FDA, 2020; Clemens et al., 2020). Pharmacokinetically, viltolarsen demonstrates dose-proportional exposure, minimal accumulation, and rapid renal elimination of unchanged drug, with low metabolic and drug–drug interaction potential (FDA, 2020). Clinically, it is generally well tolerated, with mostly mild adverse events (e.g., respiratory infections, injection-site reactions), no serious drug-related adverse events in trials, and low immunogenicity; however, renal toxicity remains a theoretical class risk, requiring ongoing monitoring (FDA, 2020; Clemens et al., 2020).

Viltolarsen is generally well tolerated, with most adverse events being mild and including upper respiratory tract infections, cough, pyrexia, and injection-site reactions (FDA, 2020). Clinical trial data show no serious drug-related adverse events or treatment discontinuations, supporting an overall favourable safety profile (Clemens et al., 2020). However, as with other antisense oligonucleotides, there is a potential risk of renal toxicity, necessitating routine monitoring during treatment (FDA, 2020). Immunogenicity appears minimal, with no significant immune responses reported (FDA, 2020).

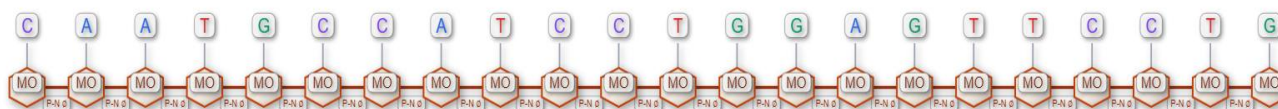
| Parameter                     | Value / Description                           |
|-------------------------------|-----------------------------------------------|
| Administration                | IV infusion (60 min)                          |
| Bioavailability               | 100%                                          |
| Tmax                          | 1 hour (end of infusion)                      |
| Exposure                      | Dose-proportional (1.25–80 mg/kg/week)        |
| Accumulation                  | Minimal with weekly dosing                    |
| Volume of distribution        | 300 mL/kg                                     |
| Plasma protein binding        | 39–40%                                        |
| Metabolism                    | Metabolically stable; no metabolites detected |
| Elimination                   | Primarily renal (unchanged drug)              |
| Half-life (t <sub>1/2</sub> ) | 2.5 hours                                     |

**Clearance**

217 mL/hr/kg

**Drug interactions**

Low potential (no major CYP/transporter effects)



Casimersen (AMONDYS 45™)

**Casimersen (AMONDYS 45)** is a phosphorodiamidate morpholino oligomer (PMO) antisense oligonucleotide approved for the treatment of Duchenne muscular dystrophy (DMD) in patients with mutations amenable to exon 45 skipping (FDA, 2021). DMD is an X-linked recessive disorder characterized by absence of functional dystrophin, leading to progressive muscle degeneration and premature mortality (Assefa et al., 2024). Casimersen received accelerated FDA approval based on its ability to increase dystrophin production, a surrogate biomarker, with confirmatory trials required to establish clinical benefit (FDA, 2021).

Casimersen acts at the level of pre-mRNA splicing. It binds specifically to exon 45 of dystrophin pre-mRNA, preventing its inclusion during mRNA processing. This restores the translational reading frame and allows synthesis of a truncated but partially functional dystrophin protein (FDA, 2021; Assefa et al., 2024). In a placebo-controlled study dystrophin increased from 0.93% to 1.74% of normal after 48 weeks

Casimersen is generally well tolerated, with most adverse events being mild to moderate and consistent with common intercurrent illnesses. The most frequently reported adverse reactions include upper respiratory tract infection (65%), cough (33%), pyrexia (33%), headache (32%), arthralgia (21%), and oropharyngeal pain (21%) (FDA, 2021). Despite this favorable tolerability profile, an important safety concern is the potential for renal toxicity, which has been observed in animal studies and is considered a class effect of antisense oligonucleotides. Consequently, careful renal monitoring is recommended during treatment, including assessment of serum cystatin C, urine protein, and urine protein-to-creatinine ratio to detect early signs of nephrotoxicity (FDA, 2021) (Assefa et al., 2024).

## PK/PD

**Key PD outcome****+0.81% dy'strophin vs +0.22% placebo****Dose**

30 mg/kg IV weekly

**Cmax**

End of infusion

**Half-life**

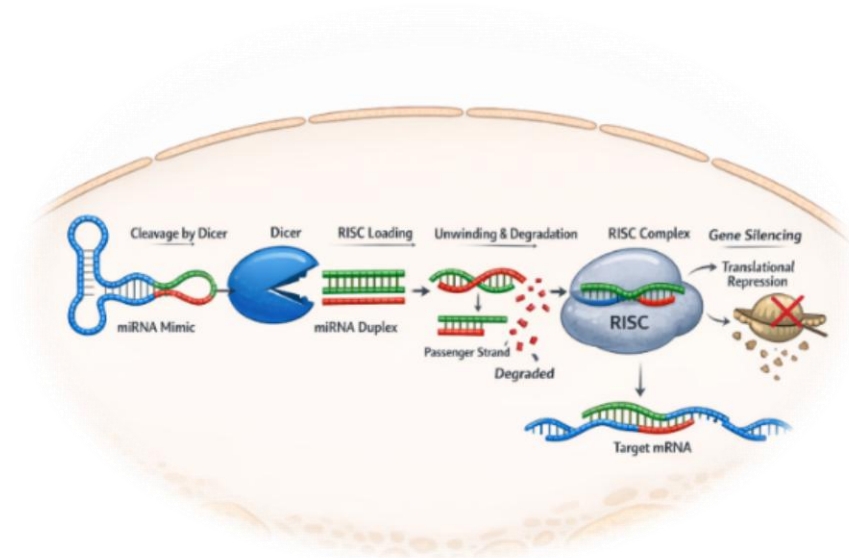
3.5 h

**Clearance**

180 mL/hr/kg

|                        |                                |
|------------------------|--------------------------------|
| <b>Vd</b>              | 367 mL/kg                      |
| <b>Protein binding</b> | 8.4–31.6%                      |
| <b>Metabolism</b>      | Minimal (non-hepatic)          |
| <b>Excretion</b>       | >90% renal (unchanged)         |
| <b>Accumulation</b>    | None                           |
| <b>Key safety risk</b> | Renal toxicity                 |
| <b>Common AEs</b>      | URTI, cough, pyrexia, headache |

## Design Strategies & Delivery



### miRNA mimics

A microRNA (miRNA) mimic is typically designed as a synthetic double-stranded RNA that resembles the endogenous miRNA duplex. It contains a guide strand (Antisense) corresponding to the mature miRNA sequence and a complementary passenger strand (Sense), forming a short duplex with characteristic features such as 2-nucleotide 3' overhangs generated by Dicer processing. After introduction into the cell, the miRNA mimic duplex is unwound during RISC loading, the passenger strand is degraded, and the guide-strand is incorporated into the RNA-induced silencing complex (RISC). Within RISC, the guide strand directs gene silencing by partial base pairing with target mRNAs, leading to translational repression or a catalytic mRNA degradation.

miRNAs are powerful regulators that can control multiple genes and entire cellular pathways, making them promising targets for therapy as well as biomarkers. However, despite strong preclinical progress and growing interest, especially following the success of mRNA vaccines, the field remains in early clinical stages, with no FDA-approved miRNA therapies and several trials halted due to toxicity and immune-related side effects. It remains unclear whether the observed toxicities in miRNA therapeutics arise from the miRNA or the delivery system.

**MRX34** was a first-in-class microRNA therapeutic composed of a synthetic double-stranded miR-34a mimic formulated within a lipid nanoparticle delivery system to restore the tumor-suppressive miR-34 pathway downstream of p53 and repress multiple oncogenic targets. Preclinical studies and early Phase 1 clinical data demonstrated successful systemic delivery, target engagement, and limited antitumor activity, supporting proof-of-concept for miRNA replacement therapy. However, the trial was terminated in 2016 due to severe immune-related

toxicities, including cytokine release syndrome and treatment-related deaths, likely caused by innate immune activation triggered by the nanoparticle. This outcome highlighted key challenges in RNA therapeutics particularly the balance between structural design, delivery efficiency, and immunogenicity despite the strong biological rationale for miR-34-based therapy (Zhang et al., 2019; Hong et al., 2020)

MRX34 is an unmodified RNA formulated using a liposomal nanoparticle delivery system (NOV340) composed of amphoteric lipids designed to encapsulate and protect the double-stranded miR-34a mimic during systemic administration. The lipid formulation typically included components such as phospholipids, cholesterol, and PEGylated lipids to enhance stability, prolong circulation time, and facilitate cellular uptake and endosomal release of the RNA cargo.

In the Phase 1 clinical trial, MRX34 was administered intravenously twice weekly (commonly on days 1 and 4 of a 3-week cycle), with dose escalation up to a maximum tolerated dose of approximately 110 mg/m<sup>2</sup> (about 40mg per dose) in non-hepatocellular carcinoma and HCC patients, before immune-related toxicities limited further dosing (Hong et al., 2020; Beg et al., 2017). The trial was terminated due to serious immune-mediated toxicities, including fatal adverse events in multiple patients (Kim and Croce, 2023). These toxicities were characterized by strong immune activation and while MRX34 successfully downregulated oncogenic target genes, the severity of these adverse effects led to trial discontinuation (Kim and Croce, 2023).

The toxicity and clinical outcomes observed with MRX34 cannot be attributed to a single factor, as multiple confounding elements were involved. The lipid nanoparticle delivery system facilitated broad systemic distribution, potentially exposing diverse tissues to the therapy, while frequent and high dosing likely led to excessive off-target accumulation. The complexity of the drug product complicates efforts to determine whether toxicity was primarily driven by the delivery system, payload miRNA, or the intrinsic biology of miRNA regulation.

An alternative to traditional, repeated systemic delivery of synthetic miRNA mimics and lipid nanoparticles (LNPs) involves a vector-mediated strategy for sustained, intracellular expression. Researchers at UCL (University College London) have utilized viral vector gene therapy to achieve this, specifically through the AAV5-delivered miRNA construct underlying AMT-130. This approach employs an engineered pri-miR-451a scaffold to express an anti-huntingtin microRNA directly within transduced neurons.

Unlike MRX34, which required frequent intravenous dosing due to the transient nature of synthetic RNA, the AAV-mediated delivery enables long-term, endogenous production of the therapeutic miRNA following a single administration. In this approach, the therapeutic sequence is not delivered as a mature, fragile duplex RNA; it is instead delivered as a DNA cassette encoding a pri-miRNA transcript, which is subsequently processed by the host native machinery into the active guide strand.

AMT-130 incorporates a CAG promoter, engineered pri-miR-451 flanking regions, a pre-miRNA hairpin containing the HTT-targeting guide strand, and an hGH polyadenylation signal cassette to support stable neuronal expression and accurate RNA processing (Evers et al., 2018). Importantly, the miR-451 scaffold was selected because it undergoes Ago2-dependent, Dicer-independent maturation, reducing passenger-strand accumulation and potentially improving strand specificity compared with conventional shRNA or duplex miRNA approaches (Cheloufi et al., 2010; Yang et al., 2010). Although AAV-mediated delivery introduces its own challenges particularly the permanence of transgene expression, limited reversibility, and concerns regarding long-term safety and immune responses to viral capsids, it avoids several limitations associated with systemic nanoparticle delivery, including repeated high-dose exposure, broad biodistribution of synthetic RNA, and dose-limiting innate immune activation. Consequently, the transition from MRX34-like nanoparticle systems to AAV-expressed miRNA therapeutics reflects a broader evolution in the RNA therapeutics field toward more durable, tissue-targeted, and biologically integrated delivery platforms.



The therapeutic cassette described in the patent consists of a fully engineered adeno-associated viral (AAV) expression designed to achieve long-term suppression of huntingtin (HTT) expression in neurons through RNA interference. In the disclosed pVD-CAG-miH12-451 configuration, the cassette begins with the synthetic CAG promoter, a strong hybrid promoter combining the cytomegalovirus (CMV) enhancer and chicken  $\beta$ -actin promoter elements, included to drive high-level transgene expression in neuronal tissue following intracerebral AAV delivery (Evers et al., 2018). Downstream of the promoter is the 5' pri-miRNA flanking region derived from endogenous miR-451a. These flanking sequences are necessary because they provide the structural and sequence context recognized by the endogenous microRNA-processing machinery, particularly the Drosha/DGCR8 complex, enabling correct cleavage of the primary transcript into a precursor miRNA hairpin (Cheloufi et al., 2010). Embedded within this framework is the engineered pre-miRNA hairpin itself, which constitutes the active therapeutic silencing element. The hairpin contains the HTT-targeting guide strand sequence 5'-AAGGACUUGAGGGACUCGAAG-3' (SEQ ID NO.5), which is fully complementary to the selected exon-1 huntingtin target sequence and mediates sequence-specific degradation of HTT mRNA through incorporation into the RNA-induced silencing complex (RISC) (Evers et al., 2018). Opposite the guide strand is the engineered passenger strand, designed not only to stabilize the stem-loop structure but also to minimize passenger-strand loading into RISC and thereby reduce off-target silencing effects. The use of the miR-451a scaffold is particularly important because miR-451 follows a non-canonical Ago2-dependent, Dicer-independent

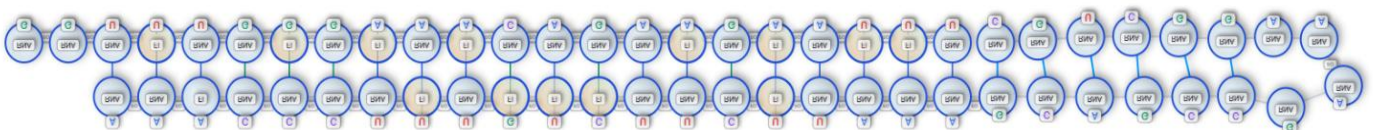


maturation pathway that preferentially produces a single dominant guide strand, improving strand specificity and reducing unintended RNAi activity (Cheloufi et al., 2010; Yang et al., 2010). Following the pre-miRNA region, the cassette contains the 3' pri-miRNA flanking sequence, which further supports accurate Drosha processing, transcript stability, and nuclear export of the precursor hairpin. Finally, the construct terminates with the human growth hormone (hGH) polyadenylation signal, which directs efficient transcriptional termination and polyadenylation, thereby stabilizing the therapeutic transcript after expression from the AAV genome. Collectively, these engineered sequence components form a highly optimized neuronal RNAi cassette intended to provide durable expression, efficient miRNA biogenesis, selective guide-strand loading, and sustained suppression of mutant and wild-type huntingtin expression within the central nervous system (Evers et al., 2018).

## The role of Dicer

RNA interference (RNAi) is a highly conserved mechanism for post-transcriptional gene regulation. In mammalian systems, this process is mediated by a series of coordinated molecular interactions involving the RNase III enzyme Dicer, which cleaves longer dsRNA into short interfering RNAs (siRNAs), and the RNA-induced silencing complex (RISC); uses the guide strand to recognize and cleave target transcripts (Hefner et al., 2008). While standard (Alnylam-style) synthetic 21–23 nucleotide siRNAs mimic the “products” of Dicer processing and can directly engage RISC, they bypass the natural Dicer processing step. This may lead to reduced efficiency and less controlled strand selection. In contrast, longer Dicer-substrate RNAs (e.g., 27mers), designed to engage the endogenous RNAi machinery more fully, may enable Dicer-mediated cleavage and directional handoff to RISC and potentially leading to enhances potency, durability, and specificity of gene silencing (Hefner et al., 2008). This is the theory behind the platform technology developed by a company called Dicerna which sold to Novo Nordisk acquired in 2021 for \$3.3 billion.

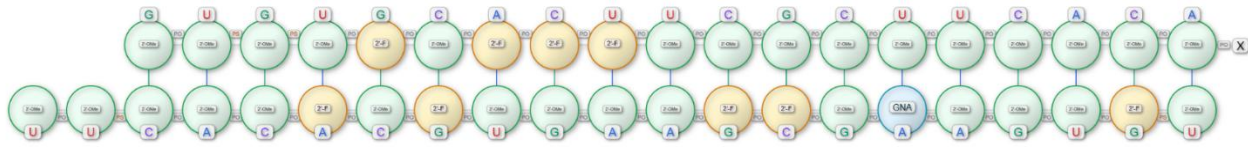
A research group from Bio-Rad Laboratories in collaboration with Integrated DNA Technologies (IDT) performed a series of experiments comparing Dicer-substrate 27mer RNAs with conventional 21mer siRNAs to evaluate the impact of engaging the Dicer processing step





on RNAi performance. In HeLa cell models across multiple gene targets (including CDK2, TP53, RAF1, and ACTB), the Dicer-substrate 27mers produced greater gene knockdown in dose–response studies, particularly at lower concentrations, indicating enhanced intrinsic potency (Hefner, E., Clark, K., Whitman, C., Behlke, M.A., Rose, S.D., Peek, A.S. and Rubio, T. (2008). They also demonstrated more sustained silencing over time, whereas the activity of 21mer siRNAs declined more rapidly. The authors attributed these improvements to Dicer-mediated processing and efficient handoff to RISC, which promotes accurate guide-strand selection and improved complex loading. Notably, the ability to achieve effective silencing at lower doses suggests a potential reduction in off-target effects and toxicity, supporting both improved efficacy and safety relative to traditional siRNA designs (Hefner et al., 2008).

## siRNA



### Length and Structure

The typical design of small interfering RNAs (siRNAs) as 21–23 nucleotide double-stranded duplexes with 2-nucleotide 3' overhangs is considered optimal because it closely mimics the natural products of Dicer processing and maximizes compatibility with the RNA interference (RNAi) machinery. Elbashir et al. (2001) demonstrated that synthetic 21-nt siRNA duplexes are sufficient to induce efficient and sequence-specific gene silencing in mammalian cells while avoiding the nonspecific responses triggered by longer double-stranded RNAs. The presence of 2-nt 3' overhangs, typically composed of UU or TT residues, further enhances silencing efficiency by facilitating Dicer recognition and promoting effective loading into the RNA-induced silencing complex (RISC), as these structural features resemble endogenous Dicer cleavage products. Additionally, the requirement for a double-stranded RNA duplex is critical for RISC assembly, as strand selection and guide strand incorporation depend on duplex structure and thermodynamic asymmetry (Sioud, 2025).

#### **21–23 nucleotides**

#### **2-nt 3' overhangs (usually “UU” or “TT”)**

#### **Double-stranded RNA duplex required for RISC loading**

### Strand Selection and Thermodynamic Asymmetry

Efficient siRNA function depends on strand selection driven by thermodynamic asymmetry within the duplex. Specifically, the 5' end of the antisense (guide) strand is designed to have lower thermodynamic stability, while the 5' end of the sense (passenger) strand is relatively more stable, creating an asymmetry that directs RISC to preferentially incorporate the correct guide strand (Schwarz et al., 2003; Sioud, 2025). This asymmetry is further reinforced by nucleotide composition, where an A/U-rich region at the 5' end of the antisense strand reduces stability, and a G/C-rich region at the 5' end of the sense strand increases stability (Ui-Tei et al., 2004). Together, these features promote accurate strand selection, ensuring efficient guide strand loading into RISC and enhancing overall gene silencing specificity and efficacy.

#### **Lower stability at the 5' end of antisense (guide) strand**

**Higher stability at the 5' end of sense (passenger) strand**  
**This promotes correct guide strand loading into RISC**

**A/U-rich at antisense 5' end**  
**G/C-rich at sense 5' end**

## GC Content

Optimal siRNA efficiency is strongly influenced by GC content, with a moderate range of approximately 30–50% providing the best balance between stability and functionality. Excessively high GC content increases duplex stability, which can hinder unwinding by RISC and reduce silencing efficiency, whereas very low GC content weakens target mRNA binding and compromises activity (Reynolds et al., 2004). Consequently, functional siRNAs typically exhibit moderate GC levels combined with relatively lower internal stability, facilitating efficient strand separation and target recognition within the RNAi pathway (Sioud, 2025).

**Optimal 30–50% GC**  
**Too high GC leads to overly stable duplex and poor RISC unwinding**  
**Too low GC results in weak target binding**  
**Functional siRNAs tend to have moderate GC and lower internal stability**

## Nucleotide Preferences

In the antisense (guide) strand, the presence of A/U at position 1 (the 5' end) reduces thermodynamic stability and promotes its incorporation into RISC, while a corresponding G/C-rich 5' end in the sense (passenger) strand increases stability and supports proper strand discrimination (Ui-Tei et al., 2004). Additionally, the seed region (positions 2–8), which is critical for target mRNA recognition, tends to be more effective when A/U-rich, improving binding specificity and silencing efficiency (Reynolds et al., 2004). Conversely, long stretches of G/C bases or repetitive sequences are generally avoided, as they can increase internal stability and reduce siRNA functionality (Reynolds et al., 2004).

**Antisense strand:**  
**A/U at position 1 (5' end)**

**Sense strand:**  
**G/C at 5' end**

## Seed Region

The seed region (positions 2–8) is crucial for target recognition and should ideally be A/U-rich while avoiding long GC stretches or repetitive sequences. Because this region largely determines binding specificity, it is also the main source of off-target effects: even partial complementarity can unintentionally silence other genes in a miRNA-like manner. To minimize this risk, sequences should be designed to avoid matching the 3' UTRs of unrelated genes.

**Seed (positions 2–8):**

**Prefer A/U**

**Avoid long GC stretches**

**Main source of off-target effects**

**Chemical modifications (e.g. 2'-O-methyl) can reduce off-target effects (Lennox and Behlke, 2016)**

## Target Site Considerations

Target sites should be accessible regions with minimal secondary structure, while also avoiding areas like the start codon and highly structured regions can hinder binding. Additionally, sequence design should exclude problematic motifs such as long repeats, palindromes, internal hairpins, and immune-stimulatory elements (e.g., GU-rich sequences) to ensure stability and reduce unwanted cellular responses.

## Gapmer (RNase-H) ASOs

Gapmer antisense oligonucleotides are single-stranded sequences typically 16–20 nucleotides long, designed with a central DNA “gap” (8–10 nt) flanked by chemically modified wings that enhance stability and binding (Bennett and Swayze, 2010; Crooke, 2017; Lennox and Behlke, 2016). The DNA gap is essential for forming an RNA–DNA hybrid, which recruits RNase H to cleave the target RNA. Chemical modifications such as LNA, 2'-O-methyl (2'-OMe), or 2'-O-methoxyethyl (2'-MOE) in the wings and a predominantly phosphorothioate (PS) backbone improve nuclease resistance, cellular uptake, and protein binding, while limited incorporation of phosphodiester linkages can fine-tune these properties without compromising activity (Kurreck, 2003; Bennett and Swayze, 2010; Geary et al., 2015). Effective gapmer design maintains 40–60% GC content and an appropriate melting temperature (55–65°C), targets accessible RNA regions, and avoids highly structured, repetitive, or protein-bound sites (Watts and Corey, 2012; Lennox and Behlke, 2016).

**GC content: 40–60%**

**Melting temperature: 55–65°C**

**Avoid:**

**Highly structured regions**

**Repetitive sequences**

**Protein-bound sites**

**Off-target & Toxicity Considerations**

**Avoid sequence motifs:**

**CpG motifs**

**Long G runs**

**Palindromes & Self-complementarity**

To minimize off-target effects and toxicity, sequences should avoid motifs such as CpG, long G runs, palindromes, and self-complementarity, be screened for transcriptome-wide specificity, and be tested as multiple candidates (Krieg, 2006; Crooke, 2017; Bennett and Swayze, 2010).

Gapmers function in both the nucleus and cytoplasm, leveraging RNase H's natural role in degrading RNA–DNA hybrids to achieve targeted gene silencing (Crooke, 2017; Wahba et al., 2011)..

## Steric-Blocking single-stranded ASOs

Steric-blocking antisense oligonucleotides are single-stranded sequences typically 15–25 nucleotides long with fully modified backbones, as they do not rely on RNase H activity (Bennett and Swayze, 2010; Lennox and Behlke, 2016). Instead, they function by physically blocking molecular interactions such as ribosome binding, spliceosome assembly, or protein–RNA interactions, enabling translation inhibition or splice modulation (Watts and Corey, 2012). These ASOs use fully modified chemistries such as 2'-OMe, 2'-MOE, LNA, or PMOs with phosphorothioate or neutral backbones to enhance stability, binding affinity, and cellular uptake (Kurreck, 2003; Summerton, 2007; Geary et al., 2015). Effective design requires moderate GC content (40–60%) and strong binding affinity, targeting accessible RNA regions such as start codons or splice regulatory elements while avoiding structured or protein-bound regions (Watts and Corey, 2012; Bennett and Swayze, 2010). To reduce toxicity and immune activation, sequences should avoid motifs like CpG, long G runs, and self-complementarity, and carefully balance high-affinity modifications (Krieg, 2006; Lennox and Behlke, 2016). These ASOs can function in both the nucleus and cytoplasm, with splice-switching applications primarily occurring in the nucleus.

### Compare Splicing Technology: PMO vs 2'-MOE

|                                    | PMO (MORPHOLINO; E.G., ETEPLIRSEN, GOLODIRSEN, VILTOLARSEN) | 2'-MOE PS ASO (E.G., NUSINERSEN)       |
|------------------------------------|-------------------------------------------------------------|----------------------------------------|
| Backbone                           | Phosphorodiamidate (neutral)                                | Phosphorothioate (negatively charged)  |
| Sugar/Scaffold                     | Morpholine ring (no ribose)                                 | Ribose with 2'-MOE modification        |
| Charge                             | Neutral                                                     | Negative                               |
| Protein binding                    | Low                                                         | High                                   |
| Nuclease resistance                | Very high                                                   | High                                   |
| Binding affinity (T <sub>m</sub> ) | High                                                        | High–very high                         |
| Cellular uptake                    | Poor (often needs high dose or conjugates)                  | Better (PS-driven uptake)              |
| Distribution                       | Limited without conjugates                                  | Broad tissue distribution              |
| Toxicity profile                   | Generally low                                               | Moderate (PS-related effects possible) |
| Mechanism                          | Steric blocking (splice modulation)                         | Steric blocking (splice modulation)    |

**Design implications:** - PMOs prioritize **safety and stability**, with tradeoffs in delivery efficiency. MOE-PS ASOs balance **delivery, potency, and pharmacokinetics**, with increased biological interactions.

## Aptamers

SELEX (Systematic Evolution of Ligands by EXponential enrichment) is an in vitro screening and evolutionary process used to isolate nucleic acid aptamers—short single-stranded DNA or RNA molecules—that bind specifically and tightly to a target molecule. The process begins with a very large library containing trillions of random nucleotide sequences, each capable of folding into different three-dimensional structures. This library is incubated with a target such as a protein, small molecule, cell, or virus, allowing some sequences to bind while nonbinding sequences remain free in solution. The bound aptamers are then separated from the unbound molecules using techniques such as affinity chromatography, magnetic beads, or filtration. After separation, the bound sequences are recovered and amplified, typically by PCR for DNA aptamers or by reverse transcription and transcription for RNA aptamers. The enriched pool is subjected to repeated rounds of binding, separation, and amplification, with increasingly stringent conditions to favor sequences with the highest affinity and specificity. Additional

counter-selection steps may be included to eliminate sequences that bind undesired targets. After several rounds, the enriched aptamer population is sequenced and characterized to identify candidates with strong and selective binding properties. Overall, SELEX functions as a form of directed molecular evolution, progressively enriching high-performance aptamers from an initially random sequence population.

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| Approach                                                                                        | Key Developers                                                                  | Technology                                                                                                            |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Nonselective ASO (total HTT lowering)                                                           | Roche + Ionis Pharmaceuticals                                                   | 2'-MOE gapmer antisense oligonucleotides with mixed PS/PO backbone (stereorandom)                                     |
| Allele-selective ASO (SNP-targeted)                                                             | Wave Life Sciences                                                              | 2'-O-Me gapmer antisense oligonucleotides with stereopure synthesis                                                   |
| Next-generation allele-selective ASO                                                            | Wave Life Sciences                                                              | PS/PO/PN backbone ASOs (phosphoramidate-modified for improved distribution and potency)                               |
| RNA interference (RNAi)                                                                         | Academic / preclinical groups                                                   | siRNA or shRNA delivered via viral vectors                                                                            |
| General ASO platform                                                                            | Multiple groups                                                                 | Chemically modified single-stranded DNA oligonucleotides targeting RNA                                                |
| RNase H-mediated degradation of both mutant and wild-type HTT mRNA after ASO binding            | Broad applicability across all patients; robust and dose-dependent HTT lowering | Loss of wild-type HTT function may cause toxicity; clinical trials halted due to safety concerns and lack of efficacy |
| Selective binding to SNP-linked mutant HTT mRNA → RNase H degradation of mutant transcript only | Preserves wild-type HTT; reduces risk of on-target toxicity                     | Limited to patients with specific SNPs; insufficient target engagement in trials; programs halted                     |
| RNase H-mediated degradation with improved pharmacokinetics and distribution                    | Enhanced stability, potency, and CNS distribution (preclinical improvements)    | Still early-stage; clinical efficacy and safety not yet established                                                   |
| RISC-mediated cleavage of HTT mRNA via siRNA/shRNA pathways                                     | Strong and sustained gene silencing in preclinical models                       | Delivery challenges; less controllable dosing; primarily preclinical                                                  |



Hybridization to RNA (mRNA/pre-mRNA) enabling degradation or modulation of RNA processing

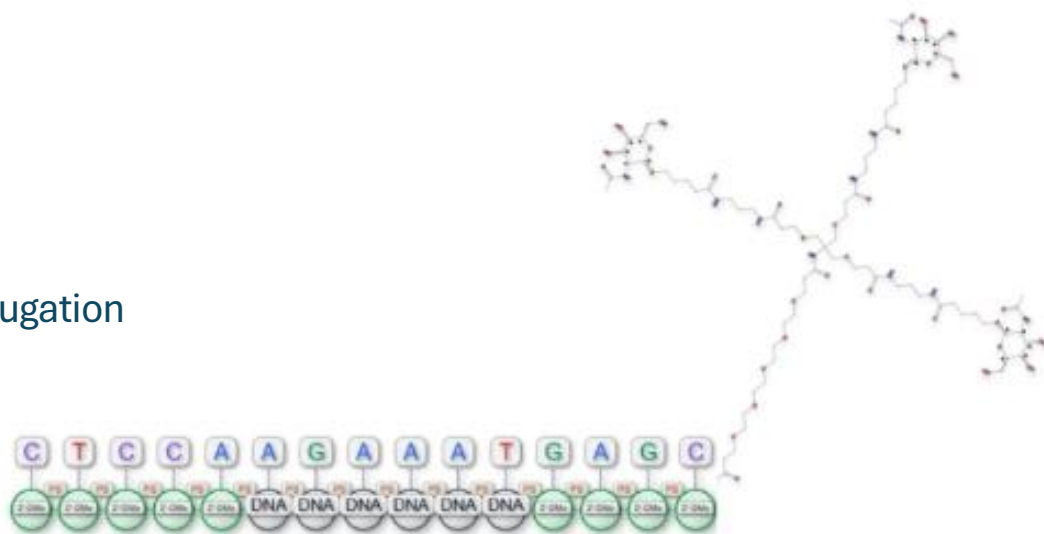
Flexible design; broad CNS distribution; no viral vector required

Requires repeated dosing; potential off-target effects and immune activation

## Conjugates and Delivery Strategies

| FEATURE                | SMALL MOLECULES                                     | GALNAC–SIRNA CONJUGATES                                      | THERAPEUTIC ANTIBODIES                                    |
|------------------------|-----------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------|
| SIZE                   | <1000 Da                                            | ~16,000 Da                                                   | >150,000 Da                                               |
| ROUTE OF DELIVERY      | Oral IV, IM, SC, inhalation                         | Subcutaneous (SC)                                            | Mostly IV and SC                                          |
| PLASMA PROTEIN BINDING | Variable (high for lipophilic, low for hydrophilic) | High (~80–90%)                                               | Low                                                       |
| TISSUE                 | Often large volume of distribution                  | Extensive liver distribution                                 | Mostly confined to circulation & extracellular fluid      |
| METABOLISM             | CYP450 enzymes (drug-metabolizing enzymes)          | Exonucleases & endonucleases (no CYP involvement)            | Proteolytic catabolism (receptor & non-specific)          |
| RENAL CLEARANCE        | Often important pathway                             | Minor (<25% for intact siRNA)                                | None for intact antibodies                                |
| PLASMA HALF-LIFE       | Short (hours)                                       | Short (<10 hours)                                            | Long (weeks)                                              |
| PK LINEARITY           | Usually linear (can be nonlinear)                   | Linear                                                       | Often nonlinear at low doses (TMDD), linear at high doses |
| PK/PD RELATIONSHIP     | PK → PD (PD depends on PK)                          | <b>PK independent of PD</b> ; PD driven by RISC-loaded siRNA | PK often depends on PD (TMDD)                             |

## GalNAc Conjugation



GalNAc–siRNA or GalNAc single stranded ASOs differ fundamentally from both small molecules and therapeutic antibodies in their size, disposition, and PK/PD behavior. Unlike traditional drugs, they exhibit linear PK with a short plasma half-life but a disconnect between plasma exposure and pharmacodynamic effect, as activity is driven by intracellular RISC-loaded siRNA rather than circulating concentrations, and they have low drug–drug interaction potential due to non–CYP-mediated metabolism.

The development of GalNAc-based liver targeting therapeutics stems from the discovery of the asialoglycoprotein receptor (ASGPR) in the 1960s by Ashwell and Morell, who showed that hepatocytes selectively clear desialylated glycoproteins via recognition of exposed terminal galactose and N-acetylgalactosamine (GalNAc) (Ashwell and Morell, 1968; Morell et al., 1966). Subsequent studies established that receptor binding is enhanced by multivalency, laying the foundation for synthetic ligand design. By optimizing ligand valency, linker design, and synthesis, Dr. Muthiah Manoharan (Alnylam) and colleagues developed the first triantennary GalNAc ligands that exist in the clinic today (Nair et al., 2014; Egli and Manoharan, 2023).



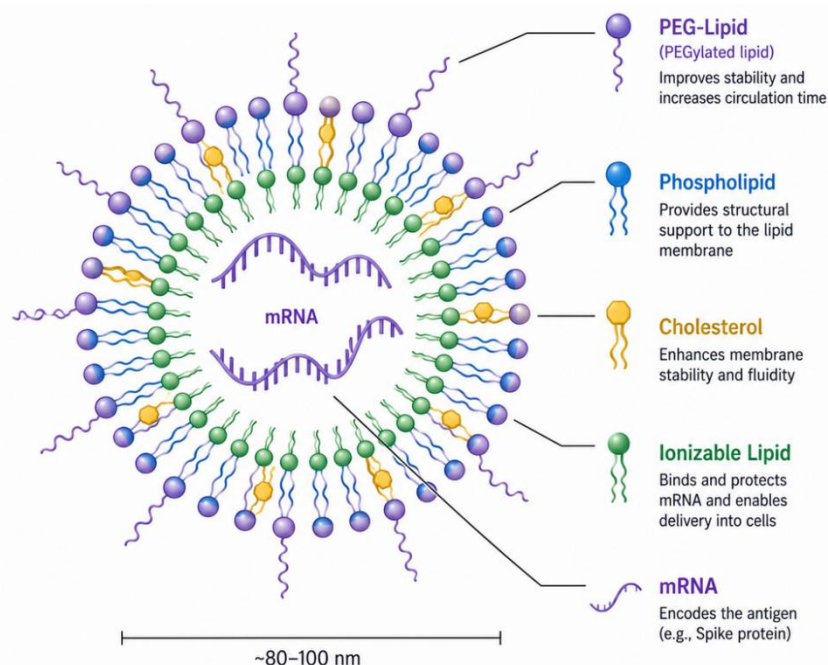
GalNA

Dicerna's GalXC platform is an RNA interference (RNAi) chemistry that uses double-stranded RNA molecules conjugated to ligands (classically GalNAc) to selectively enter cells, especially hepatocytes. The drug was mechanistically validated in preclinical work and had favorable phase 1 results demonstrated that the GalXC™ RNAi therapeutic achieved dose-dependent reductions in circulating alpha-1 antitrypsin (AAT) levels following a single subcutaneous dose. The drug was generally well tolerated with an acceptable safety profile, with

most adverse events being mild and no clinically significant changes observed in laboratory parameters or lung function. However, compound failed in the Phase II ESTRELLA study; this was terminated early after enrolling only a small fraction of planned patients, and the overall development program was discontinued by Novo Nordisk.

| DRUG                              | COMPANY              | STAGE          | TARGET              | INDICATION                   |
|-----------------------------------|----------------------|----------------|---------------------|------------------------------|
| <b>GIVOSIRAN</b>                  | Alnylam              | Approved       | ALAS1               | Acute hepatic porphyria      |
| <b>LUMASIRAN</b>                  | Alnylam              | Approved       | Glycolate oxidase   | Primary hyperoxaluria type 1 |
| <b>INCLISIRAN</b>                 | Alnylam / Novartis   | Approved       | PCSK9               | Hypercholesterolemia         |
| <b>VUTRISIRAN</b>                 | Alnylam              | Approved       | Transthyretin (TTR) | ATTR amyloidosis             |
| <b>NEDOSIRAN</b>                  | Novo Nordisk         | Approved       | LDHA                | Primary hyperoxaluria type 1 |
| <b>FAZIRSIRAN</b>                 | Arrowhead            | Phase 3        | AAT (SERPINA1)      | AAT deficiency liver disease |
| <b>FITUSIRAN</b>                  | Alnylam / Sanofi     | Phase 3        | Antithrombin        | Hemophilia                   |
| <b>OLPASIRAN</b>                  | Amgen                | Phase 3        | Lp(a)               | Cardiovascular disease       |
| <b>CEMDISIRAN</b>                 | Alnylam              | Phase 2        | Complement C5       | Complement-mediated diseases |
| <b>AB-729</b>                     | Arbutus              | Phase 2        | HBV proteins        | Hepatitis B                  |
| <b>JNJ-3989</b>                   | Arrowhead / J&J      | Phase 2        | HBV proteins        | Hepatitis B                  |
| <b>BELCESIRAN</b>                 | Alnylam / Dicerna    | Phase 2        | AAT                 | AAT liver disease            |
| <b>ZILEBESIRAN</b>                | Alnylam              | Phase 2/3      | Angiotensinogen     | Hypertension                 |
| <b>SLN-124</b>                    | Silence Therapeutics | Phase 1/2      | TMPRSS6             | Thalassemia                  |
| <b>ALN-HSD<br/>(ALN-HSD17B13)</b> | Alnylam              | Phase 1/2      | HSD17B13            | NASH                         |
| <b>ALN-HBV02</b>                  | Alnylam              | Phase 1/2      | HBV proteins        | Chronic HBV                  |
| <b>ZERLASIRAN</b>                 | —                    | Early clinical | Lp(a)               | Cardiovascular disease       |
| <b>SOLBINSIRAN</b>                | Eli Lilly            | Early clinical | ANGPTL3             | Dyslipidemia / CV disease    |

## Lipid and Liposomal Systems



Lipid nanoparticles (LNPs) are the leading non-viral platform for delivery of nucleic acid therapeutics, especially mRNA and siRNA. Their clinical rise was driven by the success of COVID-19 mRNA vaccines and by liver-directed therapies, where they have shown scalable manufacturing, reproducible formulation, and strong in vivo activity (Belaidi et al., 2025). Typically LNPs

consist of four components: an ionizable lipid, a helper phospholipid, cholesterol, and a PEG-lipid. The ionizable lipid is the key functional component because it complexes RNA during formulation at low pH, remains near-neutral in blood, and then becomes protonated in endosomes to promote endosomal escape after cellular uptake (Belaidi et al., 2025). Endosomal escape remains a major bottleneck for efficacy. (Jozić et al., 2026; Belaidi et al., 2025).

Route of administration is one of the main determinants of both efficacy and safety for LNP medicines. Intravenous dosing is the dominant route for systemic delivery, but it strongly biases biodistribution toward the liver because circulating LNPs adsorb apolipoproteins, especially ApoE, and are then taken up efficiently by hepatocytes and other liver-associated cells (Belaidi et al., 2025). This makes IV delivery attractive for hepatic disorders, but problematic for extrahepatic targets such as the brain. The same review notes that efforts to redirect LNPs away from the liver include modifying PEG desorption kinetics, adding selective organ targeting lipids, changing ionizable lipid chemistry, and decorating LNPs with receptor-binding ligands or peptides for tissue-specific entry (Belaidi et al., 2025). In the CNS setting, route becomes even more consequential because IV dosing must overcome the blood-brain barrier, whereas local routes such as intrathecal, intranasal, intracerebral, convection-enhanced delivery, or focused ultrasound-assisted systemic dosing can partly bypass or transiently open that barrier, each with different exposure profiles and safety tradeoffs (Belaidi et al., 2025).

Intramuscular administration, used for vaccines, generally prioritizes local antigen expression and immune activation rather than organ targeting. In contrast, IV administration produces systemic exposure and therefore greater relevance for infusion-related reactions, complement activation, hepatic sequestration, and off-target biodistribution (Belaidi et al., 2025). Intrathecal and intracerebral routes reduce dependence on vascular transport barriers and can improve CNS exposure, but they are invasive and can introduce procedure-related risks that differ fundamentally from those of IV administration (Belaidi et al., 2025). Focused ultrasound with microbubbles offers a less invasive adjunct to IV dosing by creating transient local BBB opening, but this route may behave differently in diseased or aged brain and could be a context sensitive safety risk (Belaidi et al., 2025). So, route of administration is not just a delivery variable; it shapes the toxicology package that matters for each product.

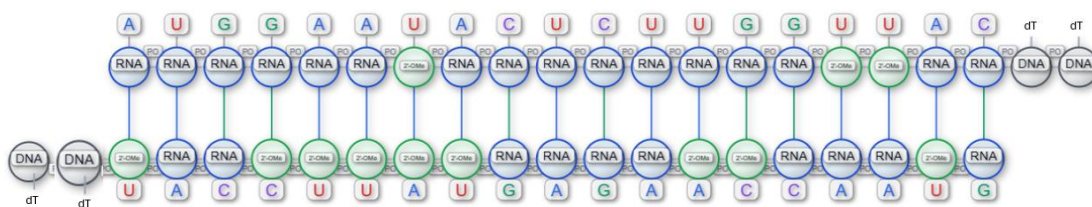
Acute toxicology of LNPs is now reasonably well characterized. The main short-term adverse effects reported across vaccine and therapeutic settings are transient flu-like symptoms, cytokine elevations, and occasional infusion-related reactions, with the severity depending on dose, infusion rate, lipid composition, and patient context (Belaidi et al., 2025). Complement activation-related pseudoallergy has been reported particularly in animal infusion models at high infusion rates, highlighting that administration kinetics can matter alongside composition (Belaidi et al., 2025). PEG-lipids, while useful for stability and circulation time, are also implicated in rare anaphylactoid or hypersensitivity-like responses and remain under scrutiny as a contributor to acute immune reactivity (Belaidi et al., 2025).

Long-term toxicology is more complicated and still less completely defined than acute reactogenicity. A central issue is repeat dosing. Chronic administration can drive anti-PEG antibody formation, accelerated blood clearance, altered biodistribution, and reduced efficacy over time, particularly when PEG-lipids shed slowly and remain associated with the particle long enough to stimulate stronger humoral responses (Belaidi et al., 2025). The review argues that this is especially relevant outside vaccines, where patients may need higher and more frequent dosing for chronic disease treatment (Belaidi et al., 2025). Long-term safety also depends on how rapidly the ionizable lipid is cleared. Persistent or slowly metabolized lipids can accumulate in the liver and other tissues, increasing the risk of chronic on-target and off-target toxicity, whereas biodegradable lipids with ester, carbonate, or disulfide linkers are intended to reduce retention and improve tolerability during repeated dosing (Belaidi et al., 2025). This is one reason the field has increasingly moved away from permanently cationic lipids toward ionizable and biodegradable chemistries.

Route of administration interacts strongly with long-term toxicology. Repeated IV dosing disproportionately stresses organs involved in nanoparticle clearance, especially the liver and cells of the mononuclear phagocyte system, so hepatic retention, Kupffer-cell uptake, and systemic immunogenicity become major concerns (Belaidi et al., 2025). Repeated intrathecal or intracerebral administration may reduce systemic burden but raises separate concerns around local inflammation, procedural burden, and acceptability for chronic treatment. For CNS

therapies, the review further notes that aging and neurodegeneration alter BBB integrity, endothelial trafficking, and neuroimmune tone, meaning that long-term safety data from young or healthy systems may overestimate tolerability and delivery performance in older patients (Belaidi et al., 2025). In other words, chronic toxicology cannot be generalized across routes or patient populations; it is route-, tissue-, and disease-context dependent.

Overall, LNPs are a highly adaptable and clinically validated delivery platform, but their risk-benefit profile is tightly linked to administration route and dosing schedule. Acute toxicology is now often manageable and increasingly predictable, especially with ionizable lipids and careful infusion control. Long-term toxicology remains the bigger developmental challenge, centered on repeat-dose immunogenicity, PEG-associated responses, tissue accumulation, and route-specific exposure burdens. The most promising path forward is the combination of lower-dose, higher-efficiency lipids, biodegradable chemistries, better control of biodistribution, and route-specific formulation strategies tailored to the target tissue and intended dosing interval (Jozić et al., 2026; Belaidi et al., 2025) .



**Patisiran** (Onpattro), an LNP-formulated RNAi therapeutic, established that chronic systemic delivery of siRNA is clinically feasible. However, its safety and administration burden reflect the inherent liabilities of lipid nanoparticles. In the APOLLO trial, patients treated with patisiran experienced higher rates of infusion-related reactions (19% vs 9% with placebo) and peripheral edema (30% vs 22%), along with systemic adverse events such as fatigue, diarrhea, and dizziness. These effects are consistent with known LNP-associated mechanisms, including complement activation, innate immune stimulation, and vascular responses. As a result, patisiran requires intravenous infusion and routine premedication (e.g., corticosteroids, antihistamines) to mitigate hypersensitivity and infusion reactions.

In contrast, vutrisiran is a next-generation RNAi therapeutic targeting the same gene (TTR) but uses a GalNAc conjugate rather than LNPs. This enables direct hepatocyte uptake via the asialoglycoprotein receptor, avoiding many of the systemic immune interactions associated with nanoparticle carriers. Consequently, vutrisiran can be administered as a subcutaneous injection once every three months, substantially reducing treatment burden.

Vutrisiran demonstrates a distinct and generally improved tolerability profile. In the HELIOS-A study, the most common adverse events in patients with hATTR-PN were pain in extremity (15%), arthralgia (11%), dyspnea (7%), and decreased vitamin A levels (7%), with no signal of infusion-related reactions. Similarly, in ATTR-CM, no new safety concerns were identified. Notably, unlike patisiran, vutrisiran does not require premedication, underscoring the advantage of eliminating LNP-associated immune activation and enabling a more convenient, repeat dosing paradigm.

ONPATTRO requires weight-based IV infusion every 3 weeks, along with premedication (corticosteroids, acetaminophen, and antihistamines) and monitoring due to infusion-related reactions and broader systemic exposure. In contrast, GalNAc-conjugated siRNAs achieve targeted hepatocyte uptake via the asialoglycoprotein receptor, enabling subcutaneous administration, fixed dosing, and much less frequent schedules (monthly to every 3–6 months or longer) without routine pre-treatment. Overall, GalNAc conjugation has enabled more precise liver targeting and more convenient, less intrusive dosing compared with earlier non-targeted LNP approaches like ONPATTRO.

| COMPOUND                | TYPE  | DOSE (MG, NORMALIZED) | FREQUENCY                           | ROUTE |
|-------------------------|-------|-----------------------|-------------------------------------|-------|
| GIVOSIRAN               | siRNA | 175 mg                | Once monthly                        | SC    |
| LUMASIRAN               | siRNA | 210–420 mg            | Monthly ×3, then q3 months          | SC    |
| INCLISIRAN              | siRNA | 800 mg                | Day 0, 3 months, then q6 months     | SC    |
| VUTRISIRAN              | siRNA | 300 mg                | Every 3 months                      | SC    |
| FITUSIRAN               | siRNA | 80 mg                 | Not specified                       | SC    |
| OLPASIRAN               | siRNA | 225 mg                | Not specified                       | SC    |
| CEMDISIRAN              | siRNA | 900 mg                | Not specified                       | SC    |
| ZILEBESIRAN             | siRNA | 10–800 mg             | Single dose; effect up to ~24 weeks | SC    |
| FAZIRSIRAN              | siRNA | 100–200 mg            | Day 1, Week 4, then q12W            | SC    |
| NEDOSIRAN               | siRNA | 160 mg                | Once monthly                        | SC    |
| AB-729 (IMDUSIRAN)      | siRNA | 60–90 mg              | q4W - q8W                           | SC    |
| JNJ-3989 / BEPIROVIRSEN | ASO   | 150–300 mg            | Weekly (12–24 weeks)                | SC    |

|                                 |       |            |                            |    |
|---------------------------------|-------|------------|----------------------------|----|
| <b>BELCESIRAN</b>               | siRNA | 200 mg     | q4W                        | SC |
| <b>ALN-HBV02<br/>(VIR-2218)</b> | siRNA | 20–200 mg  | Two doses, 4 weeks apart   | SC |
| <b>ALN-HSD</b>                  | siRNA | 25–400 mg  | q12W × 2 doses             | SC |
| <b>SLN-124</b>                  | siRNA | 70–315 mg  | Single dose (~6–8 week PD) | SC |
| <b>DCR-HBVS<br/>(RG6346)</b>    | siRNA | 105–420 mg | q4W × 4 doses              | SC |
| <b>DCR-A1AT</b>                 | siRNA | 420 mg     | Monthly (clinical study)   | SC |

|                                                                                               | <b>What Was Measured</b>                                     | <b>Reported Clearance / Persistence</b>                                                                                                                                             |
|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| “Blood Distribution of SARS-CoV-2 Lipid Nanoparticle mRNA Vaccine in Humans” (ACS Nano, 2024) | Intact vaccine mRNA + ionizable lipid in blood               | Detectable in some subjects up to <b>14–15 days</b> post-vaccination ( <a href="#">ACS PUBLICATIONS</a> )                                                                           |
| “Impact of ionizable lipid type on pharmacokinetics...” (2025)                                | Plasma pharmacokinetics of SM-102, ALC-0315 and other lipids | Different ionizable lipids showed markedly different circulation and exposure times; ALC-0315 produced more prolonged lipid exposure than SM-102 ( <a href="#">SCIENCE DIRECT</a> ) |
| “Engineering ionizable lipids for rapid biodegradation...” (2024 preprint)                    | Tissue clearance of biodegradable ionizable lipids           | Faster-degrading lipids correlated with reduced inflammation and faster tissue clearance ( <a href="#">BIO RXIV</a> )                                                               |
| “Silicon Ether Ionizable Lipids Enable Potent mRNA LNPs...”                                   | Biodegradable ionizable lipid chemistry                      | Describes “highly efficient lipid clearance profiles” through controlled hydrolysis ( <a href="#">PMC</a> )                                                                         |
| “Recent Advances in LNP-Mediated Delivery of mRNA Vaccines” (2023 review)                     | General LNP biodegradation                                   | Notes that modern ionizable lipids are specifically engineered for biodegradability and clearance after delivery ( <a href="#">MDPI</a> )                                           |
| “Advances in the study of LNPs for mRNA delivery...” (2024 review)                            | General LNP metabolism                                       | States that biodegradable lipids improve biocompatibility and reduce long-term persistence ( <a href="#">SPRINGER</a> )                                                             |



he primary challenge facing lipid nanoparticle (LNP) therapeutics is no longer demonstrating biological efficacy, which was validated during the COVID-19 pandemic, but rather achieving reproducible large-scale manufacturing while maintaining stringent critical quality attributes (CQAs). These CQAs include particle size, RNA encapsulation efficiency, sterility, stability, batch consistency, and regulatory compliance. Transitioning from laboratory-scale production to preclinical and commercial Good Manufacturing Practice (GMP) manufacturing presents substantial engineering difficulties because LNP self-assembly is highly sensitive to mixing conditions, solvent exchange kinetics, shear stress, salt concentration, temperature, and downstream purification processes.

The paper identifies manufacturing scale-up as the central industrial bottleneck in LNP production. LNPs self-assemble within milliseconds, and minor variations in flow rates, mixing geometries, or solvent dilution conditions can significantly alter particle characteristics. Commercial-scale manufacturing therefore requires the preservation of identical CQAs across substantially larger batch volumes. Major technical challenges include maintaining particle uniformity, preventing aggregation, preserving RNA encapsulation efficiency, minimizing batch-to-batch variability, and reproducing laboratory-scale performance at industrial scale. The authors note that “the most complex hurdles in the scale-up of LNP production is the scalability of the mixing technology, controlled sterility and regulatory demands.” Commercial manufacturing platforms increasingly rely on technologies such as parallel impinging jet mixers (IJMs), continuous-flow reactors, and automated GMP systems to address these issues. These systems require advanced fluid dynamic control, sterile closed-system architectures, high-precision pumps and sensors, automation, in-line analytical monitoring, and validated process control infrastructure. During COVID-19 vaccine production, manufacturing operations utilized multiple IJM units operating in parallel with industrial flow rates reaching approximately 18,000 mL/min per stream.

The economic burden associated with LNP manufacturing infrastructure is substantial. Capital expenditures include GMP facilities, cleanrooms, continuous manufacturing systems, automated fill-finish lines, analytical instrumentation, and extensive process validation systems. Operational costs arise from sterile single-use consumables, fluid handling systems, high-value ionizable lipids, RNA payload losses, and failed production batches. Consequently, manufacturing scalability remains one of the most significant barriers to widespread commercialization of RNA-LNP therapeutics.

Thermal instability and cold-chain dependence represent another major limitation of current RNA-LNP therapeutics. Existing mRNA vaccines often require storage temperatures between  $-60^{\circ}\text{C}$  and  $-90^{\circ}\text{C}$ , creating substantial logistical and economic challenges. Instability results primarily from mRNA hydrolysis, lipid oxidation, RNA-lipid chemical interactions, and

nanoparticle aggregation. The paper emphasizes that even modest increases in storage temperature can rapidly reduce therapeutic efficacy. This dependence on ultra-cold storage infrastructure contributes to high supply-chain costs, distribution bottlenecks, and limited accessibility in low-resource settings. Cold-chain logistics require ultra-low-temperature freezers, specialized thermal transport systems, continuous temperature monitoring, dry ice distribution networks, and validated storage protocols. Accordingly, the paper identifies formulation stabilization and lyophilization technologies as critical future opportunities for reducing costs and improving global accessibility.

Tangential flow filtration (TFF) is highlighted as one of the most technically challenging downstream purification processes in LNP manufacturing. TFF is widely used to remove ethanol, exchange buffers, concentrate nanoparticles, and prepare sterile formulations. However, because LNPs are mechanically fragile nanoscale structures, TFF introduces several engineering challenges, including shear-induced particle damage, RNA leakage, membrane fouling, nanoparticle aggregation, flux decline, and inconsistent product recovery. The paper states that “downstream processing with TFF remains challenging.” Common failure modes include membrane clogging, destabilization of nanoparticles, reduced encapsulation efficiency, and inconsistent filtration performance. TFF systems also contribute significantly to manufacturing costs due to the need for specialized membranes, automated pumping systems, sterile fluid paths, pressure monitoring systems, and process optimization strategies. Importantly, purification failures can compromise entire manufacturing batches.

Regulatory compliance and GMP validation are described as major time- and resource-intensive components of commercial LNP development. Manufacturers must validate all process parameters, raw materials, analytical assays, sterility controls, and comparability requirements across manufacturing scales and sites. Required characterization assays include measurements of particle size, zeta potential, RNA integrity, lipid identity, endotoxin content, potency, morphology, residual solvents, and sterility. Commercial programs further require validated analytical methods, comprehensive GMP documentation, stability studies, batch records, and comparability assessments. These requirements necessitate extensive quality assurance and quality control infrastructure, analytical laboratories, documentation systems, validation campaigns, release testing procedures, and long-term stability programs. The paper notes that regulatory programs for commercial RNA-LNP therapeutics can cost tens to hundreds of millions of dollars.

The discovery and optimization of ionizable lipids constitute a major strategic and competitive challenge within the LNP field. Ionizable lipids serve as the primary functional component responsible for intracellular delivery efficiency and endosomal escape. Key challenges include balancing delivery efficacy with toxicity, avoiding intellectual property restrictions, ensuring manufacturing reproducibility, and obtaining regulatory approval for novel lipid chemistries. The paper notes that companies frequently screen hundreds to thousands of lipid candidates during development programs. This process is resource-intensive due to the complexity of lipid synthesis, high-throughput screening requirements, uncertain regulatory pathways, and extensive patent landscapes surrounding proprietary lipid structures.

Maintaining high RNA encapsulation efficiency throughout manufacturing and downstream processing is another critical quality challenge. Encapsulation efficiency directly influences therapeutic potency, stability, and dosing accuracy, with commercial targets generally exceeding 90–95%. However, encapsulation efficiency can be compromised by variations in mixing conditions, salt concentrations, TFF-induced shear stress, ethanol exposure, and nanoparticle aggregation. Loss of encapsulation leads to RNA degradation, reduced efficacy, and failed production batches. These challenges become increasingly severe during manufacturing scale-up due to the greater difficulty of maintaining controlled self-assembly conditions across large production volumes.

A central engineering insight emphasized throughout the paper is that LNP manufacturing is fundamentally a fluid dynamics and process control problem. Successful production depends on millisecond-scale mixing, controlled solvent exchange, diffusion kinetics, shear management, and precise regulation of nanoparticle self-assembly conditions. Minor deviations in these parameters can produce substantial quality defects, underscoring the importance of advanced process engineering and real-time monitoring systems.

To address these challenges, the industry is investing heavily in several emerging technologies. Artificial intelligence and machine learning are being explored for applications including lipid discovery, predictive process modeling, digital twins, and real-time manufacturing optimization. Advanced mixing technologies such as staggered herringbone microfluidics, impinging jet mixers, and parallelized microfluidic arrays are being developed to support scalable continuous manufacturing. In addition, lyophilization is presented as a potentially transformative technology capable of enabling room-temperature-stable RNA-LNP formulations and reducing dependence on ultra-cold supply chains.

Overall, the paper characterizes the LNP industry as transitioning from biological proof-of-concept toward industrial process engineering. The dominant remaining challenges are no longer primarily biological but instead involve scalable manufacturing, downstream purification, long-term stability, regulatory reproducibility, and cost-effective commercialization. According to the authors, the two primary economic burdens are GMP manufacturing infrastructure and cold-chain logistics, while the two most significant technical bottlenecks are maintaining CQAs during manufacturing scale-up and achieving long-term stabilization of RNA-LNP formulations.

## PEGylation

PEGylation, the covalent attachment of polyethylene glycol (PEG) to therapeutic molecules. The technology was discovered in the 1960s by Frank Davis who proposed its use to reduce the immunogenicity of protein drugs by shielding them from immune recognition and prolonging circulation time (Veronese & Pasut, 2005; Pasut & Veronese, 2012). This concept was experimentally validated in a landmark study by Abuchowski et al. (1977), which demonstrated that PEGylation of bovine liver catalase significantly decreased immunogenicity while extending plasma half-life.

PEGylation transitioned from academic concept to commercial technology (Harris & Chess, 2003) with the FDA approval of Adagen in 1990, followed by Oncaspar in 1994. (Alconcel et al., 2011; Harris & Chess, 2003). From the mid-1990s onward, PEGylation rapidly expanded across diverse drug classes and architectures, including the development of PEGylated liposomes such as Doxil (Allen & Cullis, 2013), site-specific mono-PEGylated proteins like Pegintron, and branched PEG systems exemplified by Pegasys (Veronese & Mero, 2008).

The technology further diversified to include PEGylated aptamers (Macugen™), antibody fragments (Cimzia™), peptides (Omontys), and small molecules (Movantik), reflecting its broad applicability across therapeutic modalities (Keefe et al., 2010; Fishbane et al., 2013). More recently, PEGylation has played a critical role in nanomedicine, particularly in lipid nanoparticle (LNP) systems used for nucleic acid delivery; a critical component of the Onpattro LNP and the widespread deployment of PEGylated LNP-based mRNA COVID-19 vaccines (Cullis & Hope, 2017; Hou et al., 2021).

While there are many more ligand conjugate technologies used including palmitate and antibodies most pro-drug technologies for tissue specificity is for liver with various forms of GalNAc.

## Cell penetrating peptides

This study demonstrates that a cell-penetrating peptide–conjugated phosphorodiamidate morpholino oligomer (PMO-pep) can induce efficient exon skipping and restore dystrophin expression systemically in the mdx mouse model of Duchenne muscular dystrophy (Fletcher et al., 2007). Repeated low-dose systemic administration led to widespread dystrophin production across multiple skeletal muscles, with near-normal levels observed in diaphragm and improved muscle architecture, particularly when treatment was initiated early, effectively preventing the onset of dystrophic pathology. The study also showed that enhanced cellular uptake via peptide conjugation significantly improved efficacy compared to unconjugated PMOs, while maintaining a favorable safety profile, as no significant toxicity or adverse effects were detected. However, dystrophin restoration was limited in cardiac muscle and less effective in older animals,

highlighting challenges in tissue delivery and the importance of early intervention (Fletcher et al., 2007).

## RNA-targeting clinical review

[TODO]

Several trends define the next generation of oligonucleotide therapeutics:

- Increasing chemical diversity (beyond PS and standard 2' modifications)
- Greater use of noncanonical bases
- Integration of multiple backbone types within a single molecule
- Precision structural engineering for enzyme recruitment
- Continued dominance of GalNAc for liver delivery

| COMPOUND            | BASE                 | BACKBONE              | SUGAR            | CONJUGATION | ROUTE                  |
|---------------------|----------------------|-----------------------|------------------|-------------|------------------------|
| <b>FOMIVIRSEN</b>   | 21-mer antisense DNA | Full phosphorothioate | DNA              | None        | Intravitreal injection |
| <b>MIPOMERSEN</b>   | Gapmer ASO           | Phosphorothioate      | 2'-MOE + DNA gap | None        | Subcutaneous injection |
| <b>ETEPLIRSEN</b>   | 30-mer antisense     | PMO                   | Morpholino       | None        | Intravenous infusion   |
| <b>NUSINERSEN</b>   | 18-mer               | Phosphorothioate      | 2'-MOE           | None        | Intrathecal injection  |
| <b>INOTERSEN</b>    | Gapmer ASO           | Phosphorothioate      | 2'-MOE           | None        | Subcutaneous injection |
| <b>GOLODIRSEN</b>   | Antisense            | PMO                   | Morpholino       | None        | Intravenous infusion   |
| <b>VOLANESORSEN</b> | Gapmer ASO           | Phosphorothioate      | 2'-MOE           | None        | Subcutaneous injection |
| <b>VILTOLARSEN</b>  | Antisense            | PMO                   | Morpholino       | None        | Intravenous infusion   |
| <b>CASIMERSEN</b>   | Antisense            | PMO                   | Morpholino       | None        | Intravenous infusion   |
| <b>TOFERSEN</b>     | 20-mer gapmer        | Mixed PS/PO           | 2'-MOE           | None        | Intrathecal injection  |

|                           |             |                  |                 |        |                        |
|---------------------------|-------------|------------------|-----------------|--------|------------------------|
| <b>EPLONTERSEN</b>        | Gapmer ASO  | Mixed PS/PO      | 2'-MOE          | GalNAc | Subcutaneous injection |
| <b>PEGAPTANIB</b>         | RNA aptamer | Modified RNA     | Modified ribose | PEG    | Intravitreal injection |
| <b>AVACINCAPTAD PEGOL</b> | RNA aptamer | Modified RNA     | Modified ribose | PEG    | Intravitreal injection |
| <b>PATISIRAN</b>          | siRNA       | Phosphodiester   | 2'-OMe          | LNP    | Intravenous infusion   |
| <b>GIVOSIRAN</b>          | siRNA       | PS modified      | 2'-F/2'-OMe     | GalNAc | Subcutaneous injection |
| <b>LUMASIRAN</b>          | siRNA       | PS modified      | 2'-F/2'-OMe     | GalNAc | Subcutaneous injection |
| <b>INCLISIRAN</b>         | siRNA       | PS modified      | 2'-F/2'-OMe     | GalNAc | Subcutaneous injection |
| <b>VUTRISIRAN</b>         | siRNA       | PS enhanced      | 2'-F/2'-OMe     | GalNAc | Subcutaneous injection |
| <b>NEDOSIRAN</b>          | siRNA       | PS/PO            | 2'-F/2'-OMe     | GalNAc | Subcutaneous injection |
| <b>MIRAVIRSEN</b>         | Anti-miR    | Phosphorothioate | LNA/DNA         | None   | Subcutaneous injection |
| <b>RG-101</b>             | Anti-miR    | Modified ASO     | Not specified   | GalNAc | Subcutaneous injection |

## Institutions by Platform

This section maps the major chemistries described above to leading companies and institutions that have developed or commercialized them.

### **siRNA Platforms (2'-OMe / 2'-F + PS)**

Alnylam Pharmaceuticals – Active

Arrowhead Pharmaceuticals – Active

Dicerna – Active (integrated into Novo Nordisk)

Silence Therapeutics – Active

### **Gapmer ASOs (MOE / OMe / DNA gap + PS)**

Ionis Pharmaceuticals – Active

Biogen – Active

Roche / Genentech – Active

### **Steric-Blocking ASOs (Fully Modified)**

Sarepta Therapeutics – Active

Nippon Shinyaku – Active

Avidity Biosciences – Acquired by Novartis

Biogen / Ionis

### **RNA Editing Oligos (ADAR / EON Platforms)**

ProQR Therapeutics – Active

Wave Life Sciences – Active

Beam Therapeutics – Active

Academic institutions – Active (various universities and research institutes)

### **Alternative Chemistries and Platforms**

Eyetech / Pfizer (Pegaptanib) – Legacy (Eyetech acquired; program commercialized historically)

Phio Pharmaceuticals – Active

Wave Life Sciences (stereopure PS) – Active

## Industry Activity Status

This supplemental section provides a consolidated view of organizations referenced in this white paper and their current operational status in the oligonucleotide therapeutics field.

| Company / Institution                                                         | Primary Chemistry Focus                             | Notes                                                                                                |
|-------------------------------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Alnylam Pharmaceuticals                                                       | siRNA (2'-OMe / 2'-F, GalNAc)                       | Market leader in RNAi therapeutics                                                                   |
| Arrowhead Pharmaceuticals                                                     | siRNA (TRiM platform)                               | Expanding beyond liver delivery                                                                      |
| Dicerna (Novo Nordisk)                                                        | siRNA (GalXC)                                       | Fully integrated into Novo Nordisk                                                                   |
| Silence Therapeutics                                                          | siRNA                                               | Focus on mRNAi GOLD platform                                                                         |
| Ionis Pharmaceuticals<br>Biogen                                               | Gapmer ASO (MOE, cEt)<br>ASO (partnered programs)   | Pioneer in antisense chemistry<br>CNS-focused oligonucleotide drugs                                  |
| Roche / Genentech<br>Sarepta Therapeutics                                     | ASO / RNA therapeutics<br>PMO ASO                   | Broad RNA therapeutic portfolio<br>Leader in Duchenne exon skipping                                  |
| Nippon Shinyaku<br>Avidity Biosciences<br>ProQR Therapeutics                  | PMO ASO<br>Conjugated ASOs<br>RNA editing (Axiomer) | Partner in DMD therapies<br>Antibody-oligo conjugates<br>Focus on ADAR-mediated editing              |
| Wave Life Sciences                                                            | Stereopure ASO / editing                            | Chirality-controlled PS chemistry                                                                    |
| Beam Therapeutics<br>Phio Pharmaceuticals<br>Eyetechn (Pegaptanib originator) | RNA/base editing<br>Self-delivering RNAi<br>Aptamer | Expanding into RNA editing<br>Immuno-oncology focus<br>Acquired; program commercialized historically |
| Academic Institutions                                                         | Various (ASO, editing)                              | Key drivers of early innovation                                                                      |





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